

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
96	123	-0.78	14.86	-1.24	0.00	0.00	0.00		
	124	-0.78	15.64	-1.34	0.00	0.00	0.00		
	125	-0.85	14.92	-1.25	0.00	0.00	0.00		
	126	-0.85	15.70	-1.35	0.00	0.00	0.00		
	127	-0.81	14.00	-1.14	0.00	0.00	0.00		
	128	-0.83	14.02	-1.14	0.00	0.00	0.00		
	129	-0.81	16.54	-1.46	0.00	0.00	0.00		
	130	-0.83	16.56	-1.46	0.00	0.00	0.00		
	131	-0.89	15.41	-1.24	0.00	0.00	0.00		
	132	-0.89	16.01	-1.31	0.00	0.00	0.00		
	133	-0.94	15.46	-1.24	0.00	0.00	0.00		
	134	-0.94	16.06	-1.32	0.00	0.00	0.00		
	135	-0.91	14.77	-1.16	0.00	0.00	0.00		
	136	-0.93	14.79	-1.16	0.00	0.00	0.00		
	137	-0.91	16.68	-1.40	0.00	0.00	0.00		
	138	-0.93	16.69	-1.40	0.00	0.00	0.00		
	101	3.25	70.18	-13.12	0.00	0.00	0.00		
	102	2.51	57.16	-10.78	0.00	0.00	0.00		
	103	3.83	71.13	-5.22	0.00	0.00	0.00		
	104	3.32	62.86	-1.86	0.00	0.00	0.00		
	105	1.64	35.40	-1.47	0.00	0.00	0.00		
	106	0.58	18.20	2.83	0.00	0.00	0.00		
	107	3.81	77.59	-14.18	0.00	0.00	0.00		
	108	3.29	70.93	-13.07	0.00	0.00	0.00		
	109	0.41	17.88	-2.77	0.00	0.00	0.00		
	110	-0.95	-3.70	1.20	0.00	0.00	0.00		
	111	1.42	41.84	-16.39	0.00	0.00	0.00		
	112	0.30	26.25	-15.82	0.00	0.00	0.00		
	113	0.15	17.92	-8.50	0.00	0.00	0.00		
97	114	-1.27	-3.65	-5.97	0.00	0.00	0.00		
	115	-4.39	43.61	-9.16	0.00	0.00	0.00		
	116	-4.40	43.69	-7.78	0.00	0.00	0.00		
	117	7.96	44.57	-9.13	0.00	0.00	0.00		
	118	7.95	44.65	-7.74	0.00	0.00	0.00		
	119	-0.10	43.85	-10.72	0.00	0.00	0.00		
	120	3.67	44.14	-10.71	0.00	0.00	0.00		
	121	-0.11	44.12	-6.18	0.00	0.00	0.00		
	122	3.66	44.42	-6.19	0.00	0.00	0.00		
	123	-4.57	39.20	-8.32	0.00	0.00	0.00		
	124	-4.58	39.28	-6.93	0.00	0.00	0.00		
	125	7.78	40.16	-8.28	0.00	0.00	0.00		
	126	7.77	40.24	-6.90	0.00	0.00	0.00		
	127	-0.28	39.43	-9.88	0.00	0.00	0.00		
	128	3.49	39.73	-9.87	0.00	0.00	0.00		
	129	-0.29	39.71	-5.35	0.00	0.00	0.00		
	130	3.48	40.00	-5.33	0.00	0.00	0.00		
	131	-1.75	63.28	-12.50	0.00	0.00	0.00		
	132	-1.75	63.34	-11.43	0.00	0.00	0.00		
98	133	7.51	64.00	-12.47	0.00	0.00	0.00		
	134	7.51	64.06	-11.40	0.00	0.00	0.00		
	135	1.43	63.45	-13.65	0.00	0.00	0.00		
	136	4.34	63.68	-13.65	0.00	0.00	0.00		
	137	1.42	63.66	-10.25	0.00	0.00	0.00		
	138	4.33	63.89	-10.25	0.00	0.00	0.00		
	101	3.27	66.63	12.50	0.00	0.00	0.00		
	102	2.53	54.28	10.32	0.00	0.00	0.00		
	103	1.42	39.36	15.35	0.00	0.00	0.00		
	104	0.29	24.49	14.82	0.00	0.00	0.00		
	105	0.20	16.82	8.01	0.00	0.00	0.00		
	106	-1.23	-3.69	5.65	0.00	0.00	0.00		
	107	3.84	73.67	13.45	0.00	0.00	0.00		
	108	3.32	67.37	12.44	0.00	0.00	0.00		
	109	0.41	17.01	2.73	0.00	0.00	0.00		
	110	-0.98	-3.44	-0.96	0.00	0.00	0.00		
	111	3.86	67.86	5.27	0.00	0.00	0.00		
	112	3.35	60.12	2.22	0.00	0.00	0.00		
	113	1.59	33.83	1.65	0.00	0.00	0.00		
	114	0.51	17.57	-2.30	0.00	0.00	0.00		
	115	-3.95	41.52	7.47	0.00	0.00	0.00		
	116	-3.95	41.43	8.83	0.00	0.00	0.00		
	117	7.52	42.41	7.44	0.00	0.00	0.00		
	118	7.53	42.32	8.80	0.00	0.00	0.00		
	119	0.03	41.93	5.92	0.00	0.00	0.00		
	120	3.54	42.20	5.91	0.00	0.00	0.00		
	121	0.04	41.64	10.36	0.00	0.00	0.00		
	122	3.54	41.91	10.35	0.00	0.00	0.00		
	123	-4.13	37.33	6.66	0.00	0.00	0.00		
	124	-4.13	37.24	8.02	0.00	0.00	0.00		
	125	7.35	38.22	6.62	0.00	0.00	0.00		
	126	7.35	38.13	7.98	0.00	0.00	0.00		
	127	-0.15	37.74	5.10	0.00	0.00	0.00		
	128	3.36	38.01	5.09	0.00	0.00	0.00		
	129	-0.14	37.44	9.55	0.00	0.00	0.00		
	130	3.37	37.72	9.54	0.00	0.00	0.00		
	131	-1.40	60.15	10.90	0.00	0.00	0.00		
	132	-1.40	60.08	11.95	0.00	0.00	0.00		
	133	7.20	60.82	10.87	0.00	0.00	0.00		
	134	7.21	60.75	11.92	0.00	0.00	0.00		
	135	1.54	60.46	9.75	0.00	0.00	0.00		
	136	4.25	60.67	9.74	0.00	0.00	0.00		
	137	1.55	60.24	13.08	0.00	0.00	0.00		
	138	4.26	60.45	13.07	0.00	0.00	0.00		
	101	0.01	58.82	-2.93	0.00	0.00	0.00		
	102	0.01	47.94	-2.57	0.00	0.00	0.00		
	103	0.01	60.31	4.06	0.00	0.00	0.00		
	104	0.01	53.62	5.80	0.00	0.00	0.00		

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

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JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
105	0.01	30.26	3.00	0.00	0.00	0.00	0.00
106	0.01	16.05	4.46	0.00	0.00	0.00	0.00
107	0.01	64.98	-2.73	0.00	0.00	0.00	0.00
108	0.01	59.46	-2.70	0.00	0.00	0.00	0.00
109	0.01	15.22	-0.24	0.00	0.00	0.00	0.00
110	0.01	-2.74	0.41	0.00	0.00	0.00	0.00
111	0.00	34.31	-9.39	0.00	0.00	0.00	0.00
112	0.00	21.12	-11.03	0.00	0.00	0.00	0.00
113	0.00	14.70	-5.48	0.00	0.00	0.00	0.00
114	0.00	-3.40	-6.13	0.00	0.00	0.00	0.00
115	0.01	37.52	-2.57	0.00	0.00	0.00	0.00
116	0.01	37.61	-1.13	0.00	0.00	0.00	0.00
117	0.01	36.51	-3.29	0.00	0.00	0.00	0.00
118	0.01	36.60	-1.85	0.00	0.00	0.00	0.00
119	0.01	37.06	-4.46	0.00	0.00	0.00	0.00
120	0.01	36.75	-4.68	0.00	0.00	0.00	0.00
121	0.01	37.36	0.26	0.00	0.00	0.00	0.00
122	0.01	37.05	0.04	0.00	0.00	0.00	0.00
123	0.00	33.81	-2.35	0.00	0.00	0.00	0.00
124	0.00	33.90	-0.91	0.00	0.00	0.00	0.00
125	0.01	32.80	-3.07	0.00	0.00	0.00	0.00
126	0.01	32.89	-1.63	0.00	0.00	0.00	0.00
127	0.01	33.35	-4.24	0.00	0.00	0.00	0.00
128	0.01	33.05	-4.46	0.00	0.00	0.00	0.00
129	0.01	33.66	0.48	0.00	0.00	0.00	0.00
130	0.01	33.35	0.26	0.00	0.00	0.00	0.00
131	0.01	53.73	-3.04	0.00	0.00	0.00	0.00
132	0.01	53.80	-1.92	0.00	0.00	0.00	0.00
133	0.01	52.97	-3.58	0.00	0.00	0.00	0.00
134	0.01	53.04	-2.46	0.00	0.00	0.00	0.00
135	0.01	53.39	-4.43	0.00	0.00	0.00	0.00
136	0.01	53.15	-4.60	0.00	0.00	0.00	0.00
137	0.01	53.62	-0.89	0.00	0.00	0.00	0.00
138	0.01	53.38	-1.06	0.00	0.00	0.00	0.00
99	0.01	55.53	2.59	0.00	0.00	0.00	0.00
102	0.01	45.27	2.32	0.00	0.00	0.00	0.00
103	0.00	32.09	8.50	0.00	0.00	0.00	0.00
104	0.00	19.59	10.10	0.00	0.00	0.00	0.00
105	0.00	13.71	5.10	0.00	0.00	0.00	0.00
106	0.00	-3.37	5.84	0.00	0.00	0.00	0.00
107	0.01	61.35	2.30	0.00	0.00	0.00	0.00
108	0.01	56.17	2.35	0.00	0.00	0.00	0.00
109	0.00	14.43	0.26	0.00	0.00	0.00	0.00
110	0.00	-2.40	-0.20	0.00	0.00	0.00	0.00
111	0.01	57.20	-3.73	0.00	0.00	0.00	0.00
112	0.01	50.99	-5.19	0.00	0.00	0.00	0.00
113	0.01	28.77	-2.72	0.00	0.00	0.00	0.00
114	0.01	15.45	-3.93	0.00	0.00	0.00	0.00
100	0.01	58.81	-2.78	0.00	0.00	0.00	0.00
101	-0.01	47.94	-2.45	0.00	0.00	0.00	0.00
102	-0.01	60.30	4.21	0.00	0.00	0.00	0.00
103	-0.01	53.61	5.94	0.00	0.00	0.00	0.00
105	-0.01	30.25	3.08	0.00	0.00	0.00	0.00
106	-0.01	16.06	4.53	0.00	0.00	0.00	0.00
107	-0.01	64.97	-2.58	0.00	0.00	0.00	0.00
108	-0.01	59.45	-2.56	0.00	0.00	0.00	0.00
109	-0.01	15.23	-0.19	0.00	0.00	0.00	0.00
110	-0.01	-2.73	0.43	0.00	0.00	0.00	0.00
111	0.00	34.37	-9.40	0.00	0.00	0.00	0.00
112	0.00	21.20	-11.08	0.00	0.00	0.00	0.00
113	0.00	10.26	0.67	0.00	0.00	0.00	0.00
114	0.00	-8.93	1.51	0.00	0.00	0.00	0.00
115	-0.01	36.50	-3.20	0.00	0.00	0.00	0.00
116	-0.01	36.60	-1.76	0.00	0.00	0.00	0.00
117	-0.01	37.51	-2.47	0.00	0.00	0.00	0.00
118	-0.01	37.61	-1.03	0.00	0.00	0.00	0.00
119	-0.01	36.75	-4.58	0.00	0.00	0.00	0.00
120	-0.01	37.06	-4.36	0.00	0.00	0.00	0.00
121	-0.01	37.05	0.13	0.00	0.00	0.00	0.00
122	-0.01	37.36	0.36	0.00	0.00	0.00	0.00
123	-0.01	32.80	-2.99	0.00	0.00	0.00	0.00
124	-0.01	32.89	-1.55	0.00	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

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JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
101	125	0.00	33.81	-2.26	0.00	0.00	0.00
	126	0.00	33.90	-0.82	0.00	0.00	0.00
	127	-0.01	33.04	-4.37	0.00	0.00	0.00
	128	-0.01	33.35	-4.15	0.00	0.00	0.00
	129	-0.01	33.35	0.35	0.00	0.00	0.00
	130	-0.01	33.66	0.57	0.00	0.00	0.00
	131	-0.01	52.96	-3.45	0.00	0.00	0.00
	132	-0.01	53.03	-2.33	0.00	0.00	0.00
	133	-0.01	53.72	-2.90	0.00	0.00	0.00
	134	-0.01	53.79	-1.79	0.00	0.00	0.00
	135	-0.01	53.14	-4.47	0.00	0.00	0.00
	136	-0.01	53.38	-4.30	0.00	0.00	0.00
	137	-0.01	53.37	-0.93	0.00	0.00	0.00
	138	-0.01	53.61	-0.76	0.00	0.00	0.00
	101	-0.01	55.53	2.66	0.00	0.00	0.00
	102	-0.01	45.27	2.38	0.00	0.00	0.00
	103	0.00	32.09	8.59	0.00	0.00	0.00
	104	0.00	19.59	10.18	0.00	0.00	0.00
	105	0.00	13.71	5.14	0.00	0.00	0.00
	106	0.00	-3.38	5.86	0.00	0.00	0.00
103	107	-0.01	61.35	2.39	0.00	0.00	0.00
	108	-0.01	56.17	2.43	0.00	0.00	0.00
	109	0.00	14.43	0.28	0.00	0.00	0.00
	110	0.00	-2.48	-0.21	0.00	0.00	0.00
	111	-0.01	57.26	-3.61	0.00	0.00	0.00
	112	-0.01	51.06	-5.07	0.00	0.00	0.00
	113	-0.01	24.70	-8.53	0.00	0.00	0.00
	114	-0.01	10.36	-11.22	0.00	0.00	0.00
	115	-0.01	34.58	1.89	0.00	0.00	0.00
	116	-0.01	34.48	3.31	0.00	0.00	0.00
	117	0.00	35.55	0.90	0.00	0.00	0.00
	118	0.00	35.45	2.31	0.00	0.00	0.00
	119	-0.01	35.03	-0.06	0.00	0.00	0.00
	120	-0.01	35.33	-0.37	0.00	0.00	0.00
	121	-0.01	34.70	4.57	0.00	0.00	0.00
	122	-0.01	35.00	4.27	0.00	0.00	0.00
	123	-0.01	31.08	1.68	0.00	0.00	0.00
	124	-0.01	30.98	3.10	0.00	0.00	0.00
	125	0.00	32.05	0.69	0.00	0.00	0.00
	126	0.00	31.95	2.10	0.00	0.00	0.00
103	127	-0.01	31.53	-0.27	0.00	0.00	0.00
	128	-0.01	31.82	-0.58	0.00	0.00	0.00
	129	-0.01	31.20	4.36	0.00	0.00	0.00
	130	-0.01	31.50	4.06	0.00	0.00	0.00
	131	-0.01	50.07	2.34	0.00	0.00	0.00
	132	-0.01	50.00	3.44	0.00	0.00	0.00
	133	-0.01	50.81	1.60	0.00	0.00	0.00
	134	-0.01	50.73	2.69	0.00	0.00	0.00
	103	0.00	66.63	12.50	0.00	0.00	0.00
	104	-0.01	54.28	10.32	0.00	0.00	0.00
102	105	-0.01	39.36	15.35	0.00	0.00	0.00
	106	-0.31	24.49	14.82	0.00	0.00	0.00
	107	-0.21	16.82	8.01	0.00	0.00	0.00
	108	1.23	-3.69	5.65	0.00	0.00	0.00
	109	-0.43	17.88	-2.77	0.00	0.00	0.00
	110	0.96	-3.70	1.20	0.00	0.00	0.00
	111	-1.45	41.91	-16.48	0.00	0.00	0.00
	112	-0.32	26.33	-15.93	0.00	0.00	0.00
	113	-0.19	13.48	-2.40	0.00	0.00	0.00
	114	1.26	-9.20	1.66	0.00	0.00	0.00
	115	-8.01	44.57	-9.13	0.00	0.00	0.00
	116	-8.01	44.66	-7.74	0.00	0.00	0.00
	117	4.36	43.61	-9.16	0.00	0.00	0.00
	118	4.36	43.70	-7.78	0.00	0.00	0.00
	119	-3.72	44.14	-10.71	0.00	0.00	0.00
	120	0.06	43.85	-10.72	0.00	0.00	0.00
	121	-3.71	44.42	-6.18	0.00	0.00	0.00
	122	0.07	44.13	-6.19	0.00	0.00	0.00
	123	-7.83	40.16	-8.28	0.00	0.00	0.00
	124	-7.83	40.24	-6.90	0.00	0.00	0.00
	125	4.54	39.20	-8.31	0.00	0.00	0.00
103	126	4.55	39.28	-6.93	0.00	0.00	0.00
	127	-3.54	39.73	-9.87	0.00	0.00	0.00
	128	0.24	39.44	-9.88	0.00	0.00	0.00
	129	-3.53	40.01	-5.34	0.00	0.00	0.00
	130	0.25	39.71	-5.35	0.00	0.00	0.00
	131	-7.59	64.00	-12.47	0.00	0.00	0.00
	132	-7.59	64.07	-11.40	0.00	0.00	0.00
	133	1.69	63.28	-12.50	0.00	0.00	0.00
	134	1.69	63.35	-11.43	0.00	0.00	0.00
	135	-4.41	63.69	-13.64	0.00	0.00	0.00
	136	-1.49	63.46	-13.65	0.00	0.00	0.00
	137	-4.40	63.89	-10.25	0.00	0.00	0.00
	138	-1.48	63.66	-10.25	0.00	0.00	0.00
	103	-3.29	66.63	12.50	0.00	0.00	0.00
	104	-2.54	54.28	10.32	0.00	0.00	0.00
	105	-1.44	39.36	15.35	0.00	0.00	0.00
	106	-0.31	24.49	14.82	0.00	0.00	0.00
	107	-0.21	16.82	8.01	0.00	0.00	0.00
	108	1.23	-3.69	5.65	0.00	0.00	0.00
	109	-0.43	17.88	-2.77	0.00	0.00	0.00
	110	0.96	-3.70	1.20	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		

104	107	-3.86	73.67	13.45	0.00	0.00	0.00		
	108	-3.33	67.37	12.44	0.00	0.00	0.00		
	109	-0.41	17.02	2.73	0.00	0.00	0.00		
	110	0.98	-3.44	-0.96	0.00	0.00	0.00		
	111	-3.87	67.92	5.36	0.00	0.00	0.00		
	112	-3.35	60.19	2.33	0.00	0.00	0.00		
	113	-1.62	29.75	-4.19	0.00	0.00	0.00		
	114	-0.53	12.48	-9.60	0.00	0.00	0.00		
	115	-7.53	42.41	7.44	0.00	0.00	0.00		
	116	-7.54	42.32	8.80	0.00	0.00	0.00		
	117	3.94	41.52	7.47	0.00	0.00	0.00		
	118	3.94	41.43	8.83	0.00	0.00	0.00		
	119	-3.55	42.20	5.91	0.00	0.00	0.00		
	120	-0.04	41.93	5.92	0.00	0.00	0.00		
	121	-3.55	41.91	10.35	0.00	0.00	0.00		
	122	-0.05	41.64	10.36	0.00	0.00	0.00		
	123	-7.35	38.22	6.63	0.00	0.00	0.00		
	124	-7.36	38.13	7.98	0.00	0.00	0.00		
	125	4.12	37.33	6.66	0.00	0.00	0.00		
	126	4.12	37.24	8.02	0.00	0.00	0.00		
	127	-3.37	38.01	5.09	0.00	0.00	0.00		
	128	0.14	37.74	5.10	0.00	0.00	0.00		
	129	-3.37	37.72	9.54	0.00	0.00	0.00		
	130	0.13	37.45	9.55	0.00	0.00	0.00		
	131	-7.22	60.82	10.87	0.00	0.00	0.00		
	132	-7.22	60.75	11.92	0.00	0.00	0.00		
	133	1.39	60.15	10.90	0.00	0.00	0.00		
	134	1.39	60.08	11.95	0.00	0.00	0.00		
	135	-4.27	60.67	9.74	0.00	0.00	0.00		
	136	-1.56	60.46	9.75	0.00	0.00	0.00		
	137	-4.27	60.45	13.07	0.00	0.00	0.00		
	138	-1.56	60.24	13.08	0.00	0.00	0.00		
	104	101	22.34	44.19	-7.93	0.00	0.00		
	102	17.34	37.81	-7.34	0.00	0.00	0.00		
	103	25.70	50.61	-8.97	0.00	0.00	0.00		
	104	22.12	50.50	-10.04	0.00	0.00	0.00		
	105	10.82	42.47	-8.21	0.00	0.00	0.00		
	106	3.52	40.32	-9.08	0.00	0.00	0.00		
	107	25.40	56.05	-8.64	0.00	0.00	0.00		
	108	21.75	57.30	-9.62	0.00	0.00	0.00		
	109	2.52	38.06	-7.52	0.00	0.00	0.00		
	110	-6.85	34.81	-8.22	0.00	0.00	0.00		
	111	11.22	2.22	-1.33	0.00	0.00	0.00		
	112	4.02	-9.99	-0.49	0.00	0.00	0.00		
	113	2.72	2.81	-2.37	0.00	0.00	0.00		
	114	-6.60	-9.26	-1.79	0.00	0.00	0.00		
	115	2.48	30.73	-7.95	0.00	0.00	0.00		
	116	2.47	31.51	-5.63	0.00	0.00	0.00		
105	117	22.21	31.34	-7.88	0.00	0.00	0.00		
	118	22.20	32.13	-5.56	0.00	0.00	0.00		
	119	9.35	30.05	-10.56	0.00	0.00	0.00		
	120	15.38	30.24	-10.54	0.00	0.00	0.00		
	121	9.31	32.61	-2.97	0.00	0.00	0.00		
	122	15.33	32.80	-2.94	0.00	0.00	0.00		
	123	1.25	27.58	-7.27	0.00	0.00	0.00		
	124	1.24	28.37	-4.95	0.00	0.00	0.00		
	125	20.98	28.20	-7.20	0.00	0.00	0.00		
	126	20.97	28.98	-4.88	0.00	0.00	0.00		
	127	8.11	26.91	-9.89	0.00	0.00	0.00		
	128	14.14	27.10	-9.86	0.00	0.00	0.00		
	129	8.07	29.47	-2.29	0.00	0.00	0.00		
	130	14.10	29.66	-2.27	0.00	0.00	0.00		
	131	12.45	40.46	-8.56	0.00	0.00	0.00		
	132	12.44	41.07	-6.77	0.00	0.00	0.00		
	133	27.25	40.93	-8.51	0.00	0.00	0.00		
	134	27.24	41.53	-6.71	0.00	0.00	0.00		
	135	17.53	39.96	-10.49	0.00	0.00	0.00		
	136	22.19	40.11	-10.48	0.00	0.00	0.00		
	137	17.50	41.89	-4.80	0.00	0.00	0.00		
	138	22.16	42.03	-4.78	0.00	0.00	0.00		
	101	22.47	45.52	8.24	0.00	0.00	0.00		
	102	17.44	38.86	7.58	0.00	0.00	0.00		
	103	11.29	2.91	1.51	0.00	0.00	0.00		
	104	4.05	-9.69	0.56	0.00	0.00	0.00		
	105	2.71	3.02	2.39	0.00	0.00	0.00		
	106	-6.67	-9.55	1.66	0.00	0.00	0.00		
	107	25.55	57.55	9.03	0.00	0.00	0.00		
	108	21.87	58.61	9.96	0.00	0.00	0.00		
	109	2.52	38.28	7.57	0.00	0.00	0.00		
	110	-6.91	34.52	8.14	0.00	0.00	0.00		
	111	25.85	52.11	9.26	0.00	0.00	0.00		
	112	22.24	51.81	10.25	0.00	0.00	0.00		
	113	10.88	43.14	8.34	0.00	0.00	0.00		
	114	3.54	40.59	9.10	0.00	0.00	0.00		
	115	2.31	32.27	5.77	0.00	0.00	0.00		
	116	2.32	31.49	8.13	0.00	0.00	0.00		
	117	22.49	32.90	5.70	0.00	0.00	0.00		
	118	22.50	32.12	8.06	0.00	0.00	0.00		
	119	9.31	33.38	3.07	0.00	0.00	0.00		
	120	15.47	33.57	3.05	0.00	0.00	0.00		
	121	9.34	30.82	10.78	0.00	0.00	0.00		
	122	15.51	31.01	10.76	0.00	0.00	0.00		
	123	1.07	29.05	5.08	0.00	0.00	0.00		
	124	1.08	28.27	7.43	0.00	0.00	0.00		
	125	21.25	29.68	5.01	0.00	0.00	0.00		
	126	21.26	28.90	7.37	0.00	0.00	0.00		

SUPPORT REACTIONS -UNIT KN										STRUCTURE TYPE = SPACE					
METE										METE					
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JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
106	127	8.06	30.16	2.38	0.00	0.00	0.00	107	137	0.42	50.10	-9.21	0.00	0.00	0.00
	128	14.23	30.35	2.36	0.00	0.00	0.00		138	0.82	49.77	-9.97	0.00	0.00	0.00
	129	8.10	27.60	10.09	0.00	0.00	0.00		101	0.55	54.09	13.69	0.00	0.00	0.00
	130	14.27	27.79	10.07	0.00	0.00	0.00		102	0.78	46.45	12.33	0.00	0.00	0.00
	131	12.38	42.26	7.02	0.00	0.00	0.00		103	0.76	1.53	4.51	0.00	0.00	0.00
	132	12.39	41.65	8.84	0.00	0.00	0.00		104	1.41	-13.36	2.91	0.00	0.00	0.00
	133	27.52	42.73	6.97	0.00	0.00	0.00		105	1.15	3.22	4.45	0.00	0.00	0.00
	134	27.53	42.13	8.79	0.00	0.00	0.00		106	1.89	-11.25	2.84	0.00	0.00	0.00
	135	17.56	43.08	5.03	0.00	0.00	0.00		107	0.09	69.17	14.98	0.00	0.00	0.00
	136	22.32	43.23	5.01	0.00	0.00	0.00		108	0.57	71.19	16.00	0.00	0.00	0.00
	137	17.59	41.16	10.81	0.00	0.00	0.00		109	1.12	48.74	10.83	0.00	0.00	0.00
	138	22.35	41.31	10.79	0.00	0.00	0.00		110	1.86	45.65	10.80	0.00	0.00	0.00
	101	0.50	52.43	-13.66	0.00	0.00	0.00		111	0.08	61.67	14.23	0.00	0.00	0.00
	102	0.74	45.13	-12.32	0.00	0.00	0.00		112	0.56	61.81	15.06	0.00	0.00	0.00
	103	0.03	59.87	-14.13	0.00	0.00	0.00		113	0.75	53.18	11.96	0.00	0.00	0.00
107	104	0.51	60.24	-14.98	0.00	0.00	0.00	114	1.40	51.20	12.22	0.00	0.00	0.00	
	105	0.72	52.34	-11.95	0.00	0.00	0.00	115	0.15	39.96	8.02	0.00	0.00	0.00	
	106	1.37	50.82	-12.25	0.00	0.00	0.00	116	0.15	39.02	10.83	0.00	0.00	0.00	
	107	0.04	67.33	-14.90	0.00	0.00	0.00	117	1.86	38.61	11.11	0.00	0.00	0.00	
	108	0.52	69.56	-15.94	0.00	0.00	0.00	118	1.86	37.67	13.92	0.00	0.00	0.00	
	109	1.10	48.42	-10.83	0.00	0.00	0.00	119	0.75	40.56	5.90	0.00	0.00	0.00	
	110	1.85	45.92	-10.85	0.00	0.00	0.00	120	1.27	40.14	6.84	0.00	0.00	0.00	
	111	0.74	0.62	-4.60	0.00	0.00	0.00	121	0.75	37.48	15.09	0.00	0.00	0.00	
	112	1.39	-13.83	-3.06	0.00	0.00	0.00	122	1.27	37.07	16.03	0.00	0.00	0.00	
	113	1.13	2.89	-4.53	0.00	0.00	0.00	123	0.05	36.08	6.92	0.00	0.00	0.00	
	114	1.88	-10.99	-2.98	0.00	0.00	0.00	124	0.05	35.14	9.73	0.00	0.00	0.00	
	115	0.14	38.05	-10.75	0.00	0.00	0.00	125	1.76	34.73	10.01	0.00	0.00	0.00	
	116	0.14	38.99	-7.97	0.00	0.00	0.00	126	1.76	33.79	12.82	0.00	0.00	0.00	
	117	1.81	36.66	-13.99	0.00	0.00	0.00	127	0.65	36.67	4.80	0.00	0.00	0.00	
	118	1.81	37.61	-11.21	0.00	0.00	0.00	128	1.17	36.26	5.74	0.00	0.00	0.00	
108	119	0.72	36.50	-15.02	0.00	0.00	0.00	129	0.65	33.60	13.99	0.00	0.00	0.00	
	120	1.23	36.08	-16.01	0.00	0.00	0.00	130	1.17	33.19	14.94	0.00	0.00	0.00	
	121	0.72	39.58	-5.95	0.00	0.00	0.00	131	0.02	51.14	10.77	0.00	0.00	0.00	
	122	1.23	39.16	-6.94	0.00	0.00	0.00	132	0.02	50.41	12.94	0.00	0.00	0.00	
	123	0.04	34.27	-9.65	0.00	0.00	0.00	133	1.31	50.12	13.08	0.00	0.00	0.00	
	124	0.04	35.21	-6.88	0.00	0.00	0.00	134	1.31	49.40	15.26	0.00	0.00	0.00	
	125	1.71	32.88	-12.89	0.00	0.00	0.00	135	0.46	51.58	9.20	0.00	0.00	0.00	
	126	1.71	33.82	-10.12	0.00	0.00	0.00	136	0.87	51.26	9.93	0.00	0.00	0.00	
	127	0.62	32.72	-13.92	0.00	0.00	0.00	137	0.46	49.27	16.09	0.00	0.00	0.00	
	128	1.13	32.29	-14.91	0.00	0.00	0.00	138	0.87	48.96	16.82	0.00	0.00	0.00	
	129	0.62	35.80	-4.85	0.00	0.00	0.00	101	-0.50	52.43	-13.72	0.00	0.00	0.00	
	130	1.13	35.37	-5.84	0.00	0.00	0.00	102	-0.73	45.13	-12.37	0.00	0.00	0.00	
	131	-0.01	48.94	-12.85	0.00	0.00	0.00	103	-0.03	59.87	-14.19	0.00	0.00	0.00	
	132	-0.01	49.66	-10.71	0.00	0.00	0.00	104	-0.50	60.24	-15.04	0.00	0.00	0.00	
	133	1.25	47.90	-15.28	0.00	0.00	0.00	105	-0.72	52.34	-11.98	0.00	0.00	0.00	
	134	1.25	48.82	-13.14	0.00	0.00	0.00	106	-1.37	50.83	-12.27	0.00	0.00	0.00	
	135	0.42	47.79	-16.01	0.00	0.00	0.00	107	-0.03	67.33	-14.96	0.00	0.00	0.00	
	136	0.82	47.46	-16.78	0.00	0.00	0.00	108	-0.51	69.56	-16.00	0.00	0.00	0.00	

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

109	109	-1.10	48.43	-10.85	0.00	0.00	0.00		
	110	-1.85	45.94	-10.86	0.00	0.00	0.00		
	111	-0.73	0.62	-4.61	0.00	0.00	0.00		
	112	-1.39	-13.83	-3.06	0.00	0.00	0.00		
	113	-1.12	3.41	-5.70	0.00	0.00	0.00		
	114	-1.88	-10.34	-4.43	0.00	0.00	0.00		
	115	-1.79	36.68	-14.01	0.00	0.00	0.00		
	116	-1.79	37.62	-11.24	0.00	0.00	0.00		
	117	-0.14	38.05	-10.80	0.00	0.00	0.00		
	118	-0.14	38.99	-8.02	0.00	0.00	0.00		
	119	-1.22	36.08	-16.05	0.00	0.00	0.00		
110	120	-0.72	36.50	-15.07	0.00	0.00	0.00		
	121	-1.22	39.17	-6.97	0.00	0.00	0.00		
	122	-0.72	39.59	-5.99	0.00	0.00	0.00		
	123	-1.70	32.89	-12.91	0.00	0.00	0.00		
	124	-1.70	33.84	-10.14	0.00	0.00	0.00		
	125	-0.05	34.27	-9.69	0.00	0.00	0.00		
	126	-0.05	35.21	-6.92	0.00	0.00	0.00		
	127	-1.12	32.30	-14.95	0.00	0.00	0.00		
	128	-0.62	32.72	-13.96	0.00	0.00	0.00		
	129	-1.12	35.38	-5.87	0.00	0.00	0.00		
	130	-0.62	35.80	-4.88	0.00	0.00	0.00		
110	131	-1.23	47.91	-15.32	0.00	0.00	0.00		
	132	-1.23	48.63	-13.18	0.00	0.00	0.00		
	133	0.00	48.94	-12.91	0.00	0.00	0.00		
	134	0.00	49.66	-10.76	0.00	0.00	0.00		
	135	-0.81	47.47	-16.83	0.00	0.00	0.00		
	136	-0.42	47.79	-16.07	0.00	0.00	0.00		
	137	-0.81	49.78	-10.02	0.00	0.00	0.00		
	138	-0.42	50.10	-9.26	0.00	0.00	0.00		
	109	-0.55	54.06	13.76	0.00	0.00	0.00		
	102	-0.78	46.43	12.38	0.00	0.00	0.00		
	103	-0.76	1.51	4.59	0.00	0.00	0.00		
110	104	-1.41	-13.38	2.98	0.00	0.00	0.00		
	105	-1.15	3.21	4.49	0.00	0.00	0.00		
	106	-1.89	-11.26	2.86	0.00	0.00	0.00		
	107	-0.09	69.14	15.06	0.00	0.00	0.00		
	108	-0.57	71.16	16.07	0.00	0.00	0.00		
	109	-1.12	48.73	10.84	0.00	0.00	0.00		
	110	-1.86	45.65	10.80	0.00	0.00	0.00		
	111	-0.08	61.65	14.25	0.00	0.00	0.00		
	112	-0.56	61.79	15.06	0.00	0.00	0.00		
	113	-0.75	53.73	12.96	0.00	0.00	0.00		
	114	-1.40	51.89	13.45	0.00	0.00	0.00		
110	115	-1.86	38.60	11.14	0.00	0.00	0.00		
	116	-1.86	37.66	13.96	0.00	0.00	0.00		
	117	-0.15	39.94	8.06	0.00	0.00	0.00		
	118	-0.15	39.00	10.88	0.00	0.00	0.00		
	119	-1.76	34.72	10.04	0.00	0.00	0.00		
	120	-1.76	33.78	12.85	0.00	0.00	0.00		
	121	-0.05	36.06	6.96	0.00	0.00	0.00		
	122	-0.05	35.12	9.77	0.00	0.00	0.00		
	123	-1.17	36.26	5.77	0.00	0.00	0.00		
	124	-0.65	36.67	4.83	0.00	0.00	0.00		
	125	-1.17	33.17	14.99	0.00	0.00	0.00		
110	126	-0.65	33.58	14.04	0.00	0.00	0.00		
	127	-1.31	50.11	13.14	0.00	0.00	0.00		
	128	-0.02	51.12	10.83	0.00	0.00	0.00		
	129	-0.02	50.39	13.01	0.00	0.00	0.00		
	130	-0.87	51.24	9.98	0.00	0.00	0.00		
	131	-0.46	51.56	9.25	0.00	0.00	0.00		
	132	-0.87	48.93	16.89	0.00	0.00	0.00		
	133	-0.46	49.25	16.17	0.00	0.00	0.00		
	134	-22.26	44.19	-7.93	0.00	0.00	0.00		
	135	-17.27	37.81	-7.34	0.00	0.00	0.00		
	136	-25.60	50.61	-8.97	0.00	0.00	0.00		
110	137	-22.03	50.50	-10.04	0.00	0.00	0.00		
	138	-10.77	42.47	-8.21	0.00	0.00	0.00		
	139	-3.49	40.32	-9.08	0.00	0.00	0.00		
	140	-25.30	56.05	-8.64	0.00	0.00	0.00		
	141	-21.66	57.30	-9.62	0.00	0.00	0.00		
	142	-2.50	38.06	-7.52	0.00	0.00	0.00		
	143	6.84	34.81	-8.22	0.00	0.00	0.00		
	144	-11.17	2.21	-1.31	0.00	0.00	0.00		
	145	-4.00	-10.00	-0.46	0.00	0.00	0.00		
	146	-2.68	3.31	-3.55	0.00	0.00	0.00		
	147	6.62	-8.63	-3.26	0.00	0.00	0.00		
110	148	-22.09	31.34	-7.88	0.00	0.00	0.00		
	149	-22.08	32.12	-5.56	0.00	0.00	0.00		
	150	-2.49	39.73	-7.95	0.00	0.00	0.00		
	151	-2.48	31.51	-5.63	0.00	0.00	0.00		
	152	-15.30	30.24	-10.54	0.00	0.00	0.00		
	153	-9.31	30.05	-10.56	0.00	0.00	0.00		
	154	-15.26	32.80	-2.95	0.00	0.00	0.00		
	155	-9.27	32.61	-2.97	0.00	0.00	0.00		
	156	-20.86	28.20	-7.20	0.00	0.00	0.00		
	157	-20.85	28.98	-4.88	0.00	0.00	0.00		
	158	-1.27	27.58	-7.27	0.00	0.00	0.00		
110	159	-1.25	28.37	-4.95	0.00	0.00	0.00		
	160	-14.07	27.09	-9.86	0.00	0.00	0.00		
	161	-8.08	26.91	-9.88	0.00	0.00	0.00		
	162	-1.27	40.14	6.87	0.00	0.00	0.00		
	163	-0.75	40.55	5.93	0.00	0.00	0.00		
	164	-1.27	37.05	16.09	0.00	0.00	0.00		
	165	-0.75	37.46	15.15	0.00	0.00	0.00		
	166	-1.76	34.72	10.04	0.00	0.00	0.00		
	167	-1.76	33.78	12.85	0.00	0.00	0.00		
	168	-0.05	36.06	6.96	0.00	0.00	0.00		
	169	-0.05	35.12	9.77	0.00	0.00	0.00		

SUPPORT REACTIONS -UNIT KN										STRUCTURE TYPE = SPACE			
METE										METE			

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z	JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X

111	129	-14.03	29.66	-2.27	0.00	0.00	0.00	112	101	-0.01	2.30	0.41	0.00
	130	-8.04	29.47	-2.29	0.00	0.00	0.00		102	0.02	2.12	1.37	0.00
	131	-27.12	40.92	-8.51	0.00	0.00	0.00		103	-0.03	5.07	-5.31	0.00
	132	-27.11	41.53	-6.71	0.00	0.00	0.00		104	0.00	5.98	-4.71	0.00
	133	-12.42	40.46	-8.56	0.00	0.00	0.00		105	0.04	11.69	-10.32	0.00
	134	-12.41	41.07	-6.77	0.00	0.00	0.00		106	0.10	14.25	-10.97	0.00
	135	-22.09	40.11	-10.48	0.00	0.00	0.00		107	-0.02	9.27	-9.30	0.00
	136	-17.46	39.96	-10.49	0.00	0.00	0.00		108	0.01	11.24	-9.70	0.00
	137	-22.06	42.03	-4.78	0.00	0.00	0.00		109	0.09	15.65	-13.19	0.00
	138	-17.43	41.89	-4.80	0.00	0.00	0.00		110	0.15	19.21	-14.56	0.00
	101	-22.47	45.52	8.24	0.00	0.00	0.00		111	-0.01	-13.83	20.02	0.00
	102	-17.44	38.86	7.58	0.00	0.00	0.00		112	0.03	-17.65	26.96	0.00
	103	-11.29	2.91	1.51	0.00	0.00	0.00		113	0.04	-6.93	13.04	0.00
	104	-4.05	-9.69	0.56	0.00	0.00	0.00		114	0.09	-9.01	18.23	0.00
	105	-2.72	3.02	2.39	0.00	0.00	0.00		115	0.01	2.02	1.77	0.00
	106	6.67	-9.55	1.66	0.00	0.00	0.00		116	0.01	1.84	2.91	0.00
	107	-25.55	57.55	9.03	0.00	0.00	0.00		117	0.06	2.04	1.76	0.00
	113	108	-21.87	58.61	9.96	0.00	0.00		0.00	118	0.06	1.86	2.90
109		-2.52	38.28	7.57	0.00	0.00	0.00	119	0.03	2.23	0.47	0.00	
110		6.91	34.52	8.14	0.00	0.00	0.00	120	0.04	2.24	0.47	0.00	
111		-25.85	52.10	9.25	0.00	0.00	0.00	121	0.03	1.65	4.20	0.00	
112		-22.25	51.80	10.23	0.00	0.00	0.00	122	0.04	1.65	4.20	0.00	
113		-10.87	43.69	9.34	0.00	0.00	0.00	123	0.01	1.83	1.53	0.00	
114		-3.53	41.29	10.35	0.00	0.00	0.00	124	0.01	1.65	2.67	0.00	
115		-22.49	32.90	5.70	0.00	0.00	0.00	125	0.05	1.85	1.53	0.00	
116		-22.50	32.12	8.06	0.00	0.00	0.00	126	0.05	1.67	2.67	0.00	
117		-2.32	32.27	5.77	0.00	0.00	0.00	127	0.03	2.04	0.24	0.00	
118		-2.33	31.49	8.12	0.00	0.00	0.00	128	0.04	2.04	0.23	0.00	
119		-15.47	33.57	3.06	0.00	0.00	0.00	129	0.03	1.45	3.97	0.00	
120		-9.31	33.38	3.08	0.00	0.00	0.00	130	0.04	1.46	3.97	0.00	
121		-15.51	31.01	10.75	0.00	0.00	0.00	131	-0.01	2.27	0.45	0.00	
122		-9.35	30.82	10.77	0.00	0.00	0.00	132	-0.01	2.14	1.33	0.00	
123		-21.25	29.68	5.01	0.00	0.00	0.00	133	0.02	2.29	0.45	0.00	
124		-21.26	28.90	7.36	0.00	0.00	0.00	134	0.02	2.15	1.33	0.00	
125		-1.08	29.05	5.08	0.00	0.00	0.00	135	0.00	2.43	-0.51	0.00	
126	-1.09	28.27	7.43	0.00	0.00	0.00	136	0.01	2.43	-0.51	0.00		
127	-14.23	30.35	2.37	0.00	0.00	0.00	137	0.00	1.99	2.29	0.00		
128	-8.07	30.16	2.39	0.00	0.00	0.00	138	0.01	1.99	2.29	0.00		
129	-14.27	27.79	10.06	0.00	0.00	0.00	101	-0.01	2.34	-0.12	0.00		
130	-8.11	27.60	10.08	0.00	0.00	0.00	102	0.01	2.15	-1.15	0.00		
131	-27.52	42.73	6.97	0.00	0.00	0.00	103	-0.01	-13.79	-19.80	0.00		
132	-27.53	42.13	8.79	0.00	0.00	0.00	104	0.03	-17.62	-26.82	0.00		
133	-12.39	42.26	7.03	0.00	0.00	0.00	105	0.04	-6.91	-12.99	0.00		
134	-12.40	41.65	8.84	0.00	0.00	0.00	106	0.09	-9.02	-18.30	0.00		
135	-22.33	43.22	5.02	0.00	0.00	0.00	107	-0.02	9.32	9.63	0.00		
136	-17.56	43.08	5.03	0.00	0.00	0.00	108	0.01	11.27	9.97	0.00		
137	-22.35	41.31	10.79	0.00	0.00	0.00	109	0.08	15.64	13.19	0.00		
138	-17.59	41.16	10.80	0.00	0.00	0.00	110	0.15	19.17	14.42	0.00		

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
111		-0.03	5.11	5.59	0.00	0.00	0.00		
112		0.00	6.01	4.92	0.00	0.00	0.00		
113		0.04	11.70	10.41	0.00	0.00	0.00		
114		0.09	14.24	10.95	0.00	0.00	0.00		
115		0.01	1.87	-2.77	0.00	0.00	0.00		
116		0.01	2.05	-1.60	0.00	0.00	0.00		
117		0.06	1.88	-2.76	0.00	0.00	0.00		
118		0.06	2.06	-1.60	0.00	0.00	0.00		
119		0.03	1.67	-4.09	0.00	0.00	0.00		
120		0.04	1.67	-4.08	0.00	0.00	0.00		
121		0.03	2.26	-0.28	0.00	0.00	0.00		
122		0.04	2.26	-0.28	0.00	0.00	0.00		
123		0.01	1.67	-2.55	0.00	0.00	0.00		
124		0.01	1.85	-1.39	0.00	0.00	0.00		
125		0.05	1.69	-2.54	0.00	0.00	0.00		
126		0.05	1.87	-1.38	0.00	0.00	0.00		
127		0.02	1.47	-3.87	0.00	0.00	0.00		
128		0.04	1.48	-3.87	0.00	0.00	0.00		
129		0.02	2.06	-0.06	0.00	0.00	0.00		
130		0.04	2.07	-0.06	0.00	0.00	0.00		
131		-0.01	2.17	-1.08	0.00	0.00	0.00		
132		-0.01	2.31	-0.19	0.00	0.00	0.00		
133		0.02	2.19	-1.08	0.00	0.00	0.00		
134		0.02	2.33	-0.18	0.00	0.00	0.00		
135		0.00	2.03	-2.06	0.00	0.00	0.00		
136		0.01	2.03	-2.06	0.00	0.00	0.00		
137		0.00	2.47	0.79	0.00	0.00	0.00		
138		0.01	2.47	0.80	0.00	0.00	0.00		
114		-0.08	3.29	-0.38	0.00	0.00	0.00		
102		-0.08	3.04	1.13	0.00	0.00	0.00		
103		-0.08	6.78	-7.32	0.00	0.00	0.00		
104		-0.08	7.98	-6.12	0.00	0.00	0.00		
105		-0.06	15.42	-13.37	0.00	0.00	0.00		
106		-0.06	18.78	-13.69	0.00	0.00	0.00		
107		-0.08	12.36	-13.27	0.00	0.00	0.00		
108		-0.08	14.96	-13.56	0.00	0.00	0.00		
109		-0.05	20.62	-17.13	0.00	0.00	0.00		
110		-0.05	25.28	-18.38	0.00	0.00	0.00		
111		-0.05	-17.77	25.00	0.00	0.00	0.00		
112		-0.05	-22.71	34.28	0.00	0.00	0.00		
113		-0.04	-8.81	16.66	0.00	0.00	0.00		
114		-0.04	-11.51	23.85	0.00	0.00	0.00		
115		-0.11	2.78	2.77	0.00	0.00	0.00		
116		-0.11	2.57	4.14	0.00	0.00	0.00		
117		-0.03	3.01	1.14	0.00	0.00	0.00		
118		-0.03	2.80	2.51	0.00	0.00	0.00		
119		-0.08	3.10	0.65	0.00	0.00	0.00		
120		-0.06	3.17	0.15	0.00	0.00	0.00		
121		-0.08	2.41	5.13	0.00	0.00	0.00		
122		-0.06	2.48	4.64	0.00	0.00	0.00		
123		-0.10	2.50	2.51	0.00	0.00	0.00		
124		-0.10	2.29	3.88	0.00	0.00	0.00		
125		-0.02	2.73	0.88	0.00	0.00	0.00		
126		-0.02	2.52	2.25	0.00	0.00	0.00		
127		-0.07	2.82	0.39	0.00	0.00	0.00		
128		-0.05	2.89	-0.11	0.00	0.00	0.00		
129		-0.07	2.13	4.87	0.00	0.00	0.00		
130		-0.05	2.20	4.37	0.00	0.00	0.00		
131		-0.11	3.16	0.45	0.00	0.00	0.00		
132		-0.11	3.00	1.51	0.00	0.00	0.00		
133		-0.05	3.33	-0.77	0.00	0.00	0.00		
134		-0.05	3.17	0.29	0.00	0.00	0.00		
135		-0.09	3.40	-1.12	0.00	0.00	0.00		
136		-0.07	3.45	-1.50	0.00	0.00	0.00		
137		-0.09	2.88	2.25	0.00	0.00	0.00		
138		-0.07	2.93	1.86	0.00	0.00	0.00		
115		-0.08	3.31	0.55	0.00	0.00	0.00		
102		-0.07	3.06	-1.01	0.00	0.00	0.00		
103		-0.05	-17.74	-24.89	0.00	0.00	0.00		
104		-0.05	-22.68	-34.23	0.00	0.00	0.00		
105		-0.04	-8.80	-16.64	0.00	0.00	0.00		
106		-0.04	-11.51	-23.92	0.00	0.00	0.00		
107		-0.07	12.38	13.46	0.00	0.00	0.00		
108		-0.08	14.97	13.70	0.00	0.00	0.00		
109		-0.05	20.60	17.08	0.00	0.00	0.00		
110		-0.04	25.24	18.24	0.00	0.00	0.00		
111		-0.07	6.81	7.52	0.00	0.00	0.00		
112		-0.08	8.00	6.28	0.00	0.00	0.00		
113		-0.06	15.41	13.40	0.00	0.00	0.00		
114		-0.06	18.75	13.63	0.00	0.00	0.00		
115		-0.11	2.59	-4.04	0.00	0.00	0.00		
116		-0.11	2.80	-2.64	0.00	0.00	0.00		
117		-0.03	2.81	-2.49	0.00	0.00	0.00		
118		-0.02	3.02	-1.10	0.00	0.00	0.00		
119		-0.08	2.42	-5.08	0.00	0.00	0.00		
120		-0.05	2.49	-4.61	0.00	0.00	0.00		
121		-0.08	3.12	-0.52	0.00	0.00	0.00		
122		-0.05	3.19	-0.05	0.00	0.00	0.00		
123		-0.10	2.31	-3.78	0.00	0.00	0.00		
124		-0.10	2.52	-2.39	0.00	0.00	0.00		
125		-0.02	2.52	-2.23	0.00	0.00	0.00		
126		-0.02	2.74	-0.84	0.00	0.00	0.00		
127		-0.07	2.14	-4.83	0.00	0.00	0.00		
128		-0.05	2.21	-4.35	0.00	0.00	0.00		
129		-0.07	2.84	-0.27	0.00	0.00	0.00		
130		-0.05	2.91	0.21	0.00	0.00	0.00		

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE									

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z		
116	131	-0.11	3.02	-1.35	0.00	0.00	0.00		
	132	-0.11	3.18	-0.27	0.00	0.00	0.00		
	133	-0.05	3.18	-0.19	0.00	0.00	0.00		
	134	-0.05	3.35	0.89	0.00	0.00	0.00		
	135	-0.09	2.90	-2.12	0.00	0.00	0.00		
	136	-0.07	2.95	-1.76	0.00	0.00	0.00		
	137	-0.09	3.42	1.30	0.00	0.00	0.00		
	138	-0.07	3.47	1.67	0.00	0.00	0.00		
	101	0.08	3.29	-0.42	0.00	0.00	0.00		
	102	0.08	3.04	1.10	0.00	0.00	0.00		
	103	0.08	6.78	-7.36	0.00	0.00	0.00		
117	104	0.08	7.98	-6.16	0.00	0.00	0.00		
	105	0.06	15.42	-13.40	0.00	0.00	0.00		
	106	0.06	18.78	-13.71	0.00	0.00	0.00		
	107	0.08	12.36	-13.31	0.00	0.00	0.00		
	108	0.08	14.96	-13.59	0.00	0.00	0.00		
	109	0.05	20.62	-17.14	0.00	0.00	0.00		
	110	0.05	25.28	-18.39	0.00	0.00	0.00		
	111	0.05	-17.77	24.99	0.00	0.00	0.00		
	112	0.05	-22.71	34.27	0.00	0.00	0.00		
	113	0.04	-8.72	15.87	0.00	0.00	0.00		
	114	0.04	-11.39	22.88	0.00	0.00	0.00		
118	115	0.03	3.01	1.12	0.00	0.00	0.00		
	116	0.03	2.80	2.49	0.00	0.00	0.00		
	117	0.11	2.79	2.74	0.00	0.00	0.00		
	118	0.11	2.57	4.11	0.00	0.00	0.00		
	119	0.06	3.18	0.13	0.00	0.00	0.00		
	120	0.08	3.11	0.62	0.00	0.00	0.00		
	121	0.06	2.48	4.61	0.00	0.00	0.00		
	122	0.08	2.41	5.11	0.00	0.00	0.00		
	123	0.02	2.73	0.86	0.00	0.00	0.00		
	124	0.02	2.52	2.23	0.00	0.00	0.00		
	125	0.10	2.51	2.48	0.00	0.00	0.00		
119	126	0.10	2.30	3.85	0.00	0.00	0.00		
	127	0.05	2.90	-0.13	0.00	0.00	0.00		
	128	0.08	2.83	0.36	0.00	0.00	0.00		
	129	0.05	2.20	4.35	0.00	0.00	0.00		
	130	0.08	2.13	4.85	0.00	0.00	0.00		
	131	0.05	3.33	-0.80	0.00	0.00	0.00		
	132	0.05	3.17	0.26	0.00	0.00	0.00		
	133	0.11	3.16	0.42	0.00	0.00	0.00		
	134	0.11	3.00	1.48	0.00	0.00	0.00		
	135	0.07	3.45	-1.53	0.00	0.00	0.00		
	136	0.09	3.40	-1.15	0.00	0.00	0.00		
117	137	0.07	2.93	1.83	0.00	0.00	0.00		
	138	0.09	2.88	2.21	0.00	0.00	0.00		
	101	0.08	3.32	0.58	0.00	0.00	0.00		
	102	0.07	3.06	-0.98	0.00	0.00	0.00		
	103	0.05	3.03	5.07	0.00	0.00	0.00		
	104	0.00	5.98	-4.71	0.00	0.00	0.00		
	105	-0.04	11.69	-10.32	0.00	0.00	0.00		
	106	-0.10	14.25	-10.97	0.00	0.00	0.00		
	107	0.02	9.27	-9.30	0.00	0.00	0.00		
	108	-0.01	11.23	-9.70	0.00	0.00	0.00		
	109	-0.09	15.65	-13.19	0.00	0.00	0.00		
118	110	-0.15	19.21	-14.56	0.00	0.00	0.00		
	111	0.01	-13.84	20.03	0.00	0.00	0.00		
	112	-0.03	-17.65	26.97	0.00	0.00	0.00		
	103	0.05	-17.73	-24.85	0.00	0.00	0.00		
	104	0.05	-22.67	-34.19	0.00	0.00	0.00		
	105	0.04	-8.80	-16.62	0.00	0.00	0.00		
	106	0.04	-11.51	-23.91	0.00	0.00	0.00		
	107	0.07	12.39	13.49	0.00	0.00	0.00		
	108	0.08	14.97	13.74	0.00	0.00	0.00		
	109	0.05	20.60	17.09	0.00	0.00	0.00		
	110	0.04	25.24	18.23	0.00	0.00	0.00		

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
113	-0.04	-6.83	12.25	0.00	0.00	0.00	0.00
114	-0.09	-8.90	17.24	0.00	0.00	0.00	0.00
115	-0.06	2.84	1.76	0.00	0.00	0.00	0.00
116	-0.06	1.86	2.90	0.00	0.00	0.00	0.00
117	-0.01	2.02	1.77	0.00	0.00	0.00	0.00
118	-0.01	1.84	2.91	0.00	0.00	0.00	0.00
119	-0.04	2.24	0.47	0.00	0.00	0.00	0.00
120	-0.03	2.23	0.47	0.00	0.00	0.00	0.00
121	-0.04	1.65	4.20	0.00	0.00	0.00	0.00
122	-0.03	1.65	4.20	0.00	0.00	0.00	0.00
123	-0.05	1.85	1.53	0.00	0.00	0.00	0.00
124	-0.05	1.67	2.67	0.00	0.00	0.00	0.00
125	-0.01	1.83	1.53	0.00	0.00	0.00	0.00
126	-0.01	1.65	2.67	0.00	0.00	0.00	0.00
127	-0.04	2.04	0.23	0.00	0.00	0.00	0.00
128	-0.03	2.04	0.24	0.00	0.00	0.00	0.00
129	-0.04	1.46	3.97	0.00	0.00	0.00	0.00
130	-0.02	1.45	3.97	0.00	0.00	0.00	0.00
131	-0.02	2.29	0.45	0.00	0.00	0.00	0.00
132	-0.02	2.15	1.33	0.00	0.00	0.00	0.00
133	0.01	2.27	0.45	0.00	0.00	0.00	0.00
134	0.01	2.14	1.33	0.00	0.00	0.00	0.00
135	-0.01	2.43	-0.51	0.00	0.00	0.00	0.00
136	0.00	2.43	-0.51	0.00	0.00	0.00	0.00
137	-0.01	1.99	2.29	0.00	0.00	0.00	0.00
138	0.00	1.99	2.29	0.00	0.00	0.00	0.00
119	0.01	2.34	-0.12	0.00	0.00	0.00	0.00
102	-0.01	2.15	-1.15	0.00	0.00	0.00	0.00
103	0.01	-13.79	-19.80	0.00	0.00	0.00	0.00
104	-0.03	-17.62	-26.82	0.00	0.00	0.00	0.00
105	-0.04	-6.91	-12.99	0.00	0.00	0.00	0.00
106	-0.09	-9.02	-18.30	0.00	0.00	0.00	0.00
107	0.02	9.32	9.63	0.00	0.00	0.00	0.00
108	-0.01	11.27	9.97	0.00	0.00	0.00	0.00
109	-0.08	15.64	13.19	0.00	0.00	0.00	0.00
110	-0.15	19.17	14.42	0.00	0.00	0.00	0.00
111	0.03	5.11	5.58	0.00	0.00	0.00	0.00
112	0.00	6.01	4.90	0.00	0.00	0.00	0.00
113	-0.04	11.77	11.11	0.00	0.00	0.00	0.00
114	-0.09	14.34	11.81	0.00	0.00	0.00	0.00
115	-0.06	1.88	-2.76	0.00	0.00	0.00	0.00
116	-0.06	2.06	-1.60	0.00	0.00	0.00	0.00
117	-0.01	1.87	-2.77	0.00	0.00	0.00	0.00
118	-0.01	2.05	-1.61	0.00	0.00	0.00	0.00
119	-0.04	1.67	-4.08	0.00	0.00	0.00	0.00
120	-0.03	1.67	-4.08	0.00	0.00	0.00	0.00
121	-0.04	2.26	-0.28	0.00	0.00	0.00	0.00
122	-0.03	2.26	-0.28	0.00	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
123	-0.05	1.69	-2.54	0.00	0.00	0.00	0.00
124	-0.05	1.87	-1.38	0.00	0.00	0.00	0.00
125	-0.01	1.67	-2.55	0.00	0.00	0.00	0.00
126	-0.01	1.85	-1.39	0.00	0.00	0.00	0.00
127	-0.04	1.48	-3.86	0.00	0.00	0.00	0.00
128	-0.02	1.47	-3.86	0.00	0.00	0.00	0.00
129	-0.04	2.07	-0.06	0.00	0.00	0.00	0.00
130	-0.02	2.06	-0.07	0.00	0.00	0.00	0.00
131	-0.02	2.19	-1.08	0.00	0.00	0.00	0.00
132	-0.02	2.33	-0.18	0.00	0.00	0.00	0.00
133	0.01	2.17	-1.08	0.00	0.00	0.00	0.00
134	0.01	2.31	-0.19	0.00	0.00	0.00	0.00
135	-0.01	2.03	-2.06	0.00	0.00	0.00	0.00
136	0.00	2.03	-2.06	0.00	0.00	0.00	0.00
137	-0.01	2.47	0.79	0.00	0.00	0.00	0.00
138	0.00	2.47	0.79	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

591. LOAD LIST 301 TO 316

592. PRINT JOINT DISPLACEMENTS

JOINT DISPLACE

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN
1	301	0.0000	0.0000	0.0000	0.0001	-0.0003	0.0004		
	302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	303	0.0000	0.0000	0.0000	0.0002	-0.0004	0.0000		
	304	0.0000	0.0000	0.0000	0.0008	-0.0015	-0.0002		
	305	0.0000	0.0000	0.0000	0.0006	-0.0011	-0.0001		
	306	0.0000	0.0000	0.0000	0.0000	-0.0021	-0.0003		
	307	0.0000	0.0000	0.0000	-0.0012	0.0023	0.0003		
	308	0.0000	0.0000	0.0000	-0.0007	0.0012	0.0002		
	309	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001		
	310	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001		
	311	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001		
	312	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001		
	313	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	314	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	315	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	316	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
2	301	0.0003	-0.0038	-0.0011	0.0000	0.0000	-0.0005		
	302	-0.0004	0.0006	0.0006	0.0000	0.0000	0.0001		
	303	-0.0008	0.0027	0.0136	-0.0001	-0.0002	0.0001		
	304	-0.0005	0.0057	0.0440	-0.0005	-0.0009	0.0000		
	305	-0.0010	0.0050	0.0322	-0.0003	-0.0006	0.0001		
	306	-0.0004	0.0075	0.0621	-0.0007	-0.0013	-0.0001		
	307	0.0007	-0.0086	-0.0670	0.0007	0.0014	0.0000		
	308	0.0007	-0.0051	-0.0362	0.0004	0.0007	-0.0001		
	309	-0.0174	0.0004	0.0005	0.0000	0.0000	0.0001		
	310	-0.0174	0.0006	0.0005	0.0000	0.0000	0.0001		
	311	0.0167	0.0004	0.0005	0.0000	0.0000	0.0001		
	312	0.0167	0.0006	0.0005	0.0000	0.0000	0.0001		
	313	-0.0054	0.0001	0.0004	0.0000	0.0000	0.0001		
	314	0.0048	0.0001	0.0004	0.0000	0.0000	0.0001		
	315	-0.0054	0.0009	0.0006	0.0000	0.0000	0.0001		
	316	0.0047	0.0009	0.0006	0.0000	0.0000	0.0001		
3	301	0.0135	-0.0028	-0.0012	0.0000	0.0002	0.0000		
	302	-0.0159	0.0007	0.0003	0.0000	-0.0007	0.0003		
	303	-0.0230	0.0016	0.0015	-0.0001	-0.0010	0.0004		
	304	0.0062	0.0015	0.0037	-0.0002	0.0002	-0.0001		
	305	-0.0193	0.0021	0.0030	-0.0003	0.0008	0.0004		
	306	0.0226	0.0015	0.0050	-0.0003	-0.0009	-0.0003		
	307	-0.0139	-0.0021	-0.0056	0.0003	-0.0005	0.0002		
	308	0.0052	-0.0017	-0.0032	0.0002	0.0003	-0.0001		
	309	-0.0267	0.0004	0.0002	0.0000	-0.0009	0.0002		
	310	-0.0268	0.0007	0.0002	0.0000	-0.0009	0.0002		
	311	0.0013	0.0004	0.0002	0.0000	-0.0002	0.0002		
	312	0.0012	0.0007	0.0002	0.0000	-0.0002	0.0002		
	313	-0.0168	0.0001	0.0002	0.0000	-0.0006	0.0002		
	314	-0.0085	0.0001	0.0002	0.0000	-0.0004	0.0002		
	315	-0.0170	0.0011	0.0002	0.0000	-0.0006	0.0002		
	316	-0.0087	0.0011	0.0002	0.0000	-0.0004	0.0002		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE										JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN				
7	301	-0.0012	-0.1451	0.0001	0.0000	0.0000	0.0001	10	301	0.0147	-0.0006	0.0001	0.0001	-0.0005	-0.0009				
	302	-0.0006	-0.0857	0.0001	0.0000	0.0000	0.0000		302	0.0099	-0.0003	0.0009	0.0001	-0.0002	-0.0006				
	303	-0.0004	-0.0692	-0.0004	0.0000	0.0000	0.0000		303	0.0006	0.0005	-0.0003	0.0000	0.0001	0.0000				
	304	0.0002	0.0295	-0.0004	0.0000	0.0000	0.0000		304	-0.0075	0.0005	-0.0008	-0.0001	0.0002	0.0005				
	305	-0.0010	-0.1388	0.0001	0.0000	0.0000	0.0000		305	0.0158	-0.0006	0.0015	0.0001	-0.0003	-0.0010				
	306	0.0004	0.0592	0.0000	0.0000	0.0000	0.0000		306	-0.0066	-0.0002	-0.0003	0.0000	0.0000	0.0004				
	307	-0.0006	-0.0713	0.0005	0.0000	0.0000	0.0000		307	0.0159	-0.0005	0.0014	0.0001	-0.0003	-0.0010				
	308	0.0012	0.0283	0.0004	0.0000	0.0000	0.0000		308	0.0014	-0.0003	0.0003	0.0000	0.0000	-0.0001				
	309	0.0276	-0.0685	0.0005	0.0000	0.0000	0.0000		309	0.0146	-0.0002	0.0009	0.0000	-0.0001	-0.0007				
	310	0.0276	-0.0686	-0.0005	0.0000	0.0000	0.0000		310	0.0146	-0.0002	0.0004	0.0000	-0.0001	-0.0007				
	311	-0.0286	-0.0686	0.0005	0.0000	0.0000	0.0000		311	0.0013	-0.0002	0.0009	0.0000	-0.0002	-0.0003				
	312	-0.0286	-0.0687	-0.0005	0.0000	0.0000	0.0000		312	0.0013	-0.0002	0.0005	0.0000	-0.0002	-0.0003				
	313	0.0079	-0.0684	0.0017	0.0000	0.0000	0.0000		313	0.0100	-0.0002	0.0015	0.0001	-0.0001	-0.0006				
	314	-0.0089	-0.0685	0.0017	0.0000	0.0000	0.0000		314	0.0060	-0.0002	0.0015	0.0001	-0.0002	-0.0004				
	315	0.0079	-0.0687	-0.0016	0.0000	0.0000	0.0000		315	0.0099	-0.0002	-0.0001	0.0000	-0.0001	-0.0006				
	8	316	-0.0089	-0.0687	-0.0016	0.0000	0.0000		0.0000	316	0.0060	-0.0002	-0.0001	0.0000	-0.0002	-0.0004			
301		0.0012	-0.0021	0.0013	0.0001	0.0000	-0.0001	11	301	0.0129	-0.0027	0.0012	0.0000	-0.0002	0.0000				
302		0.0010	-0.0012	0.0011	0.0001	0.0000	-0.0001		302	-0.0158	0.0007	-0.0003	0.0000	0.0007	0.0003				
303		0.0001	-0.0003	0.0003	0.0001	0.0000	0.0000		303	-0.0136	-0.0021	0.0056	-0.0003	0.0004	0.0002				
304		-0.0008	-0.0008	-0.0008	0.0000	0.0000	0.0001		304	0.0052	-0.0017	0.0032	-0.0002	-0.0003	-0.0001				
305		0.0016	-0.0020	0.0019	0.0001	0.0000	-0.0001		305	-0.0193	0.0021	-0.0030	0.0002	0.0009	0.0004				
306		-0.0007	0.0008	-0.0006	-0.0001	0.0000	0.0001		306	0.0223	0.0014	-0.0050	0.0003	-0.0008	-0.0003				
307		0.0015	-0.0017	0.0014	0.0000	0.0000	-0.0001		307	-0.0228	0.0016	-0.0016	0.0001	0.0010	0.0004				
308		0.0004	0.0000	0.0001	-0.0001	0.0000	0.0000		308	0.0060	0.0015	-0.0037	0.0002	-0.0002	-0.0001				
309		0.0271	-0.0010	0.0014	0.0001	0.0000	-0.0004		309	-0.0267	0.0007	-0.0002	0.0000	0.0009	0.0002				
310		0.0271	-0.0010	0.0004	0.0001	0.0000	-0.0004		310	-0.0266	0.0004	-0.0002	0.0000	0.0009	0.0002				
311		-0.0255	-0.0010	0.0014	0.0001	0.0000	0.0003		311	0.0013	0.0007	-0.0002	0.0000	0.0002	0.0002				
312		-0.0255	-0.0010	0.0004	0.0001	0.0000	0.0003		312	0.0013	0.0004	-0.0002	0.0000	0.0002	0.0002				
313		0.0087	-0.0010	0.0025	0.0001	0.0000	-0.0002		313	-0.0169	0.0011	-0.0002	0.0000	0.0006	0.0002				
314		-0.0070	-0.0010	0.0025	0.0001	0.0000	0.0000		314	-0.0086	0.0011	-0.0002	0.0000	0.0004	0.0002				
315		0.0086	-0.0010	-0.0008	0.0000	0.0000	-0.0002		315	-0.0168	0.0001	-0.0002	0.0000	0.0006	0.0002				
9	316	-0.0071	-0.0010	-0.0007	0.0000	0.0000	0.0000		316	-0.0084	0.0001	-0.0002	0.0000	0.0004	0.0002				
	301	-0.0092	-0.0158	0.0027	0.0002	0.0000	0.0011	12	301	0.0002	-0.0037	0.0011	0.0000	0.0000	-0.0005				
	302	0.0062	-0.0098	0.0025	0.0001	-0.0001	-0.0001		302	-0.0005	0.0007	-0.0007	0.0000	0.0000	0.0001				
	303	0.0069	0.0043	-0.0008	0.0000	0.0000	-0.0005		303	0.0007	-0.0085	0.0069	-0.0007	-0.0014	0.0000				
	304	-0.0012	0.0101	-0.0024	-0.0001	0.0000	-0.0001		304	0.0007	-0.0051	0.0362	-0.0004	-0.0007	-0.0001				
	305	0.0069	-0.0168	0.0042	0.0002	-0.0001	0.0000		305	-0.0011	0.0050	-0.0323	0.0003	0.0006	0.0001				
	306	-0.0101	0.0037	-0.0010	-0.0001	0.0001	0.0005		306	-0.0003	0.0075	0.0621	0.0007	0.0013	-0.0001				
	307	0.0086	-0.0181	0.0043	0.0002	-0.0001	-0.0001		307	-0.0009	0.0027	-0.0136	0.0001	0.0002	0.0001				
	308	-0.0034	-0.0046	0.0010	0.0000	0.0000	0.0003		308	-0.0005	0.0057	-0.0439	0.0005	0.0009	0.0000				
	309	0.0194	-0.0081	0.0023	0.0001	0.0000	-0.0004		309	-0.0175	0.0006	-0.0005	0.0000	0.0000	0.0000				
	310	0.0193	-0.0075	0.0016	0.0001	0.0000	-0.0004		310	-0.0175	0.0004	-0.0005	0.0000	0.0000	0.0000				
	311	-0.0094	-0.0081	0.0023	0.0001	-0.0001	0.0002		311	0.0167	0.0007	-0.0006	0.0000	0.0000	0.0001				
	312	-0.0095	-0.0075	0.0016	0.0001	-0.0001	0.0002		312	0.0167	0.0004	-0.0005	0.0000	0.0000	0.0001				
	313	0.0094	-0.0088	0.0031	0.0001	0.0000	-0.0002		313	-0.0055	0.0010	-0.0006	0.0000	0.0000	0.0001				
	314	0.0008	-0.0088	0.0031	0.0001	0.0000	0.0000		314	0.0047	0.0010	-0.0006	0.0000	0.0000	0.0001				
	315	0.0091	-0.0068	0.0009	0.0001	0.0000	-0.0002		315	-0.0055	0.0001	-0.0005	0.0000	0.0000	0.0001				
316	0.0005	-0.0068	0.0009	0.0001	0.0000	0.0000	316		0.0047	0.0001	-0.0005	0.0000	0.0000	0.0001					

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE										JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE										
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN			JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN			
13	301	0.0000	0.0000	0.0000	-0.0001	0.0003	0.0004			16	301	-0.0251	-0.0043	-0.0017	0.0000	-0.0009	0.0004			
	302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000				302	-0.0041	0.0007	0.0003	0.0000	-0.0002	0.0001	0.0001		
	303	0.0000	0.0000	0.0000	0.0012	-0.0023	0.0003				303	-0.0068	0.0018	0.0020	-0.0001	-0.0003	0.0000	0.0000		
	304	0.0000	0.0000	0.0000	0.0007	-0.0012	0.0002				304	-0.0013	0.0019	0.0048	-0.0003	0.0000	0.0000	0.0000		
	305	0.0000	0.0000	0.0000	-0.0006	0.0011	-0.0001				305	-0.0070	0.0024	0.0039	-0.0002	-0.0003	0.0001	0.0001		
	306	0.0000	0.0000	0.0000	-0.0012	0.0021	-0.0003				306	0.0017	0.0020	0.0065	-0.0004	0.0001	0.0000	0.0000		
	307	0.0000	0.0000	0.0000	-0.0002	0.0004	0.0000				307	0.0008	-0.0027	-0.0073	0.0004	0.0000	0.0000	0.0000		
	308	0.0000	0.0000	0.0000	-0.0008	0.0015	-0.0002				308	0.0037	-0.0021	-0.0041	0.0002	0.0000	-0.0001	-0.0001		
	309	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001				309	-0.0244	0.0006	0.0003	0.0000	-0.0008	0.0002	0.0002		
	310	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001				310	-0.0244	0.0010	0.0003	0.0000	-0.0008	0.0002	0.0002		
	311	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001				311	0.0178	0.0002	0.0003	0.0000	0.0005	-0.0001	-0.0003		
	312	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001				312	0.0178	0.0006	0.0003	0.0000	0.0005	-0.0001	-0.0001		
	313	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000				313	-0.0096	0.0000	0.0003	0.0000	-0.0003	0.0000	0.0000		
	314	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000				314	0.0030	-0.0001	0.0003	0.0000	0.0000	0.0000	0.0000		
	315	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000				315	-0.0096	0.0012	0.0003	0.0000	-0.0003	0.0001	0.0001		
	14	361	0.0000	0.0000	0.0000	0.0001	0.0044	0.0006				17	361	0.0042	-0.0008	0.0006	-0.0001	-0.0002	-0.0002	-0.0002
362		0.0000	0.0000	0.0000	0.0000	0.0000	0.0000			362	0.0028		-0.0003	-0.0008	-0.0001	0.0000	0.0000	0.0000		
363		0.0000	0.0000	0.0000	0.0003	0.0001	0.0000			363	0.0044		-0.0006	-0.0014	-0.0001	0.0000	-0.0002	0.0000		
364		0.0000	0.0000	0.0000	0.0011	0.0003	0.0000			364	0.0004		-0.0004	-0.0005	0.0000	0.0000	0.0000	0.0000		
365		0.0000	0.0000	0.0000	0.0008	0.0002	0.0000			365	0.0044		-0.0007	-0.0014	-0.0001	0.0000	-0.0002	0.0000		
366		0.0000	0.0000	0.0000	0.0015	0.0004	0.0001			366	-0.0018		-0.0004	0.0002	0.0000	0.0000	0.0001	0.0001		
367		0.0000	0.0000	0.0000	-0.0016	-0.0005	-0.0001			367	0.0002		0.0006	0.0005	0.0001	0.0000	0.0000	0.0000		
368		0.0000	0.0000	0.0000	-0.0009	-0.0003	0.0000			368	-0.0021		0.0005	0.0009	0.0001	0.0000	0.0001	0.0001		
369		0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000			369	0.0123		-0.0002	-0.0006	0.0000	0.0001	-0.0006	0.0000		
310		0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000			310	0.0123		-0.0003	0.0012	-0.0001	0.0001	-0.0001	-0.0006		
311		0.0000	0.0000	0.0000	0.0000	0.0003	0.0000			311	-0.0079		-0.0002	0.0000	0.0000	0.0000	0.0000	0.0004		
312		0.0000	0.0000	0.0000	0.0000	0.0003	0.0000			312	-0.0079		-0.0002	-0.0006	0.0000	0.0000	0.0000	0.0004		
313		0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000			313	0.0052		-0.0002	0.0002	0.0000	0.0000	-0.0003	0.0000		
314		0.0000	0.0000	0.0000	0.0000	0.0001	0.0000			314	-0.0008		-0.0002	0.0004	0.0000	0.0000	0.0000	0.0000		
315		0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000			315	0.0052		-0.0003	-0.0016	-0.0001	0.0000	-0.0003	0.0000		
15		361	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000			18		361	0.0200	-0.0179	-0.0024	-0.0002	0.0001	-0.0012	0.0000
	362	-0.0001	0.0007	0.0007	0.0000	0.0002	0.0000			362		0.0033	-0.0114	-0.0026	-0.0001	0.0000	-0.0001	-0.0001		
	363	-0.0002	0.0033	0.0175	-0.0002	0.0003	0.0000			363		0.0056	-0.0207	-0.0048	-0.0002	0.0000	-0.0002	-0.0002		
	364	-0.0002	0.0074	0.0573	-0.0006	0.0000	0.0000			364		0.0013	-0.0053	-0.0013	0.0000	0.0000	-0.0001	-0.0001		
	365	-0.0003	0.0062	0.0419	-0.0004	0.0003	0.0000			365		0.0059	-0.0199	-0.0046	-0.0002	0.0000	-0.0002	-0.0002		
	366	-0.0001	0.0099	0.0811	-0.0009	-0.0001	0.0000			366		-0.0010	0.0039	0.0009	0.0001	0.0000	0.0000	0.0000		
	367	0.0002	-0.0112	-0.0873	0.0009	0.0000	0.0000			367		-0.0011	0.0049	0.0012	0.0000	0.0000	0.0000	0.0001		
	368	0.0002	-0.0065	-0.0471	0.0005	-0.0002	0.0000			368		-0.0033	0.0117	0.0027	0.0001	0.0000	0.0000	0.0001		
	369	-0.0172	0.0005	0.0006	0.0000	0.0002	0.0000			369		0.0186	-0.0091	-0.0021	-0.0001	0.0000	-0.0003	-0.0003		
	310	-0.0172	0.0009	0.0006	0.0000	0.0002	0.0000			310		0.0186	-0.0098	-0.0028	-0.0001	0.0000	-0.0003	-0.0003		
	311	0.0170	0.0002	0.0005	0.0000	0.0000	0.0000			311		-0.0133	-0.0085	-0.0014	-0.0001	0.0000	0.0000	0.0001		
	312	0.0170	0.0005	0.0006	0.0000	0.0000	0.0000			312		-0.0133	-0.0092	-0.0021	-0.0001	0.0000	0.0000	0.0002		
	313	-0.0052	0.0001	0.0005	0.0000	0.0002	0.0000			313		0.0074	-0.0080	-0.0009	-0.0001	0.0000	-0.0002	-0.0002		
	314	0.0050	0.0000	0.0005	0.0000	0.0001	0.0000			314		-0.0021	-0.0079	-0.0007	-0.0001	0.0000	-0.0001	-0.0002		
	315	-0.0052	0.0011	0.0007	0.0000	0.0002	0.0000			315		0.0074	-0.0104	-0.0034	-0.0001	0.0000	-0.0002	-0.0002		
	316	0.0050	0.0010	0.0006	0.0000	0.0001	0.0000			316		-0.0021	-0.0102	-0.0032	-0.0001	0.0000	0.0000	0.0000		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN

19	361	-0.0002	-0.0018	-0.0007	0.0000	0.0000	0.0000		
	362	-0.0001	-0.0011	-0.0008	0.0000	0.0000	0.0000		
	363	-0.0001	-0.0015	-0.0014	0.0000	0.0000	0.0000		
	364	0.0000	0.0000	-0.0004	0.0000	0.0000	0.0000		
	365	-0.0002	-0.0018	-0.0014	0.0000	0.0000	0.0000		
	366	0.0000	0.0007	0.0003	0.0000	0.0000	0.0000		
	367	-0.0001	-0.0002	0.0004	-0.0001	0.0000	0.0000		
	368	0.0003	0.0007	0.0009	0.0000	0.0000	0.0000		
	369	0.0291	-0.0009	-0.0004	0.0000	0.0000	-0.0006		
	310	0.0291	-0.0009	-0.0014	0.0000	0.0000	-0.0006		
	311	-0.0292	-0.0008	0.0002	0.0000	0.0000	0.0006		
	312	-0.0292	-0.0008	-0.0009	0.0000	0.0000	0.0006		
	313	-0.0086	-0.0009	0.0011	0.0000	0.0000	-0.0002		
	314	-0.0088	-0.0009	0.0012	0.0000	0.0000	0.0002		
	315	0.0086	-0.0009	-0.0025	0.0000	0.0000	-0.0002		
	316	-0.0088	-0.0009	-0.0023	0.0000	0.0000	0.0002		
	20	361	-0.0003	-0.1010	0.0003	0.0000	0.0001		
	362	-0.0002	-0.0595	0.0002	0.0000	0.0000	0.0000		
23	363	-0.0001	-0.0477	-0.0008	0.0000	0.0000	0.0000		
	364	0.0001	0.0209	-0.0008	0.0000	0.0000	0.0000		
	365	-0.0003	-0.0962	0.0002	0.0000	0.0000	0.0000		
	366	0.0001	0.0417	-0.0001	0.0000	0.0000	0.0000		
	367	-0.0002	-0.0501	0.0011	0.0000	0.0000	0.0000		
	368	0.0013	0.0195	0.0007	0.0000	0.0000	0.0000		
	369	0.0281	-0.0478	0.0007	0.0000	0.0000	-0.0004		
	310	0.0281	-0.0479	-0.0004	0.0000	0.0000	-0.0004		
	311	-0.0283	-0.0473	0.0006	0.0000	0.0000	0.0005		
	312	-0.0283	-0.0473	-0.0004	0.0000	0.0000	0.0005		
	313	0.0083	-0.0475	0.0019	0.0000	0.0000	-0.0001		
	314	-0.0085	-0.0474	0.0018	0.0000	0.0000	0.0001		
	315	0.0083	-0.0478	-0.0016	0.0000	0.0000	-0.0001		
	316	-0.0085	-0.0476	-0.0016	0.0000	0.0000	0.0001		
	21	361	-0.0004	-0.0017	0.0000	0.0000	0.0000		
	362	-0.0002	-0.0010	0.0011	0.0000	0.0000	0.0000		
	363	-0.0001	-0.0002	0.0003	0.0001	0.0000	0.0000		
24	364	0.0001	0.0007	-0.0011	0.0000	0.0000	0.0000		
	365	-0.0003	-0.0017	0.0018	0.0000	0.0000	0.0000		
	366	0.0001	0.0007	-0.0005	0.0000	0.0000	0.0000		
	367	-0.0003	-0.0015	0.0018	0.0000	0.0000	0.0000		
	368	0.0003	0.0000	0.0005	0.0000	0.0000	0.0000		
	369	0.0270	-0.0008	0.0017	0.0000	0.0000	-0.0005		
	310	0.0270	-0.0008	0.0007	0.0000	0.0000	-0.0005		
	311	-0.0273	-0.0008	0.0010	0.0000	0.0000	0.0005		
	312	-0.0273	-0.0008	0.0000	0.0000	0.0000	0.0005		
	313	0.0000	-0.0000	0.0027	0.0000	0.0000	-0.0002		
	314	-0.0082	-0.0008	0.0024	0.0000	0.0000	0.0002		
	315	-0.0079	-0.0008	-0.0007	0.0000	0.0000	-0.0002		
	316	-0.0083	-0.0008	-0.0010	0.0000	0.0000	0.0002		
22	361	0.0176	-0.0186	0.0026	0.0002	0.0000	0.0001		
	362	0.0032	-0.0119	0.0028	0.0001	0.0000	0.0000		
	363	-0.0009	0.0046	-0.0011	0.0000	0.0000	0.0001		
	364	-0.0030	0.0119	-0.0028	-0.0001	0.0000	0.0001		
	365	0.0055	-0.0206	0.0048	0.0002	0.0000	-0.0002		
	366	-0.0011	0.0044	-0.0010	-0.0001	0.0000	0.0000		
	367	0.0052	-0.0213	0.0049	0.0002	0.0000	-0.0002		
	368	0.0011	-0.0052	0.0012	0.0000	0.0000	-0.0001		
	369	0.0183	-0.0102	0.0029	0.0001	0.0000	-0.0004		
	310	0.0184	-0.0095	0.0022	0.0001	0.0000	-0.0004		
	311	-0.0133	-0.0096	0.0022	0.0001	0.0000	0.0000		
	312	-0.0133	-0.0089	0.0015	0.0001	0.0000	0.0002		
	313	-0.0072	-0.0108	0.0036	0.0001	0.0000	-0.0002		
	314	-0.0022	-0.0106	0.0033	0.0001	0.0000	0.0000		
	315	0.0073	-0.0085	0.0011	0.0001	0.0000	-0.0002		
	316	-0.0022	-0.0083	0.0009	0.0001	0.0000	0.0000		
	23	361	0.0041	-0.0006	-0.0006	0.0001	0.0001		
	362	0.0027	-0.0003	0.0008	0.0001	0.0000	-0.0001		
	363	0.0002	0.0006	-0.0004	-0.0001	0.0000	0.0000		
	364	-0.0020	0.0006	-0.0009	-0.0001	0.0000	0.0001		
	365	0.0043	-0.0008	0.0015	0.0001	0.0000	-0.0002		
	366	-0.0018	-0.0004	-0.0002	0.0000	0.0000	0.0001		
	367	0.0043	-0.0006	0.0015	0.0001	0.0000	-0.0002		
	368	0.0004	-0.0004	0.0004	0.0000	0.0000	0.0000		
	369	0.0124	-0.0003	0.0012	0.0001	-0.0001	-0.0001		
	310	0.0124	-0.0002	0.0007	0.0000	-0.0001	-0.0006		
	311	-0.0081	-0.0002	0.0006	0.0000	0.0000	0.0004		
	312	-0.0081	-0.0002	0.0001	0.0000	0.0000	0.0004		
	313	0.0052	-0.0003	0.0017	0.0001	0.0000	-0.0003		
	314	-0.0009	-0.0003	0.0015	0.0001	0.0000	0.0000		
	315	0.0052	-0.0002	-0.0002	0.0000	0.0000	-0.0003		
	316	-0.0009	-0.0002	-0.0004	0.0000	0.0000	0.0000		
	24	361	-0.0239	0.0017	0.0000	0.0008	0.0003		
	362	-0.0040	0.0008	-0.0004	0.0000	0.0002	0.0001		
	363	0.0007	-0.0027	0.0072	-0.0004	0.0000	0.0000		
	364	0.0035	-0.0021	0.0041	-0.0002	-0.0001	-0.0001		
	365	-0.0068	0.0025	-0.0039	0.0002	0.0003	0.0001		
	366	0.0018	0.0019	-0.0065	0.0004	-0.0001	0.0000		
	367	-0.0065	0.0018	-0.0020	0.0001	0.0003	0.0001		
	368	-0.0012	0.0019	-0.0048	0.0003	0.0000	0.0000		
	369	-0.0245	0.0010	-0.0003	0.0000	0.0008	0.0002		
	310	-0.0245	0.0006	-0.0003	0.0000	0.0008	0.0002		
	311	0.0181	0.0006	-0.0003	0.0000	-0.0005	-0.0001		
	312	-0.0181	0.0003	-0.0003	0.0000	-0.0005	-0.0001		
	313	-0.0096	0.0013	-0.0003	0.0000	0.0000	0.0001		
	314	0.0032	0.0012	-0.0003	0.0000	-0.0001	0.0000		
	315	-0.0095	0.0001	-0.0003	0.0000	0.0003	0.0001		
	316	0.0032	0.0000	-0.0003	0.0000	-0.0001	0.0000		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN
25	301	0.0001	-0.0055	0.0011	0.0000	-0.0005	0.0002		
	302	-0.0002	0.0007	-0.0008	0.0000	-0.0002	0.0000		
	303	0.0002	-0.0111	0.0873	-0.0009	0.0000	0.0000		
	304	0.0002	-0.0065	0.0471	-0.0005	0.0001	0.0000		
	305	-0.0004	0.0063	-0.0420	0.0004	-0.0003	0.0000		
	306	-0.0001	0.0099	-0.0811	0.0009	0.0001	0.0000		
	307	-0.0003	0.0033	-0.0176	0.0002	-0.0003	0.0000		
	308	-0.0002	0.0074	-0.0573	0.0006	0.0000	0.0000		
	309	-0.0173	0.0009	-0.0007	0.0000	-0.0002	0.0000		
	310	-0.0173	0.0006	-0.0006	0.0000	-0.0002	0.0000		
	311	0.0171	0.0006	-0.0006	0.0000	0.0000	0.0000		
	312	0.0171	0.0002	-0.0006	0.0000	0.0000	0.0000		
	313	-0.0053	0.0011	-0.0007	0.0000	-0.0002	0.0000		
	314	0.0050	0.0010	-0.0007	0.0000	-0.0001	0.0000		
	315	-0.0053	0.0001	-0.0006	0.0000	-0.0002	0.0000		
	316	0.0050	0.0000	-0.0005	0.0000	-0.0001	0.0000		
26	301	0.0000	0.0000	0.0000	-0.0001	-0.0043	0.0006		
	302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	303	0.0000	0.0000	0.0000	0.0016	0.0005	-0.0001		
	304	0.0000	0.0000	0.0000	0.0009	0.0003	0.0000		
	305	0.0000	0.0000	-0.0008	-0.0000	-0.0002	0.0000		
	306	0.0000	0.0000	0.0000	-0.0015	-0.0004	0.0001		
	307	0.0000	0.0000	-0.0003	-0.0003	-0.0001	0.0000		
	308	0.0000	0.0000	0.0000	-0.0011	-0.0003	0.0000		
	309	0.0000	0.0000	0.0000	0.0000	0.0003	0.0000		
	310	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000		
	311	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000		
	312	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000		
	313	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000		
	314	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000		
	315	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000		
	316	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
27	301	0.0000	0.0000	0.0000	0.0001	-0.0043	-0.0006		
	302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	303	0.0000	0.0000	0.0000	0.0003	-0.0001	0.0000		
	304	0.0000	0.0000	0.0000	0.0011	-0.0003	0.0000		
	305	0.0000	0.0000	0.0000	0.0000	0.0008	0.0000		
	306	0.0000	0.0000	0.0000	0.0015	-0.0004	-0.0001		
	307	0.0000	0.0000	0.0000	-0.0016	0.0005	0.0001		
	308	0.0000	0.0000	0.0000	-0.0009	0.0003	0.0000		
	309	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000		
	310	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000		
	311	0.0000	0.0000	0.0000	0.0000	0.0003	0.0000		
	312	0.0000	0.0000	0.0000	0.0000	0.0003	0.0000		
	313	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000		
	314	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000		
	315	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000		
	316	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
28	301	0.0000	-0.0056	-0.0011	0.0000	-0.0006	-0.0002		
	302	0.0002	0.0007	0.0007	0.0000	-0.0002	0.0000		
	303	0.0004	0.0033	0.0175	-0.0002	-0.0003	0.0000		
	304	0.0002	0.0074	0.0573	-0.0006	0.0000	0.0000		
	305	0.0005	0.0062	0.0419	-0.0004	-0.0003	0.0000		
	306	0.0000	0.0099	0.0811	-0.0009	0.0001	0.0000		
	307	-0.0002	-0.0112	-0.0873	0.0009	0.0000	0.0000		
	308	-0.0003	-0.0065	-0.0470	0.0005	0.0001	0.0000		
	309	-0.0169	0.0002	0.0005	0.0000	0.0000	0.0000		
	310	-0.0169	0.0005	0.0006	0.0000	0.0000	0.0000		
	311	0.0173	0.0005	0.0006	0.0000	-0.0002	0.0000		
	312	0.0173	0.0009	0.0006	0.0000	-0.0002	0.0000		
	313	-0.0049	0.0000	0.0005	0.0000	-0.0001	0.0000		
	314	0.0053	0.0001	0.0005	0.0000	-0.0002	0.0000		
	315	-0.0049	0.0010	0.0006	0.0000	-0.0001	0.0000		
	316	0.0053	0.0011	0.0007	0.0000	-0.0002	0.0000		
29	301	0.0253	-0.0043	-0.0017	0.0000	0.0009	-0.0004		
	302	0.0043	0.0007	0.0003	0.0000	0.0000	-0.0001		
	303	0.0071	0.0018	0.0020	-0.0001	0.0003	-0.0001		
	304	0.0013	0.0019	0.0048	-0.0003	0.0000	0.0000		
	305	0.0073	0.0024	0.0039	-0.0002	0.0000	-0.0001		
	306	-0.0019	0.0020	0.0065	-0.0004	-0.0001	0.0000		
	307	-0.0008	-0.0027	-0.0073	0.0004	0.0000	0.0000		
	308	-0.0038	-0.0021	-0.0041	0.0002	-0.0002	0.0001		
	309	-0.0176	0.0002	0.0003	0.0000	-0.0005	0.0001		
	310	-0.0176	0.0006	0.0003	0.0000	-0.0005	0.0001		
	311	0.0244	0.0006	0.0003	0.0000	0.0008	-0.0002		
	312	0.0244	0.0010	0.0003	0.0000	0.0000	-0.0002		
	313	-0.0028	-0.0001	0.0003	0.0000	0.0000	0.0000		
	314	0.0097	0.0000	0.0003	0.0000	0.0000	-0.0001		
	315	-0.0028	0.0011	0.0003	0.0000	0.0000	0.0000		
	316	0.0097	0.0012	0.0003	0.0000	0.0003	-0.0001		
30	301	-0.0043	-0.0008	0.0006	-0.0001	0.0002	0.0002		
	302	-0.0028	-0.0003	-0.0008	-0.0001	0.0000	0.0001		
	303	-0.0045	-0.0006	-0.0014	-0.0001	0.0000	0.0002		
	304	-0.0004	-0.0004	-0.0005	0.0000	0.0000	0.0000		
	305	-0.0045	-0.0007	-0.0014	-0.0001	0.0000	0.0000		
	306	0.0019	-0.0004	0.0001	0.0000	0.0000	-0.0001		
	307	-0.0002	0.0006	0.0005	0.0001	0.0000	0.0000		
	308	0.0021	0.0005	0.0010	0.0001	0.0000	-0.0001		
	309	0.0077	-0.0002	0.0000	0.0000	0.0000	-0.0004		
	310	0.0077	-0.0002	-0.0006	0.0000	0.0000	-0.0000		
	311	-0.0122	-0.0002	-0.0006	0.0000	-0.0001	0.0006		
	312	-0.0122	-0.0003	-0.0012	-0.0001	-0.0001	0.0006		
	313	0.0007	-0.0002	0.0004	0.0000	0.0000	0.0000		
	314	-0.0052	-0.0002	0.0002	0.0000	0.0000	0.0003		
	315	0.0007	-0.0003	-0.0014	-0.0001	0.0000	0.0000		
	316	-0.0052	-0.0003	-0.0016	-0.0001	0.0000	0.0003		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE								JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE							
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
31	301	-0.0204	-0.0180	-0.0024	-0.0002	-0.0001	0.0012	34	301	0.0005	-0.0017	0.0010	0.0000	0.0000	0.0000
	302	-0.0036	-0.0114	-0.0026	-0.0001	0.0000	0.0001		302	0.0002	-0.0010	0.0010	0.0000	0.0000	0.0000
	303	-0.0059	-0.0207	-0.0048	-0.0002	0.0000	0.0002		303	0.0001	-0.0002	-0.0003	0.0001	0.0000	0.0000
	304	-0.0014	-0.0053	-0.0013	0.0000	0.0000	0.0001		304	-0.0001	0.0007	-0.0011	0.0000	0.0000	0.0000
	305	-0.0063	-0.0199	-0.0046	-0.0002	0.0000	0.0002		305	0.0004	-0.0017	0.0018	0.0000	0.0000	0.0000
	306	0.0012	0.0039	0.0009	0.0001	0.0000	0.0000		306	-0.0001	0.0007	-0.0005	0.0000	0.0000	0.0000
	307	0.0011	0.0049	0.0012	0.0000	0.0000	-0.0001		307	0.0003	0.0015	0.0018	0.0000	0.0000	0.0000
	308	0.0034	0.0103	0.0027	0.0001	0.0000	-0.0001		308	0.0001	0.0002	0.0000	-0.0001	0.0000	0.0000
	309	0.0130	-0.0085	-0.0014	-0.0001	0.0000	-0.0001		309	0.0274	-0.0008	0.0010	0.0000	0.0000	-0.0005
	310	0.0130	-0.0092	-0.0021	-0.0001	0.0000	-0.0001		310	0.0274	-0.0008	-0.0001	0.0000	0.0000	-0.0005
	311	-0.0187	-0.0091	-0.0021	-0.0001	0.0000	0.0004		311	-0.0270	-0.0008	0.0017	0.0000	0.0000	0.0005
	312	-0.0187	-0.0098	-0.0028	-0.0001	0.0000	0.0003		312	-0.0270	-0.0008	0.0007	0.0000	0.0000	0.0005
	313	0.0018	-0.0079	-0.0007	-0.0001	0.0000	0.0000		313	-0.0083	-0.0008	0.0024	0.0000	0.0000	-0.0002
	314	-0.0076	-0.0080	-0.0009	-0.0001	0.0000	0.0002		314	-0.0079	-0.0008	0.0026	0.0000	0.0000	0.0002
	315	0.0019	-0.0102	-0.0032	-0.0001	0.0000	0.0000		315	0.0083	-0.0008	-0.0019	0.0000	0.0000	-0.0002
	316	-0.0075	-0.0104	-0.0034	-0.0001	0.0000	0.0002		316	-0.0079	-0.0008	-0.0007	0.0000	0.0000	0.0002
32	301	0.0005	-0.0018	-0.0007	0.0000	0.0000	0.0000	35	301	-0.0176	0.0186	0.0026	0.0002	0.0001	0.0011
	302	0.0003	-0.0011	-0.0008	0.0000	0.0000	0.0000		302	-0.0032	-0.0119	0.0027	0.0001	0.0000	0.0001
	303	0.0004	-0.0015	-0.0015	0.0000	0.0000	0.0000		303	0.0009	0.0046	-0.0011	0.0000	0.0000	-0.0001
	304	0.0000	-0.0000	-0.0004	0.0000	0.0000	0.0000		304	0.0030	0.0119	-0.0028	-0.0001	0.0000	-0.0001
	305	0.0005	-0.0018	-0.0014	0.0000	0.0000	0.0000		305	-0.0055	-0.0206	0.0048	0.0002	0.0000	0.0002
	306	-0.0002	0.0007	0.0004	0.0000	0.0000	0.0000		306	0.0011	0.0044	-0.0010	-0.0001	0.0000	0.0000
	307	0.0001	-0.0002	0.0003	-0.0001	0.0000	0.0000		307	-0.0052	-0.0213	0.0049	0.0002	0.0000	0.0002
	308	0.0000	0.0010	0.0016	0.0001	0.0000	0.0000		308	-0.0011	-0.0066	0.0013	0.0001	0.0000	0.0001
	309	0.0295	-0.0008	0.0001	0.0000	0.0000	-0.0006		309	0.0133	-0.0096	0.0022	0.0001	0.0000	-0.0002
	310	0.0295	-0.0008	-0.0009	0.0000	0.0000	-0.0006		310	0.0133	-0.0089	0.0015	0.0001	0.0000	-0.0002
	311	-0.0290	-0.0009	-0.0004	0.0000	0.0000	0.0006		311	-0.0183	-0.0102	0.0029	0.0001	0.0000	0.0004
	312	-0.0290	-0.0009	-0.0015	0.0000	0.0000	0.0006		312	-0.0184	-0.0095	0.0022	0.0001	0.0000	0.0004
	313	0.0090	-0.0009	0.0012	0.0000	0.0000	-0.0002		313	0.0022	-0.0106	0.0033	0.0001	0.0000	0.0000
	314	-0.0085	-0.0009	0.0010	0.0000	0.0000	0.0002		314	-0.0072	-0.0107	0.0036	0.0001	0.0000	0.0002
	315	0.0089	-0.0009	-0.0023	0.0000	0.0000	-0.0002		315	0.0022	-0.0083	0.0008	0.0001	0.0000	0.0000
	316	-0.0085	-0.0009	-0.0025	0.0000	0.0000	0.0002		316	-0.0073	-0.0085	0.0011	0.0001	0.0000	0.0002
33	301	0.0005	-0.0101	0.0002	0.0000	0.0000	-0.0001	36	301	-0.0041	-0.0008	0.0006	0.0001	-0.0001	0.0002
	302	0.0003	-0.0595	0.0001	0.0000	0.0000	0.0000		302	-0.0027	-0.0003	0.0008	0.0001	0.0000	0.0001
	303	0.0002	-0.0478	-0.0009	0.0000	0.0000	0.0000		303	-0.0002	0.0006	-0.0004	-0.0001	0.0000	0.0000
	304	-0.0001	0.0209	-0.0008	0.0000	0.0000	0.0000		304	0.0020	0.0006	-0.0009	-0.0001	0.0000	-0.0001
	305	0.0004	-0.0963	0.0002	0.0000	0.0000	0.0000		305	-0.0043	-0.0008	0.0014	0.0001	0.0000	0.0002
	306	-0.0002	0.0417	-0.0001	0.0000	0.0000	0.0000		306	0.0018	-0.0004	-0.0002	0.0000	0.0000	-0.0001
	307	0.0002	-0.0508	0.0011	0.0000	0.0000	0.0000		307	-0.0043	-0.0006	0.0015	0.0001	0.0000	0.0002
	308	0.0015	0.0702	0.0008	0.0000	0.0000	0.0000		308	-0.0004	-0.0005	0.0004	0.0001	0.0000	0.0000
	309	0.0284	-0.0473	0.0006	0.0000	0.0000	-0.0005		309	0.0081	-0.0002	0.0006	0.0000	0.0000	-0.0004
	310	0.0284	-0.0474	-0.0005	0.0000	0.0000	-0.0005		310	0.0081	-0.0002	0.0001	0.0000	0.0000	-0.0004
	311	-0.0280	-0.0479	0.0006	0.0000	0.0000	0.0004		311	-0.0124	-0.0003	0.0012	0.0001	0.0000	0.0006
	312	-0.0280	-0.0479	-0.0004	0.0000	0.0000	0.0004		312	-0.0124	-0.0002	0.0007	0.0000	0.0000	0.0006
	313	0.0086	-0.0474	0.0018	0.0000	0.0000	-0.0001		313	0.0009	-0.0003	0.0015	0.0001	0.0000	0.0000
	314	-0.0082	-0.0476	0.0018	0.0000	0.0000	0.0000		314	-0.0052	-0.0003	0.0017	0.0001	0.0000	0.0003
	315	0.0086	-0.0477	-0.0016	0.0000	0.0000	-0.0001		315	0.0009	-0.0002	-0.0004	0.0000	0.0000	0.0000
	316	-0.0082	-0.0478	-0.0016	0.0000	0.0000	0.0001		316	-0.0052	-0.0002	-0.0002	0.0000	0.0000	0.0003

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE										JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE											
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN			JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN				
37	301	0.0239	-0.0042	0.0017	0.0000	-0.0008	-0.0003			40	301	0.0000	0.0000	0.0000	0.0001	0.0003	-0.0004				
	302	0.0040	0.0008	-0.0004	0.0000	-0.0002	-0.0001				302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000			
	303	-0.0007	-0.0027	0.0072	-0.0004	0.0000	0.0000				303	0.0000	0.0000	0.0000	0.0000	0.0002	0.0000	0.0000			
	304	-0.0035	-0.0021	0.0041	-0.0002	0.0001	0.0001				304	0.0000	0.0000	0.0000	0.0000	0.0008	0.0015	0.0002	0.0002		
	305	0.0068	0.0024	-0.0039	0.0002	-0.0003	-0.0001				305	0.0000	0.0000	0.0000	0.0000	0.0006	0.0011	0.0001	0.0001		
	306	-0.0018	0.0019	-0.0065	0.0004	0.0001	0.0000				306	0.0000	0.0000	0.0000	0.0000	0.0012	0.0021	0.0003	0.0003		
	307	0.0065	0.0018	-0.0020	0.0001	-0.0003	-0.0001				307	0.0000	0.0000	0.0000	0.0000	-0.0012	-0.0023	-0.0003	-0.0003		
	308	0.0012	0.0019	-0.0049	0.0003	0.0000	0.0000				308	0.0000	0.0000	0.0000	0.0000	-0.0007	-0.0012	-0.0002	-0.0002		
	309	-0.0181	0.0006	-0.0003	0.0000	0.0005	0.0001				309	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0001		
	310	-0.0181	0.0002	-0.0003	0.0000	0.0005	0.0001				310	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001		
	311	0.0245	0.0010	-0.0003	0.0000	-0.0008	-0.0002				311	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001		
	312	0.0245	0.0006	-0.0003	0.0000	-0.0008	-0.0002				312	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001		
	313	-0.0032	0.0012	-0.0003	0.0000	0.0001	0.0000				313	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	314	0.0096	0.0013	-0.0003	0.0000	-0.0003	-0.0001				314	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
	315	-0.0032	0.0000	-0.0003	0.0000	0.0001	0.0000				315	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000		
38	316	0.0095	0.0001	-0.0003	0.0000	-0.0003	-0.0001			316	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000			
	301	-0.0001	-0.0055	0.0011	0.0000	0.0005	-0.0002			41	301	-0.0001	-0.0038	-0.0011	0.0000	0.0000	0.0000	0.0005			
	302	0.0002	0.0007	-0.0008	0.0000	0.0002	0.0000				302	0.0006	0.0006	0.0006	0.0000	0.0000	0.0000	-0.0001	-0.0001		
	303	-0.0002	-0.0111	0.0873	-0.0009	0.0000	0.0000				303	0.0010	0.0027	0.0136	-0.0001	0.0002	0.0000	0.0000	0.0000		
	304	-0.0002	-0.0065	0.0471	-0.0005	-0.0001	0.0000				304	0.0005	0.0057	0.0440	-0.0005	0.0009	0.0000	0.0000	0.0000		
	305	0.0004	0.0063	-0.0420	0.0004	0.0003	0.0000				305	0.0012	0.0050	0.0322	-0.0003	0.0006	-0.0001	-0.0001	-0.0001		
	306	0.0001	0.0099	-0.0811	0.0009	-0.0001	0.0000				306	0.0003	0.0075	0.0621	-0.0007	0.0013	0.0001	0.0001	0.0001		
	307	0.0003	0.0033	-0.0176	0.0002	0.0003	0.0000				307	-0.0007	-0.0086	-0.0670	0.0007	-0.0014	0.0000	0.0000	0.0000		
	308	0.0001	0.0074	-0.0574	0.0006	0.0000	0.0000				308	-0.0008	-0.0051	-0.0361	0.0004	-0.0007	0.0001	0.0001	0.0001		
	309	-0.0171	0.0006	-0.0006	0.0000	0.0000	0.0000				309	-0.0166	0.0004	0.0005	0.0000	0.0000	0.0000	-0.0001	-0.0001		
	310	-0.0171	0.0002	-0.0006	0.0000	0.0000	0.0000				310	-0.0166	0.0006	0.0006	0.0005	0.0000	0.0000	-0.0001	-0.0001		
	311	0.0173	0.0009	-0.0007	0.0000	0.0002	0.0000				311	0.0175	0.0006	0.0005	0.0005	0.0000	0.0000	0.0000	-0.0001		
	312	0.0173	0.0006	-0.0006	0.0000	0.0002	0.0000				312	0.0175	0.0006	0.0006	0.0005	0.0000	0.0000	0.0000	-0.0001		
	313	-0.0050	0.0010	-0.0007	0.0000	0.0001	0.0000				313	-0.0046	0.0001	0.0004	0.0000	0.0000	0.0000	0.0000	-0.0001		
	314	0.0053	0.0011	-0.0007	0.0000	0.0002	0.0000				314	0.0055	0.0001	0.0004	0.0000	0.0000	0.0000	-0.0001	-0.0001		
315	-0.0050	0.0000	-0.0005	0.0000	0.0001	0.0000			315		-0.0046	0.0009	0.0006	0.0000	0.0000	0.0000	-0.0001	-0.0001			
39	316	0.0053	0.0001	-0.0006	0.0000	0.0002	0.0000			316	0.0055	0.0009	0.0006	0.0000	0.0000	0.0000	-0.0001	-0.0001			
	301	0.0000	0.0000	0.0000	-0.0001	0.0043	-0.0006			42	301	-0.0133	-0.0028	-0.0012	0.0000	-0.0002	0.0000	0.0000	0.0000		
	302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000				302	0.0161	0.0007	0.0003	0.0000	0.0000	0.0007	-0.0003	-0.0003		
	303	0.0000	0.0000	0.0000	0.0016	-0.0005	0.0001				303	0.0231	0.0016	0.0015	-0.0001	0.0010	-0.0004	-0.0004	-0.0004		
	304	0.0000	0.0000	0.0000	0.0009	-0.0003	0.0000				304	-0.0062	0.0015	0.0037	-0.0002	-0.0002	0.0001	0.0001	0.0001		
	305	0.0000	0.0000	0.0000	-0.0008	0.0002	0.0000				305	-0.0195	0.0021	0.0030	-0.0002	0.0009	-0.0004	-0.0004	-0.0004		
	306	0.0000	0.0000	0.0000	-0.0015	0.0004	-0.0001				306	-0.0227	0.0015	0.0050	-0.0003	-0.0008	0.0003	0.0003	0.0003		
	307	0.0000	0.0000	0.0000	-0.0003	0.0001	0.0000				307	0.0139	-0.0021	-0.0056	0.0003	0.0005	-0.0002	-0.0002	0.0001		
	308	0.0000	0.0000	0.0000	-0.0011	0.0003	0.0000				308	-0.0054	-0.0017	-0.0032	0.0002	0.0003	0.0000	0.0001	0.0001		
	309	0.0000	0.0000	0.0000	0.0000	0.0003	0.0000				309	-0.0012	0.0004	0.0002	0.0000	0.0000	0.0002	-0.0002	-0.0002		
	310	0.0000	0.0000	0.0000	0.0000	0.0003	0.0000				310	-0.0011	0.0007	0.0002	0.0000	0.0000	0.0002	-0.0002	-0.0002		
	311	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000				311	0.0268	0.0004	0.0002	0.0000	0.0009	-0.0002	-0.0002	-0.0002		
	312	0.0000	0.0000	0.0000	0.0000	-0.0003	0.0000				312	0.0269	0.0007	0.0002	0.0000	0.0009	-0.0002	-0.0002	-0.0002		
	313	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000				313	0.0066	0.0001	0.0002	0.0000	0.0004	-0.0002	-0.0002	-0.0002		
	314	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000				314	0.0169	0.0001	0.0002	0.0000	0.0006	-0.0002	-0.0002	-0.0002		
315	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000			315		0.0088	0.0011	0.0002	0.0000	0.0000	0.0004	-0.0002	-0.0002			
316	0.0000	0.0000	0.0000	0.0000	-0.0001	0.0000			316	0.0171	0.0011	0.0002	0.0000	0.0006	-0.0002	-0.0002	-0.0002				

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN
43	301	-0.0147	-0.0066	-0.0001	-0.0001	-0.0005	0.0009		
	302	-0.0100	-0.0003	-0.0008	-0.0001	-0.0002	0.0006		
	303	-0.0160	-0.0005	-0.0014	-0.0001	-0.0003	0.0010		
	304	-0.0014	-0.0003	-0.0003	0.0000	-0.0001	0.0001		
	305	-0.0159	-0.0006	-0.0015	-0.0001	-0.0003	0.0010		
	306	0.0066	-0.0003	0.0003	0.0000	0.0000	-0.0004		
	307	-0.0006	0.0005	0.0003	0.0000	0.0001	0.0000		
	308	0.0076	0.0004	0.0009	0.0001	0.0002	-0.0005		
	309	-0.0015	-0.0002	-0.0004	0.0000	-0.0002	0.0003		
	310	-0.0015	-0.0002	-0.0009	0.0000	-0.0002	0.0003		
	311	-0.0145	-0.0002	-0.0004	0.0000	-0.0001	0.0007		
	312	-0.0145	-0.0002	-0.0009	0.0000	-0.0001	0.0007		
	313	-0.0061	-0.0002	0.0001	0.0000	-0.0002	0.0000		
	314	-0.0099	-0.0002	0.0001	0.0000	-0.0001	0.0006		
	315	-0.0061	-0.0002	-0.0014	-0.0001	-0.0002	0.0004		
	316	-0.0099	-0.0002	-0.0014	-0.0001	-0.0001	0.0006		
44	301	0.0112	-0.0153	-0.0027	-0.0002	0.0000	-0.0012		
	302	-0.0065	-0.0093	-0.0024	-0.0001	0.0000	0.0000		
	303	-0.0089	-0.0175	-0.0042	-0.0002	-0.0001	0.0001		
	304	0.0039	-0.0046	-0.0010	0.0000	0.0000	-0.0003		
	305	-0.0070	-0.0161	-0.0041	-0.0002	-0.0001	0.0000		
	306	0.0111	0.0033	0.0009	0.0000	0.0001	-0.0005		
	307	-0.0077	0.0046	0.0009	0.0000	0.0000	0.0005		
	308	0.0012	0.0085	0.0023	0.0001	0.0000	0.0002		
	309	0.0095	-0.0071	-0.0016	-0.0001	-0.0001	-0.0002		
	310	0.0094	-0.0078	-0.0022	-0.0001	-0.0001	-0.0002		
	311	-0.0198	-0.0071	-0.0015	-0.0001	0.0000	0.0004		
	312	-0.0199	-0.0077	-0.0022	-0.0001	0.0000	0.0000		
	313	-0.0007	-0.0064	-0.0008	-0.0001	0.0000	0.0000		
	314	-0.0094	-0.0064	-0.0008	-0.0001	0.0000	0.0002		
	315	-0.0010	-0.0085	-0.0030	-0.0001	0.0000	0.0000		
	316	-0.0097	-0.0085	-0.0030	-0.0001	0.0000	0.0002		
45	301	-0.0011	-0.0022	-0.0013	-0.0001	0.0000	0.0001		
	302	-0.0009	-0.0013	-0.0010	-0.0001	0.0000	0.0001		
	303	-0.0014	-0.0018	-0.0012	0.0000	0.0000	0.0001		
	304	-0.0001	0.0000	0.0000	0.0001	0.0000	0.0000		
	305	-0.0015	-0.0021	-0.0018	-0.0001	0.0000	0.0001		
	306	0.0006	0.0009	0.0006	0.0001	0.0000	-0.0001		
	307	-0.0001	-0.0003	-0.0004	-0.0001	0.0000	0.0000		
	308	0.0006	0.0011	0.0014	0.0001	0.0000	-0.0001		
	309	0.0276	-0.0011	-0.0003	-0.0001	0.0000	-0.0003		
	310	0.0276	-0.0011	-0.0014	-0.0001	0.0000	-0.0003		
	311	-0.0291	-0.0010	-0.0003	-0.0001	0.0000	0.0004		
	312	-0.0291	-0.0010	-0.0013	-0.0001	0.0000	0.0004		
	313	0.0077	-0.0010	0.0009	0.0000	0.0000	0.0000		
	314	-0.0092	-0.0010	0.0009	0.0000	0.0000	0.0002		
	315	0.0077	-0.0010	-0.0025	-0.0001	0.0000	0.0000		
	316	-0.0092	-0.0010	-0.0025	-0.0001	0.0000	0.0002		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN
49	301	-0.0147	-0.0006	0.0001	0.0001	0.0005	0.0009		
	302	-0.0099	-0.0003	0.0009	0.0001	0.0002	0.0006		
	303	-0.0006	0.0005	-0.0003	0.0000	-0.0001	0.0000		
	304	0.0075	0.0005	-0.0008	-0.0001	-0.0002	-0.0005		
	305	-0.0158	-0.0006	0.0015	0.0001	0.0003	0.0010		
	306	0.0066	-0.0002	-0.0003	0.0000	-0.0000	-0.0004		
	307	-0.0159	-0.0005	0.0014	0.0001	0.0003	0.0010		
	308	-0.0014	-0.0004	0.0003	0.0000	0.0001	0.0001		
	309	-0.0013	-0.0002	0.0009	0.0000	0.0002	0.0003		
	310	-0.0013	-0.0002	0.0005	0.0000	0.0002	0.0003		
	311	-0.0146	-0.0002	0.0009	0.0000	0.0001	0.0007		
	312	-0.0146	-0.0002	0.0005	0.0000	0.0001	0.0007		
	313	-0.0060	-0.0002	0.0015	0.0001	0.0002	0.0004		
	314	-0.0100	-0.0002	0.0015	0.0001	0.0001	0.0006		
	315	-0.0060	-0.0002	-0.0001	0.0000	0.0002	0.0004		
	316	-0.0099	-0.0002	-0.0001	0.0000	0.0001	0.0006		
50	301	-0.0129	-0.0027	0.0012	0.0000	0.0002	0.0000		
	302	0.0158	0.0007	-0.0003	0.0000	-0.0007	-0.0003		
	303	0.0136	-0.0021	0.0056	-0.0003	-0.0004	-0.0002		
	304	-0.0052	-0.0017	0.0032	-0.0002	0.0003	0.0004		
	305	-0.0192	0.0021	-0.0030	0.0002	-0.0009	-0.0004		
	306	-0.0223	0.0014	-0.0050	0.0003	0.0008	0.0003		
	307	0.0228	0.0016	-0.0016	0.0001	-0.0010	-0.0004		
	308	-0.0061	0.0015	-0.0038	0.0002	0.0002	0.0001		
	309	-0.0013	0.0007	-0.0002	0.0000	-0.0002	-0.0002		
	310	-0.0013	0.0004	-0.0002	0.0000	-0.0002	-0.0002		
	311	0.0267	0.0007	-0.0002	0.0000	-0.0009	-0.0002		
	312	0.0266	0.0004	-0.0002	0.0000	-0.0002	-0.0002		
	313	0.0086	0.0011	-0.0002	0.0000	-0.0004	-0.0002		
	314	0.0169	0.0011	-0.0002	0.0000	-0.0006	-0.0004		
	315	0.0084	0.0001	-0.0002	0.0000	-0.0004	-0.0002		
	316	0.0168	0.0001	-0.0002	0.0000	-0.0006	-0.0002		
51	301	-0.0002	-0.0037	0.0011	0.0000	0.0000	0.0005		
	302	0.0005	0.0007	-0.0007	0.0000	0.0000	-0.0001		
	303	-0.0007	-0.0085	0.0669	0.0007	0.0014	0.0000		
	304	-0.0007	-0.0051	0.0362	-0.0004	0.0007	0.0001		
	305	0.0011	0.0050	-0.0323	0.0003	-0.0006	-0.0001		
	306	0.0003	0.0075	-0.0621	0.0007	-0.0013	0.0001		
	307	0.0009	0.0027	-0.0136	0.0001	-0.0002	-0.0001		
	308	0.0005	0.0057	-0.0440	0.0005	-0.0009	0.0000		
	309	-0.0167	0.0007	-0.0006	0.0000	0.0000	-0.0001		
	310	-0.0167	0.0004	-0.0005	0.0000	0.0000	-0.0001		
	311	0.0175	0.0006	-0.0005	0.0000	0.0000	0.0000		
	312	0.0175	0.0004	-0.0005	0.0000	0.0000	0.0000		
	313	-0.0047	0.0010	-0.0006	0.0000	0.0000	-0.0001		
	314	0.0055	0.0010	-0.0006	0.0000	0.0000	-0.0001		
	315	-0.0047	0.0001	-0.0005	0.0000	0.0000	-0.0001		
	316	0.0055	0.0001	-0.0005	0.0000	0.0000	-0.0001		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN		
55	301	0.0147	0.1182	0.0181	-0.0001	-0.0001	-0.0008		
	302	0.0100	0.0345	0.0111	-0.0001	0.0000	-0.0006		
	303	0.0159	0.1344	-0.0184	-0.0001	0.0001	0.0009		
	304	0.0014	0.0108	0.0020	0.0000	0.0000	-0.0001		
	305	0.0159	0.1333	0.0177	-0.0001	0.0000	-0.0009		
	306	-0.0066	-0.0580	-0.0073	0.0000	-0.0001	0.0004		
	307	0.0066	0.0070	0.0000	0.0000	0.0000	0.0000		
	308	-0.0076	-0.0030	-0.0087	0.0001	0.0000	0.0004		
	309	0.0145	0.1024	0.0047	0.0000	0.0000	-0.0007		
	310	0.0145	0.1024	0.0039	0.0000	0.0000	-0.0007		
	311	0.0015	0.0328	0.0139	0.0000	0.0001	-0.0002		
	312	0.0015	0.0328	0.0131	0.0000	0.0001	-0.0002		
	313	0.0099	0.0780	0.0089	0.0000	0.0000	-0.0005		
	314	0.0060	0.0572	0.0116	0.0000	0.0000	-0.0004		
	315	0.0099	0.0780	0.0062	-0.0001	0.0000	-0.0005		
	316	0.0061	0.0572	0.0089	-0.0001	0.0000	-0.0004		
	317	0.0147	0.1185	-0.0198	0.0001	0.0000	-0.0008		
56	301	0.0100	0.0844	-0.0123	0.0001	0.0000	-0.0006		
	302	0.0070	0.0070	-0.0002	0.0000	0.0000	0.0000		
	303	-0.0075	-0.0029	0.0096	-0.0001	0.0000	0.0004		
	304	0.0158	0.1332	-0.0195	0.0001	-0.0001	-0.0009		
	305	-0.0066	-0.0579	0.0081	0.0000	0.0001	0.0004		
	306	0.0159	0.1343	-0.0202	0.0001	-0.0001	-0.0009		
	307	0.0014	0.0108	-0.0021	0.0000	0.0000	-0.0001		
	308	0.0146	0.1031	-0.0055	0.0000	0.0000	-0.0007		
	309	0.0146	0.1031	-0.0062	0.0000	0.0000	-0.0004		
	310	0.0013	0.0320	-0.0135	0.0000	-0.0001	-0.0002		
	311	0.0013	0.0320	-0.0142	0.0000	-0.0001	-0.0002		
	312	0.0100	0.0782	-0.0074	0.0001	0.0000	-0.0005		
	313	0.0060	0.0569	-0.0098	0.0001	0.0000	-0.0004		
	314	0.0099	0.0781	-0.0099	0.0000	0.0000	-0.0005		
	315	0.0060	0.0569	-0.0123	0.0000	0.0000	-0.0004		
	316	-0.0011	-0.1729	0.0022	-0.0001	0.0000	-0.0008		
	317	-0.0009	-0.1017	0.0015	0.0000	0.0000	-0.0004		
57	301	-0.0014	-0.1654	0.0006	0.0000	0.0000	-0.0007		
	302	-0.0001	-0.0140	-0.0009	0.0001	0.0000	-0.0001		
	303	-0.0015	-0.1653	0.0024	0.0000	0.0000	-0.0007		
	304	-0.0006	-0.0500	-0.0009	0.0001	0.0000	-0.0003		
	305	-0.0001	-0.0005	0.0018	-0.0001	0.0000	0.0000		
	306	0.0006	0.0041	0.0017	0.0001	0.0000	0.0004		
	307	0.0276	-0.1868	-0.0006	-0.0001	0.0000	-0.0007		
	308	0.0276	-0.1868	-0.0006	-0.0001	0.0000	-0.0007		
	309	0.0276	-0.1868	-0.0006	-0.0001	0.0000	-0.0007		
	310	-0.0291	0.0242	0.0019	-0.0001	0.0000	0.0000		
	311	-0.0291	0.0242	0.0019	-0.0001	0.0000	0.0000		
	312	-0.0291	0.0242	0.0019	-0.0001	0.0000	0.0000		
	313	-0.0077	-0.1129	0.0024	0.0000	0.0000	-0.0005		
	314	-0.0092	-0.0500	0.0032	0.0000	0.0000	-0.0003		
	315	0.0077	-0.1127	-0.0008	-0.0001	0.0000	-0.0005		
	316	-0.0092	-0.0498	-0.0001	-0.0001	0.0000	-0.0003		

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN

64	301	-0.0001	-0.0506	0.0025	0.0000	-0.0001	-0.0007	-0.0001	-0.0007
	302	0.0000	-0.0309	0.0024	0.0000	0.0000	-0.0001	0.0000	-0.0001
	303	0.0001	0.0078	-0.0008	0.0000	0.0000	0.0000	0.0000	0.0000
	304	0.0000	0.0287	-0.0024	0.0000	0.0000	0.0001	0.0000	0.0001
	305	0.0000	-0.0510	0.0040	0.0000	0.0000	-0.0002	0.0000	-0.0002
	306	-0.0001	0.0162	-0.0010	0.0000	0.0000	0.0000	0.0000	0.0000
	307	0.0000	-0.0555	0.0042	0.0000	0.0000	-0.0002	0.0000	-0.0002
	308	-0.0001	-0.0106	0.0010	0.0000	0.0000	0.0000	0.0000	0.0000
	309	0.0191	-0.0251	0.0027	0.0000	0.0000	-0.0005	0.0000	-0.0005
	310	0.0191	-0.0248	0.0018	0.0000	0.0000	-0.0005	0.0000	-0.0005
	311	-0.0191	-0.0247	0.0019	0.0000	0.0000	0.0003	0.0000	0.0003
	312	-0.0191	-0.0244	0.0011	0.0000	0.0000	0.0003	0.0000	0.0003
	313	0.0057	-0.0254	0.0034	0.0000	0.0000	-0.0002	0.0000	-0.0002
	314	-0.0057	-0.0253	0.0032	0.0000	0.0000	0.0000	0.0000	0.0000
	315	0.0057	-0.0242	0.0006	0.0000	0.0000	-0.0002	0.0000	-0.0002
	316	-0.0057	-0.0241	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000
65	301	0.0001	-0.0506	0.0024	0.0000	0.0001	0.0007	0.0001	0.0007
	302	0.0000	-0.0309	0.0023	0.0000	0.0000	0.0001	0.0000	0.0001
	303	0.0000	0.0078	-0.0008	0.0000	0.0000	0.0000	0.0000	0.0000
	304	0.0000	0.0287	-0.0024	0.0000	0.0000	-0.0001	0.0000	-0.0001
	305	0.0000	-0.0510	0.0040	0.0000	0.0000	0.0002	0.0000	0.0002
	306	0.0000	0.0162	-0.0010	0.0000	0.0000	0.0000	0.0000	0.0000
	307	0.0000	-0.0554	0.0042	0.0000	0.0000	0.0002	0.0000	0.0002
	308	0.0000	-0.0162	0.0009	0.0000	0.0000	0.0000	0.0000	0.0000
	309	0.0191	-0.0247	0.0019	0.0000	0.0000	-0.0003	0.0000	-0.0003
	310	0.0191	-0.0244	0.0011	0.0000	0.0000	-0.0003	0.0000	-0.0003
	311	-0.0191	-0.0251	0.0027	0.0000	0.0000	0.0005	0.0000	0.0005
	312	-0.0191	-0.0248	0.0018	0.0000	0.0000	0.0005	0.0000	0.0005
	313	0.0057	-0.0253	0.0032	0.0000	0.0000	0.0000	0.0000	0.0000
	314	-0.0057	-0.0254	0.0034	0.0000	0.0000	0.0002	0.0000	0.0002
	315	0.0057	-0.0241	0.0003	0.0000	0.0000	0.0000	0.0000	0.0000
	316	-0.0057	-0.0242	0.0006	0.0000	0.0000	0.0002	0.0000	0.0002
66	301	0.0004	-0.0407	0.0025	0.0000	0.0000	0.0013	0.0000	0.0013
	302	0.0000	-0.0243	0.0021	0.0000	0.0000	0.0009	0.0000	0.0009
	303	-0.0002	0.0094	-0.0004	0.0000	0.0000	0.0000	0.0000	0.0000
	304	-0.0001	0.0244	-0.0019	0.0000	0.0000	-0.0007	0.0000	-0.0007
	305	0.0001	-0.0399	0.0036	0.0000	0.0000	0.0014	0.0000	0.0014
	306	0.0002	0.0126	-0.0009	0.0000	0.0000	-0.0006	0.0000	-0.0006
	307	0.0000	-0.0468	0.0035	0.0000	0.0001	0.0014	0.0000	0.0014
	308	0.0001	-0.0160	0.0006	0.0000	0.0000	0.0001	0.0000	0.0001
	309	0.0191	-0.0196	0.0021	0.0000	0.0000	0.0007	0.0000	0.0007
	310	0.0191	-0.0193	0.0013	0.0000	0.0000	0.0007	0.0000	0.0007
	311	-0.0191	-0.0195	0.0021	0.0000	-0.0001	0.0007	0.0000	0.0007
	312	-0.0191	-0.0193	0.0013	0.0000	-0.0001	0.0007	0.0000	0.0007
	313	0.0057	-0.0199	0.0030	0.0000	0.0000	0.0007	0.0000	0.0007
	314	-0.0057	-0.0199	0.0030	0.0000	0.0000	0.0007	0.0000	0.0007
	315	0.0057	-0.0190	0.0004	0.0000	0.0000	0.0007	0.0000	0.0007
	316	-0.0057	-0.0189	0.0004	0.0000	0.0000	0.0007	0.0000	0.0007

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN		

67	301	-0.0006	-0.0440	-0.0024	0.0001	0.0000	-0.0013		
	302	-0.0001	-0.0258	-0.0020	0.0000	0.0000	-0.0009		
	303	-0.0002	-0.0511	-0.0033	0.0001	0.0000	-0.0014		
	304	-0.0001	-0.0117	-0.0007	0.0000	0.0000	-0.0001		
	305	-0.0003	-0.0424	-0.0034	0.0001	0.0000	-0.0014		
	306	-0.0001	0.0135	0.0009	0.0000	0.0000	0.0006		
	307	0.0002	0.0111	0.0004	0.0000	0.0000	0.0000		
	308	0.0001	0.0265	0.0019	0.0000	0.0000	0.0007		
	309	0.0212	-0.0206	-0.0012	0.0000	0.0000	-0.0007		
	310	0.0212	-0.0207	-0.0020	0.0000	0.0000	-0.0007		
	311	-0.0214	-0.0206	-0.0012	0.0000	0.0001	-0.0007		
	312	-0.0214	-0.0208	-0.0020	0.0000	0.0001	-0.0007		
	313	0.0063	-0.0204	-0.0002	0.0000	0.0000	-0.0007		
	314	-0.0065	-0.0204	-0.0002	0.0000	0.0000	-0.0007		
	315	0.0063	-0.0209	-0.0030	0.0000	0.0000	-0.0007		
	68	316	-0.0064	-0.0209	-0.0030	0.0000	0.0000	-0.0007	
301		-0.0003	-0.0556	-0.0022	0.0001	0.0001	-0.0006		
302		-0.0001	-0.0336	-0.0021	0.0000	0.0000	-0.0001		
303		-0.0002	-0.0610	-0.0039	0.0001	0.0000	-0.0002		
304		-0.0001	-0.0118	-0.0010	0.0000	0.0000	0.0000		
305		-0.0002	-0.0552	-0.0037	0.0001	0.0000	-0.0002		
306		0.0000	0.0179	0.0008	0.0000	0.0000	0.0000		
307		0.0001	0.0089	0.0009	0.0000	0.0000	0.0000		
308		0.0001	0.0314	0.0022	0.0000	0.0000	0.0001		
309		0.0212	-0.0270	-0.0016	0.0000	0.0000	-0.0006		
310		0.0212	-0.0272	-0.0024	0.0000	0.0000	-0.0006		
311		-0.0214	-0.0266	-0.0009	0.0000	0.0000	0.0004		
312		-0.0214	-0.0268	-0.0018	0.0000	0.0000	0.0004		
313		0.0062	-0.0266	-0.0003	0.0000	0.0000	-0.0002		
314		-0.0065	-0.0265	-0.0001	0.0000	0.0000	-0.0001		
69		315	0.0063	-0.0273	-0.0033	0.0000	0.0000	-0.0002	
	316	-0.0064	-0.0272	-0.0031	0.0000	0.0000	0.0001		
	301	-0.0002	-0.0557	-0.0022	0.0001	-0.0001	0.0006		
	302	-0.0001	-0.0336	-0.0021	0.0000	0.0000	0.0001		
	303	-0.0002	-0.0609	-0.0039	0.0001	0.0000	0.0002		
	304	0.0000	-0.0118	-0.0010	0.0000	0.0000	0.0000		
	305	-0.0002	-0.0551	-0.0037	0.0001	0.0000	0.0002		
	306	0.0001	0.0179	0.0008	0.0000	0.0000	0.0000		
	307	0.0000	0.0090	0.0009	0.0000	0.0000	0.0000		
	308	0.0000	0.0246	0.0025	0.0000	0.0000	-0.0001		
	309	0.0212	-0.0266	-0.0010	0.0000	0.0000	-0.0004		
	310	0.0212	-0.0268	-0.0018	0.0000	0.0000	-0.0004		
	311	-0.0214	-0.0269	-0.0016	0.0000	0.0000	0.0006		
	312	-0.0214	-0.0271	-0.0025	0.0000	0.0000	0.0006		
	313	0.0062	-0.0264	-0.0001	0.0000	0.0000	-0.0000		
	314	-0.0065	-0.0265	-0.0003	0.0000	0.0000	-0.0002		
315	0.0063	-0.0272	-0.0031	0.0000	0.0000	-0.0001			
316	-0.0065	-0.0273	-0.0033	0.0000	0.0000	0.0000			

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN		

70	301	0.0001	-0.0440	-0.0024	0.0001	0.0000	-0.0013		
	302	-0.0001	-0.0258	-0.0020	0.0000	0.0000	-0.0009		
	303	-0.0002	-0.0511	-0.0033	0.0001	0.0000	-0.0014		
	304	0.0001	-0.0117	-0.0007	0.0000	0.0000	-0.0001		
	305	-0.0002	-0.0423	-0.0034	0.0001	0.0000	-0.0014		
	306	0.0002	0.0135	0.0009	0.0000	0.0000	0.0006		
	307	-0.0001	0.0111	0.0004	0.0000	0.0000	0.0000		
	308	0.0000	0.0198	0.0021	0.0000	0.0000	0.0007		
	309	0.0212	-0.0206	-0.0012	0.0000	0.0000	-0.0007		
	310	0.0212	-0.0208	-0.0020	0.0000	0.0000	-0.0007		
	311	-0.0215	-0.0206	-0.0012	0.0000	0.0000	-0.0007		
	312	-0.0214	-0.0207	-0.0020	0.0000	0.0000	-0.0007		
	313	0.0062	-0.0204	-0.0002	0.0000	0.0000	-0.0007		
	314	-0.0065	-0.0204	-0.0002	0.0000	0.0000	-0.0007		
	315	0.0063	-0.0209	-0.0030	0.0000	0.0000	-0.0007		
	71	316	-0.0065	-0.0209	-0.0030	0.0000	0.0000	-0.0007	
301		0.0004	0.1697	-0.0229	0.0000	0.0000	0.0002		
302		0.0000	0.1155	-0.0147	0.0000	0.0001	-0.0008		
303		-0.0002	0.0147	-0.0017	0.0000	0.0000	-0.0000		
304		-0.0001	-0.0831	0.0106	0.0000	0.0000	-0.0001		
305		0.0000	0.1838	-0.0232	0.0000	0.0012	0.0012		
306		0.0002	-0.0769	0.0099	0.0000	0.0000	-0.0001		
307		0.0000	0.1763	-0.0227	0.0000	0.0000	0.0002		
308		0.0001	0.0060	-0.0012	0.0000	0.0000	0.0001		
309		0.0191	0.0921	-0.0196	0.0000	0.0002	0.0006		
310		0.0191	0.0923	-0.0201	0.0000	0.0001	0.0006		
311		-0.0191	0.0925	-0.0034	0.0000	0.0000	0.0006		
312		-0.0191	0.0928	-0.0039	0.0000	0.0000	0.0006		
313		0.0057	0.0919	-0.0133	0.0000	0.0001	0.0006		
314		-0.0057	0.0921	-0.0085	0.0000	0.0001	0.0006		
315		0.0057	0.0928	-0.0150	0.0000	0.0001	0.0006		
72	316	-0.0057	0.0929	-0.0101	0.0000	0.0001	0.0006		
	301	-0.0004	0.1697	-0.0229	0.0000	-0.0002	-0.0011		
	302	0.0000	0.1155	-0.0147	0.0000	0.0000	-0.0001		
	303	0.0002	0.0147	-0.0017	0.0000	0.0000	0.0000		
	304	0.0001	-0.0831	0.0106	0.0000	0.0001	0.0006		
	305	0.0000	0.1838	-0.0232	0.0000	0.0000	-0.0002		
	306	-0.0002	-0.0769	0.0099	0.0000	0.0000	-0.0001		
	307	0.0000	0.1763	-0.0227	0.0000	0.0000	0.0002		
	308	-0.0001	0.0116	-0.0016	0.0000	0.0000	-0.0001		
	309	0.0191	0.0925	-0.0034	0.0000	0.0000	-0.0006		
	310	0.0191	0.0928	-0.0039	0.0000	0.0000	-0.0006		
	311	-0.0191	0.0921	-0.0196	0.0000	0.0000	0.0001		
	312	-0.0191	0.0923	-0.0201	0.0000	0.0000	0.0001		
	313	0.0057	0.0921	-0.0085	0.0000	0.0000	0.0001		
	314	-0.0057	0.0928	-0.0133	0.0000	0.0000	0.0001		
	315	0.0057	0.0930	-0.0101	0.0000	0.0000	0.0001		
316	-0.0057	0.0928	-0.0150	0.0000	0.0000	0.0001			

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE										JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									
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JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
73	301	0.0001	0.1664	0.0201	0.0001	-0.0002	0.0011	0.0001	-0.0002	0.0201	0.0001	-0.0002	0.0011	0.0001	-0.0002	0.0201	0.0001	-0.0002	0.0011
	302	-0.0001	0.1144	0.0129	0.0000	-0.0001	0.0008	0.0000	-0.0001	0.0129	0.0000	-0.0001	0.0008	0.0000	-0.0001	0.0129	0.0000	-0.0001	0.0008
	303	-0.0002	0.1725	0.0198	0.0001	-0.0002	0.0012	0.0001	-0.0002	0.0198	0.0001	-0.0002	0.0012	0.0001	-0.0002	0.0198	0.0001	-0.0002	0.0012
	304	0.0001	0.0103	0.0012	0.0000	0.0000	0.0001	0.0000	0.0000	0.0012	0.0000	0.0000	0.0001	0.0000	0.0000	0.0012	0.0000	0.0000	0.0001
	305	-0.0002	0.1820	0.0204	0.0001	-0.0002	0.0012	0.0001	-0.0002	0.0204	0.0001	-0.0002	0.0012	0.0001	-0.0002	0.0204	0.0001	-0.0002	0.0012
	306	0.0002	-0.0761	-0.0088	0.0000	0.0001	-0.0005	0.0000	0.0001	-0.0088	0.0000	0.0001	-0.0005	0.0000	0.0001	-0.0088	0.0000	0.0001	-0.0005
	307	-0.0001	0.0165	0.0017	0.0000	0.0000	0.0000	0.0000	0.0000	0.0017	0.0000	0.0000	0.0000	0.0000	0.0000	0.0017	0.0000	0.0000	0.0000
	308	0.0000	-0.0879	-0.0096	0.0000	0.0001	-0.0006	0.0000	0.0001	-0.0096	0.0000	0.0001	-0.0006	0.0000	0.0001	-0.0096	0.0000	0.0001	-0.0006
	309	0.0212	0.0912	0.0190	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0190	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0190	0.0000	-0.0001	0.0006
	310	0.0212	0.0912	0.0185	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0185	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0185	0.0000	-0.0001	0.0006
	311	-0.0215	0.0918	0.0021	0.0000	0.0000	0.0006	0.0000	0.0000	0.0021	0.0000	0.0000	0.0006	0.0000	0.0000	0.0021	0.0000	0.0000	0.0006
	312	-0.0215	0.0916	0.0016	0.0000	0.0000	0.0006	0.0000	0.0000	0.0016	0.0000	0.0000	0.0006	0.0000	0.0000	0.0016	0.0000	0.0000	0.0006
	313	0.0062	0.0916	0.0136	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0136	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0136	0.0000	-0.0001	0.0006
	314	-0.0065	0.0918	0.0086	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0086	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0086	0.0000	-0.0001	0.0006
	315	0.0063	0.0912	0.0120	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0120	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0120	0.0000	-0.0001	0.0006
74	316	-0.0005	0.0913	0.0070	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0070	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0070	0.0000	-0.0001	0.0006
	301	-0.0005	0.1664	0.0203	0.0001	0.0002	-0.0011	0.0001	0.0002	0.0203	0.0001	0.0002	-0.0011	0.0001	0.0002	0.0203	0.0001	0.0002	-0.0011
	302	-0.0001	0.1144	0.0130	0.0000	0.0001	-0.0008	0.0000	0.0001	0.0130	0.0000	0.0001	-0.0008	0.0000	0.0001	0.0130	0.0000	0.0001	-0.0008
	303	-0.0002	0.1725	0.0200	0.0001	0.0002	-0.0012	0.0001	0.0002	0.0200	0.0001	0.0002	-0.0012	0.0001	0.0002	0.0200	0.0001	0.0002	-0.0012
	304	-0.0001	0.0103	0.0013	0.0000	0.0000	0.0001	0.0000	0.0000	0.0013	0.0000	0.0000	0.0001	0.0000	0.0000	0.0013	0.0000	0.0000	0.0001
	305	-0.0003	0.1820	0.0206	0.0001	-0.0002	0.0005	0.0001	-0.0002	0.0206	0.0001	-0.0002	0.0005	0.0001	-0.0002	0.0206	0.0001	-0.0002	0.0005
	306	-0.0001	-0.0761	-0.0089	0.0000	-0.0001	0.0005	0.0000	-0.0001	-0.0089	0.0000	-0.0001	0.0005	0.0000	-0.0001	-0.0089	0.0000	-0.0001	0.0005
	307	0.0002	0.0165	0.0017	0.0000	0.0000	0.0000	0.0000	0.0000	0.0017	0.0000	0.0000	0.0000	0.0000	0.0000	0.0017	0.0000	0.0000	0.0000
	308	0.0001	-0.0812	-0.0093	0.0000	-0.0001	0.0006	0.0000	-0.0001	-0.0093	0.0000	-0.0001	0.0006	0.0000	-0.0001	-0.0093	0.0000	-0.0001	0.0006
	309	0.0212	0.0918	0.0022	0.0000	0.0000	-0.0006	0.0000	0.0000	0.0022	0.0000	0.0000	-0.0006	0.0000	0.0000	0.0022	0.0000	0.0000	-0.0006
	310	0.0213	0.0916	0.0018	0.0000	0.0000	-0.0006	0.0000	0.0000	0.0018	0.0000	0.0000	-0.0006	0.0000	0.0000	0.0018	0.0000	0.0000	-0.0006
	311	-0.0214	0.0913	0.0190	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0190	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0190	0.0000	0.0001	-0.0006
	312	-0.0214	0.0912	0.0186	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0186	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0186	0.0000	0.0001	-0.0006
	313	0.0063	0.0918	0.0087	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0087	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0087	0.0000	0.0001	-0.0006
	314	-0.0065	0.0917	0.0137	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0137	0.0000	-0.0001	0.0006	0.0000	-0.0001	0.0137	0.0000	-0.0001	0.0006
	315	0.0063	0.0913	0.0071	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0071	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0071	0.0000	0.0001	-0.0006
75	316	-0.0064	0.0912	0.0121	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0121	0.0000	0.0001	-0.0006	0.0000	0.0001	0.0121	0.0000	0.0001	-0.0006
	301	-0.0011	-0.0951	0.0106	0.0000	-0.0001	0.0008	0.0000	-0.0001	0.0106	0.0000	-0.0001	0.0008	0.0000	-0.0001	0.0106	0.0000	-0.0001	0.0008
	302	0.0001	0.0903	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0000	0.0000	0.0000
	303	0.0003	-0.0929	0.0421	0.0000	-0.0004	-0.0001	0.0000	-0.0004	-0.0421	0.0000	-0.0004	-0.0001	0.0000	-0.0004	-0.0421	0.0000	-0.0004	-0.0001
	304	0.0002	-0.0119	0.1436	0.0000	-0.0015	-0.0001	0.0000	-0.0015	0.1436	0.0000	-0.0015	-0.0001	0.0000	-0.0015	0.1436	0.0000	-0.0015	-0.0001
	305	0.0003	-0.0981	0.1034	0.0000	-0.0011	-0.0001	0.0000	-0.0011	0.1034	0.0000	-0.0011	-0.0001	0.0000	-0.0011	0.1034	0.0000	-0.0011	-0.0001
	306	0.0002	-0.0173	0.2044	0.0000	-0.0021	-0.0002	0.0000	-0.0021	0.2044	0.0000	-0.0021	-0.0002	0.0000	-0.0021	0.2044	0.0000	-0.0021	-0.0002
	307	-0.0003	0.0183	-0.2190	0.0000	0.0023	0.0002	0.0000	0.0023	-0.2190	0.0000	0.0023	0.0002	0.0000	0.0023	-0.2190	0.0000	0.0023	0.0002
	308	-0.0002	0.0995	-0.1173	0.0000	0.0012	0.0001	0.0000	0.0012	-0.1173	0.0000	0.0012	0.0001	0.0000	0.0012	-0.1173	0.0000	0.0012	0.0001
	309	-0.0127	0.0902	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000
	310	-0.0127	0.0903	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000
	311	0.0129	0.0903	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000
	312	0.0129	0.0903	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000
	313	-0.0037	0.0903	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000
	314	0.0039	0.0900	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000
	315	-0.0037	0.0904	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0000	0.0000	0.0000
	316	0.0039	0.0904	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0000	0.0000	0.0000	0.0000	0.0000	0.0007	0.0000	0.0000	0.0000

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	Y-ROTAN	Z-ROTAN
79	301	0.0013	0.0030	-0.0032	0.0000	0.0001	0.0006		
	302	0.0000	0.0020	0.0008	0.0000	0.0000	0.0000	-0.0044	0.0005
	303	-0.0001	0.0053	-0.0110	0.0000	0.0004	0.0001	0.0000	0.0000
	304	-0.0002	0.0075	-0.0434	0.0000	0.0014	0.0001	0.0005	-0.0001
	305	-0.0002	0.0083	-0.0298	0.0000	0.0010	0.0001	0.0003	0.0000
	306	-0.0002	0.0088	-0.0628	0.0000	0.0021	0.0002	-0.0002	0.0000
	307	0.0003	-0.0108	0.0663	0.0000	-0.0022	-0.0002	0.0001	0.0001
	308	0.0002	-0.0074	0.0348	0.0000	-0.0012	-0.0001	-0.0003	0.0000
	309	-0.0128	0.0048	0.0008	0.0000	0.0000	0.0000	0.0003	0.0000
	310	-0.0128	0.0049	0.0009	0.0000	0.0000	0.0000	0.0003	0.0000
	311	0.0128	-0.0016	0.0004	0.0000	0.0000	0.0000	-0.0003	0.0000
	312	0.0128	-0.0015	0.0005	0.0000	0.0000	0.0000	-0.0003	0.0000
	313	-0.0038	0.0024	0.0006	0.0000	0.0000	0.0000	0.0001	0.0000
	314	-0.0038	0.0005	0.0005	0.0000	0.0000	0.0000	-0.0001	0.0000
	315	-0.0038	0.0028	0.0008	0.0000	0.0000	0.0000	0.0001	0.0000
	316	-0.0038	0.0009	0.0007	0.0000	0.0000	0.0000	-0.0001	0.0000
80	301	-0.0011	0.0031	-0.0032	0.0000	-0.0001	-0.0006	0.0000	-0.0005
	302	0.0001	0.0021	0.0008	0.0000	0.0000	0.0000	0.0000	0.0000
	303	0.0003	0.0054	-0.0110	0.0000	-0.0004	-0.0001	-0.0005	0.0001
	304	0.0002	0.0075	-0.0434	0.0000	-0.0014	-0.0001	-0.0003	0.0000
	305	0.0003	0.0084	-0.0298	0.0000	-0.0010	-0.0001	0.0002	0.0000
	306	0.0002	0.0088	-0.0628	0.0000	-0.0021	-0.0002	0.0004	-0.0001
	307	-0.0003	-0.0108	0.0663	0.0000	0.0022	0.0002	0.0001	0.0000
	308	-0.0002	-0.0074	0.0347	0.0000	0.0012	0.0001	0.0003	0.0000
	309	-0.0127	-0.0016	0.0004	0.0000	0.0000	0.0000	0.0000	0.0000
	310	-0.0127	-0.0015	0.0005	0.0000	0.0000	0.0000	0.0003	0.0000
	311	0.0129	0.0048	0.0008	0.0000	0.0000	0.0000	-0.0003	0.0000
	312	0.0129	0.0049	0.0009	0.0000	0.0000	0.0000	-0.0003	0.0000
	313	-0.0037	0.0005	0.0005	0.0000	0.0000	0.0000	0.0001	0.0000
	314	0.0039	0.0024	0.0006	0.0000	0.0000	0.0000	-0.0001	0.0000
	315	-0.0037	0.0009	0.0007	0.0000	0.0000	0.0000	0.0001	0.0000
	316	0.0039	0.0028	0.0008	0.0000	0.0000	0.0000	-0.0001	0.0000
81	301	-0.0012	-0.0051	-0.0106	0.0000	0.0001	-0.0008	-0.0001	0.0008
	302	0.0001	0.0003	-0.0008	0.0000	0.0000	0.0000	0.0000	0.0000
	303	-0.0003	0.0183	0.2190	0.0000	-0.0023	0.0002	0.0000	-0.0002
	304	-0.0002	0.0095	0.1173	0.0000	-0.0012	0.0001	0.0012	-0.0001
	305	0.0003	-0.0081	-0.1034	0.0000	0.0011	-0.0001	-0.0011	0.0001
	306	0.0002	-0.0173	-0.2043	0.0000	0.0021	-0.0002	0.0021	0.0002
	307	0.0002	-0.0029	-0.0422	0.0000	0.0004	-0.0001	-0.0004	0.0001
	308	0.0002	-0.0119	-0.1436	0.0000	0.0015	-0.0001	-0.0015	0.0001
	309	-0.0128	0.0003	-0.0006	0.0000	0.0000	0.0000	0.0000	0.0000
	310	-0.0128	0.0002	-0.0006	0.0000	0.0000	0.0000	0.0000	0.0000
	311	0.0130	0.0003	-0.0007	0.0000	0.0000	0.0000	0.0000	0.0000
	312	0.0130	0.0002	-0.0006	0.0000	0.0000	0.0000	0.0000	0.0000
	313	-0.0038	0.0004	-0.0007	0.0000	0.0000	0.0000	0.0000	0.0000
	314	0.0039	0.0005	-0.0007	0.0000	0.0000	0.0000	0.0000	0.0000
	315	-0.0038	0.0000	-0.0005	0.0000	0.0000	0.0000	0.0000	0.0000
	316	0.0039	0.0000	-0.0005	0.0000	0.0000	0.0000	0.0000	0.0000

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE									

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	X-ROTAN	Y-ROTAN

85	301	0.0012	0.0031	0.0032	0.0000	-0.0001	0.0006		
	302	-0.0001	0.0021	-0.0008	0.0000	0.0000	0.0000		
	303	0.0003	-0.0108	-0.0664	0.0000	0.0022	-0.0002		
	304	0.0002	-0.0347	0.0000	0.0000	0.0000	-0.0000		
	305	-0.0003	0.0084	0.0297	0.0000	-0.0010	0.0001		
	306	-0.0002	0.0089	0.0628	0.0000	-0.0021	0.0002		
	307	-0.0002	0.0053	0.0109	0.0000	-0.0004	0.0001		
	308	-0.0002	0.0075	0.0433	0.0000	-0.0014	0.0001		
	309	-0.0130	0.0049	-0.0009	0.0000	0.0000	0.0000		
	310	-0.0130	0.0048	-0.0008	0.0000	0.0000	0.0000		
	311	0.0128	-0.0015	-0.0005	0.0000	0.0000	0.0000		
	312	0.0128	-0.0016	-0.0004	0.0000	0.0000	0.0000		
	313	-0.0039	0.0028	-0.0008	0.0000	0.0000	0.0000		
	314	-0.0038	0.0009	-0.0007	0.0000	0.0000	0.0000		
	315	-0.0039	0.0024	-0.0006	0.0000	0.0000	0.0000		
	316	0.0038	0.0005	0.0000	0.0000	0.0000	0.0000		
86	301	-0.0012	0.0031	0.0032	0.0000	0.0001	-0.0006		
	302	0.0001	0.0021	-0.0008	0.0000	0.0000	0.0000		
	303	-0.0003	-0.0108	-0.0664	0.0000	-0.0022	0.0002		
	304	-0.0002	-0.0074	-0.0347	0.0000	-0.0012	0.0001		
	305	0.0003	0.0084	0.0297	0.0000	0.0010	-0.0001		
	306	0.0002	0.0089	0.0628	0.0000	0.0021	-0.0002		
	307	0.0002	0.0053	0.0109	0.0000	0.0004	-0.0001		
	308	0.0002	0.0075	0.0434	0.0000	0.0014	-0.0001		
	309	-0.0128	-0.0015	-0.0005	0.0000	0.0000	0.0000		
	310	-0.0128	-0.0016	-0.0004	0.0000	0.0000	0.0000		
	311	0.0130	0.0049	-0.0009	0.0000	0.0000	0.0000		
	312	0.0130	0.0048	-0.0008	0.0000	0.0000	0.0000		
	313	-0.0038	0.0009	-0.0007	0.0000	0.0000	0.0000		
	314	0.0039	0.0028	-0.0008	0.0000	0.0000	0.0000		
	315	-0.0038	0.0005	-0.0005	0.0000	0.0000	0.0000		
	316	0.0039	0.0024	-0.0006	0.0000	0.0000	0.0000		
87	301	-0.0001	0.0413	-0.0014	0.0000	0.0000	0.0003		
	302	0.0006	-0.0096	0.0024	0.0000	0.0000	-0.0001		
	303	0.0010	-0.0134	-0.0156	-0.0001	0.0002	-0.0001		
	304	0.0005	0.0052	-0.0680	-0.0005	0.0009	0.0000		
	305	0.0012	-0.0107	-0.0451	-0.0003	0.0006	-0.0001		
	306	0.0003	0.0157	-0.0993	-0.0007	0.0012	0.0001		
	307	-0.0007	-0.0106	0.1048	0.0007	-0.0013	0.0000		
	308	-0.0008	0.0020	0.0541	0.0004	-0.0007	0.0001		
	309	-0.0166	-0.0095	0.0025	0.0000	0.0000	-0.0001		
	310	-0.0166	-0.0092	0.0025	0.0000	0.0000	-0.0001		
	311	0.0175	-0.0062	0.0013	0.0000	0.0000	-0.0001		
	312	0.0175	-0.0060	0.0014	0.0000	0.0000	-0.0001		
	313	-0.0046	-0.0076	-0.0017	0.0000	0.0000	-0.0001		
	314	0.0055	-0.0078	0.0022	0.0000	0.0000	-0.0001		
	315	-0.0046	-0.0075	-0.0017	0.0000	0.0000	-0.0001		
	316	0.0055	-0.0068	0.0018	0.0000	0.0000	-0.0001		
88	301	-0.0002	0.0410	0.0013	0.0000	0.0000	0.0000		
	302	0.0005	-0.0095	-0.0024	0.0000	0.0000	0.0000		
	303	-0.0007	-0.0104	-0.1048	-0.0007	0.0013	0.0000		
	304	-0.0007	0.0019	-0.0540	-0.0004	-0.0006	0.0001		
	305	0.0011	-0.0106	0.0451	0.0003	-0.0006	-0.0001		
	306	0.0003	0.0155	0.0993	0.0007	-0.0012	0.0001		
	307	0.0009	-0.0133	0.0156	0.0001	-0.0002	-0.0001		
	308	0.0005	0.0052	0.0679	0.0005	-0.0009	0.0000		
	309	-0.0167	-0.0092	-0.0025	0.0000	0.0000	-0.0001		
	310	-0.0167	-0.0094	-0.0025	0.0000	0.0000	-0.0001		
	311	0.0175	-0.0058	-0.0014	0.0000	0.0000	0.0000		
	312	0.0175	-0.0060	-0.0013	0.0000	0.0000	0.0000		
	313	-0.0047	-0.0077	-0.0022	0.0000	0.0000	-0.0001		
	314	0.0055	-0.0067	-0.0018	0.0000	0.0000	-0.0001		
	315	-0.0047	-0.0085	-0.0020	0.0000	0.0000	-0.0001		
	316	0.0003	0.0413	-0.0014	0.0000	0.0000	-0.0003		
89	301	0.0003	0.0413	-0.0014	0.0000	0.0000	-0.0003		
	302	-0.0004	-0.0097	0.0024	0.0000	0.0000	0.0001		
	303	-0.0008	-0.0134	-0.0156	-0.0001	-0.0002	0.0001		
	304	-0.0005	0.0052	-0.0680	-0.0005	-0.0009	0.0000		
	305	-0.0010	-0.0107	-0.0451	-0.0003	-0.0006	0.0001		
	306	-0.0004	0.0157	-0.0993	-0.0007	-0.0012	-0.0001		
	307	0.0007	-0.0106	0.1048	0.0007	-0.0013	0.0000		
	308	0.0007	0.0019	0.0540	0.0004	0.0007	-0.0001		
	309	-0.0174	-0.0062	0.0013	0.0000	0.0000	0.0001		
	310	-0.0174	-0.0060	0.0014	0.0000	0.0000	0.0001		
	311	0.0167	-0.0095	0.0025	0.0000	0.0000	0.0001		
	312	0.0167	-0.0092	0.0025	0.0000	0.0000	0.0001		
	313	-0.0054	-0.0076	0.0017	0.0000	0.0000	0.0001		
	314	0.0048	-0.0086	0.0020	0.0000	0.0000	0.0001		
	315	-0.0054	-0.0069	0.0018	0.0000	0.0000	0.0001		
	316	0.0047	-0.0078	0.0022	0.0000	0.0000	0.0001		
90	301	0.0002	0.0410	0.0013	0.0000	0.0000	-0.0003		
	302	-0.0005	-0.0095	-0.0024	0.0000	0.0000	0.0001		
	303	0.0007	-0.0104	-0.1048	-0.0007	-0.0013	0.0000		
	304	0.0007	0.0019	-0.0540	-0.0004	-0.0007	-0.0001		
	305	-0.0011	-0.0106	0.0451	0.0003	0.0006	0.0001		
	306	-0.0003	0.0155	0.0993	0.0007	0.0012	-0.0001		
	307	-0.0009	-0.0133	0.0156	0.0001	0.0002	0.0001		
	308	-0.0005	0.0051	0.0680	0.0005	0.0009	0.0000		
	309	-0.0175	-0.0058	-0.0014	0.0000	0.0000	0.0000		
	310	-0.0175	-0.0060	-0.0013	0.0000	0.0000	0.0000		
	311	0.0167	-0.0092	-0.0025	0.0000	0.0000	0.0001		
	312	0.0167	-0.0094	-0.0025	0.0000	0.0000	0.0001		
	313	-0.0055	-0.0067	-0.0018	0.0000	0.0000	0.0001		
	314	0.0047	-0.0077	-0.0022	0.0000	0.0000	0.0001		
	315	-0.0055	-0.0075	-0.0017	0.0000	0.0000	0.0001		
	316	0.0047	-0.0085	-0.0020	0.0000	0.0000	0.0001		

JOINT DISPLACEMENT (CM RADIANS)							STRUCTURE TYPE = SPACE						
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JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
91	301	-0.0002	-0.3556	-0.0181	0.0001	0.0000	0.0002						
	302	-0.0001	-0.2160	-0.0118	0.0000	0.0000	0.0001						
	303	-0.0002	-0.3527	-0.0194	0.0001	0.0000	0.0002						
	304	0.0000	-0.0408	-0.0024	0.0000	0.0000	0.0000						
	305	-0.0002	-0.3471	-0.0191	0.0001	0.0000	0.0002						
	306	0.0001	0.1347	0.0072	0.0000	0.0000	-0.0001						
	307	0.0000	0.0025	0.0004	0.0000	0.0000	0.0000						
	308	0.0001	0.1685	0.0097	0.0000	0.0000	-0.0001						
	309	0.0212	-0.1727	-0.0094	0.0000	0.0000	0.0001						
	310	0.0212	-0.1731	-0.0102	0.0000	0.0000	0.0001						
	311	-0.0214	-0.1726	-0.0087	0.0000	0.0000	0.0001						
	312	-0.0214	-0.1730	-0.0095	0.0000	0.0000	0.0001						
	313	0.0062	-0.1722	-0.0082	0.0000	0.0000	0.0001						
	314	-0.0065	-0.1722	-0.0080	0.0000	0.0000	0.0001						
	315	0.0063	-0.1735	-0.0109	0.0000	0.0000	0.0001						
	316	-0.0064	-0.1734	-0.0107	0.0000	0.0000	0.0001						
	317	0.0001	-0.4674	0.0008	0.0000	0.0000	0.0000						
93	301	0.0001	-0.2871	0.0005	0.0000	0.0000	0.0000						
	302	0.0001	-0.2335	-0.0005	0.0000	0.0000	0.0000						
	303	0.0000	0.0965	-0.0009	0.0000	0.0000	0.0000						
	304	0.0000	-0.4649	0.0008	0.0000	0.0000	0.0000						
	305	0.0001	-0.1956	-0.0003	0.0000	0.0000	0.0000						
	306	0.0000	-0.2361	0.0014	0.0000	0.0000	0.0000						
	307	0.0000	0.1203	0.0005	0.0000	0.0000	0.0001						
	308	0.0013	-0.2296	0.0010	0.0000	0.0000	0.0000						
	309	0.0283	-0.2297	-0.0001	0.0000	0.0000	0.0000						
	310	0.0283	-0.2297	-0.0001	0.0000	0.0000	0.0000						
	311	-0.0282	-0.2296	0.0009	0.0000	0.0000	0.0000						
	312	-0.0282	-0.2297	-0.0001	0.0000	0.0000	0.0000						
	313	0.0085	-0.2295	0.0022	0.0000	0.0000	0.0000						
	314	-0.0084	-0.2295	0.0022	0.0000	0.0000	0.0000						
	315	0.0085	-0.2298	-0.0013	0.0000	0.0000	0.0000						
	316	-0.0084	-0.2298	-0.0013	0.0000	0.0000	0.0000						
95	301	0.0000	-0.3496	0.0187	0.0000	0.0000	0.0000						
	302	0.0000	-0.2130	0.0122	0.0000	0.0000	0.0000						
	303	0.0000	0.0014	-0.0003	0.0000	0.0000	0.0000						
	304	0.0000	0.1689	-0.0098	0.0000	0.0000	0.0000						
	305	0.0000	-0.3422	0.0197	0.0000	0.0000	0.0000						
	306	0.0000	0.1327	-0.0075	0.0000	0.0000	0.0000						
	307	0.0000	-0.3465	0.0199	0.0000	0.0000	0.0000						
	308	0.0000	-0.0422	0.0024	0.0000	0.0000	0.0000						
	309	0.0191	-0.1706	0.0102	0.0000	0.0000	0.0000						
	310	0.0191	-0.1701	0.0094	0.0000	0.0000	0.0000						
	311	-0.0191	-0.1706	0.0102	0.0000	0.0000	0.0000						
	312	-0.0191	-0.1701	0.0094	0.0000	0.0000	0.0000						
	313	0.0057	-0.1712	0.0111	0.0000	0.0000	0.0000						
	314	-0.0057	-0.1712	0.0111	0.0000	0.0000	0.0000						
	315	0.0057	-0.1695	0.0084	0.0000	0.0000	0.0000						
	316	-0.0057	-0.1695	0.0084	0.0000	0.0000	0.0000						

JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE
-----JOINT DISPLACEMENT (CM) RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
105	301	0.0000	0.0000	0.0000	0.0000	-0.0005	-0.0008
	302	0.0000	0.0000	0.0000	0.0001	-0.0002	-0.0005
	303	0.0000	0.0000	0.0000	0.0000	0.0001	0.0000
	304	0.0000	0.0000	0.0000	-0.0001	0.0002	0.0000
	305	0.0000	0.0000	0.0000	0.0001	-0.0003	-0.0008
	306	0.0000	0.0000	0.0000	0.0000	0.0000	0.0004
	307	0.0000	0.0000	0.0000	0.0001	-0.0003	-0.0008
	308	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0001
	309	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0007
	310	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0007
	311	0.0000	0.0000	0.0000	0.0000	-0.0002	-0.0001
	312	0.0000	0.0000	0.0000	0.0000	-0.0002	-0.0001
	313	0.0000	0.0000	0.0000	0.0001	-0.0001	-0.0001
	314	0.0000	0.0000	0.0000	0.0001	-0.0002	-0.0003
	315	0.0000	0.0000	0.0000	0.0000	-0.0001	-0.0005
	316	0.0000	0.0000	0.0000	0.0000	-0.0002	-0.0003
106	301	0.0000	0.0000	0.0000	0.0000	-0.0002	-0.0002
	302	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0001
	303	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0002
	304	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	305	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0001
	306	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	307	0.0000	0.0000	0.0000	0.0001	0.0000	0.0000
	308	0.0000	0.0000	0.0000	0.0001	0.0000	0.0001
	309	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	310	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	311	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	312	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	313	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	314	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	315	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	316	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
107	301	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002
	302	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	303	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	304	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	305	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	306	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	307	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	308	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	309	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	310	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	311	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	312	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	313	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	314	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	315	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	316	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
131	301	-0.0092	-0.0014	0.0001	0.0000	0.0002	0.0001
	302	0.0088	0.0006	-0.0001	0.0000	-0.0002	0.0001
	303	0.0083	0.0008	0.0008	-0.0002	-0.0002	0.0000
	304	-0.0025	-0.0001	0.0005	0.0001	-0.0001	-0.0001
	305	0.0103	0.0006	-0.0005	0.0001	-0.0002	0.0002
	306	-0.0131	-0.0011	-0.0006	0.0001	0.0003	-0.0001
	307	0.0125	0.0009	-0.0003	0.0001	-0.0003	0.0002
	308	-0.0039	-0.0004	-0.0006	0.0001	0.0001	0.0000
	309	-0.0002	0.0006	-0.0001	0.0000	0.0000	0.0000
	310	-0.0002	0.0003	-0.0001	0.0000	0.0000	0.0000
	311	0.0143	0.0006	-0.0001	0.0000	-0.0003	0.0002
	312	0.0142	0.0003	-0.0001	0.0000	-0.0003	0.0002
	313	0.0049	0.0009	0.0000	0.0000	-0.0001	0.0001
	314	0.0092	0.0000	0.0000	0.0000	-0.0002	0.0002
	315	0.0048	0.0000	-0.0001	0.0000	-0.0001	0.0001
	316	0.0091	0.0000	-0.0001	0.0000	-0.0002	0.0002
132	301	0.0096	-0.0014	-0.0001	0.0000	0.0002	-0.0001
	302	-0.0088	0.0006	0.0001	0.0000	-0.0002	0.0001
	303	-0.0126	0.0008	0.0003	-0.0001	-0.0003	-0.0002
	304	0.0039	-0.0004	0.0005	-0.0001	0.0001	0.0000
	305	-0.0104	0.0006	-0.0005	-0.0001	-0.0002	0.0002
	306	0.0133	-0.0011	0.0007	-0.0001	0.0003	0.0001
	307	-0.0085	-0.0007	-0.0008	0.0002	-0.0002	0.0000
	308	0.0025	-0.0001	-0.0005	0.0001	0.0001	-0.0001
	309	-0.0144	0.0003	0.0001	0.0000	-0.0003	0.0002
	310	-0.0144	0.0006	0.0000	0.0000	-0.0003	0.0002
	311	0.0002	0.0003	0.0001	0.0000	0.0000	0.0000
	312	0.0002	0.0006	0.0000	0.0000	-0.0002	0.0000
	313	-0.0092	0.0000	0.0001	0.0000	-0.0002	0.0002
	314	-0.0048	0.0000	0.0001	0.0000	-0.0001	0.0000
	315	-0.0093	0.0009	0.0000	0.0000	-0.0002	0.0001
	316	-0.0050	0.0009	-0.0001	0.0000	0.0000	0.0000
133	301	-0.0153	-0.0024	-0.0001	0.0000	-0.0004	-0.0001
	302	-0.0022	0.0006	0.0001	0.0000	0.0000	0.0000
	303	-0.0036	0.0009	0.0004	-0.0001	-0.0001	0.0001
	304	-0.0008	-0.0005	0.0007	-0.0001	0.0000	0.0002
	305	-0.0038	0.0004	0.0007	-0.0001	0.0000	0.0000
	306	0.0008	-0.0014	0.0008	-0.0002	0.0000	0.0000
	307	0.0006	0.0009	-0.0010	0.0002	0.0000	0.0000
	308	0.0020	0.0000	-0.0006	0.0001	0.0000	0.0000
	309	-0.0132	0.0005	0.0001	0.0000	-0.0003	-0.0002
	310	-0.0132	0.0008	0.0001	0.0000	-0.0003	-0.0002
	311	0.0097	0.0001	0.0002	0.0000	0.0002	0.0001
	312	0.0097	0.0004	0.0001	0.0000	0.0002	0.0001
	313	-0.0051	0.0000	0.0002	0.0000	-0.0001	0.0000
	314	0.0017	-0.0001	0.0002	0.0000	0.0000	0.0000
	315	-0.0051	0.0011	0.0000	0.0000	-0.0001	-0.0001
	316	0.0017	0.0010	0.0000	0.0000	0.0000	0.0000

***** END OF LATEST ANALYSIS RESULT *****

593. LOAD LIST 201 TO 236

594. PARAMETER 1

595. CODE BSS950

596. LZ 9.3 MEMB 1 TO 3 10 TO 15 22 TO 27 34 TO 39 46 TO 48 95 TO 102 562 564 566 -

597. 567 572 TO 575

598. LZ 6.65 MEMB 6 7 18 19 30 31 42 43

599. LZ 5.6 MEMB 4 5 8 9 16 17 20 21 28 29 32 33 40 41 44 45 52 TO 59

600. LZ 6 MEMB 208 218 228 236 238 240

601. UNL 1.5 MEMB 501 TO 218 220 TO 236 238 240 241 301 TO 320

602. KY 0.65 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

603. KZ 0.65 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

604. MAIN 3 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

605. PY 345000 ALL

606. SBLT 1 ALL

607. SBLT 0 MEMB 501 TO 516 522 TO 525 530 TO 533 538 TO 541

608. TRACK 2 ALL

609. CHECK CODE ALL

STEEL DESIGN

STAAD.Pro CODE CHECKING - (BSI)

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
1	TAPERED	PASS	BS-4.7 (C)	0.262	214
		93.79 C	3.69	-37.46	2.70

CALCULATED CAPACITIES FOR MEMB 1 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1421.2 PT= 2063.1 MB= 210.2					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 KN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

2	TAPERED	PASS	BS-4.8.2.2	0.154	211
		80.47 T	0.25	36.49	1.02

CALCULATED CAPACITIES FOR MEMB 2 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 KN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

3	TAPERED	PASS	BS-4.8.2.2	0.153	211
		79.54 T	-0.24	-36.49	0.00

CALCULATED CAPACITIES FOR MEMB 3 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 KN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

4	TAPERED	PASS	BS-4.7 (C)	0.073	201
		19.01 C	0.74	16.15	0.00

CALCULATED CAPACITIES FOR MEMB 4 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1902.8 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 KN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

5	TAPERED	PASS	BS-4.7 (C)	0.095	201
		16.42 C	0.26	26.30	2.18

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 5 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1642.4	PT= 2166.6	MB= 315.5
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

6	TAPERED	PASS	BS-4.7 (C)	0.196	207
32.01 C -0.10 56.14 3.19					

CALCULATED CAPACITIES FOR MEMB 6 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1301.0	PT= 2166.6	MB= 244.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

7	TAPERED	PASS	BS-4.7 (C)	0.197	207
32.05 C 0.10 -56.14 0.00					

CALCULATED CAPACITIES FOR MEMB 7 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1207.8	PT= 2166.6	MB= 228.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

8	TAPERED	PASS	BS-4.7 (C)	0.127	207
13.00 C -0.14 37.91 0.00					

CALCULATED CAPACITIES FOR MEMB 8 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1488.2	PT= 2166.6	MB= 280.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

9	TAPERED	PASS	BS-4.7 (C)	0.077	213
20.27 C -0.85 16.67 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 9 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1902.8	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

10	TAPERED	PASS	BS-4.8.2.2	0.153	211
79.43 T 0.24 -36.54 0.00					

CALCULATED CAPACITIES FOR MEMB 10 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1663.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

11	TAPERED	PASS	BS-4.8.2.2	0.154	211
80.34 T -0.24 36.54 1.02					

CALCULATED CAPACITIES FOR MEMB 11 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1663.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

12	TAPERED	PASS	BS-4.7 (C)	0.262	202
93.71 C -3.69 -37.46 2.70					

CALCULATED CAPACITIES FOR MEMB 12 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY= 56.6	PC= 1421.2	PT= 2063.1	MB= 210.2
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

13	TAPERED	PASS	BS-4.7 (C)	0.259	214
124.52 C 0.14 -48.35 2.70					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 13 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.7 PC= 1421.2 PT= 2063.1 MB= 202.9					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

14 TAPERED PASS BS-4.8.2.2 0.199 211					
107.05 T -0.07 48.26 1.02					

CALCULATED CAPACITIES FOR MEMB 14 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

15 TAPERED PASS BS-4.8.2.2 0.198 211					
105.62 T 0.06 -48.26 0.00					

CALCULATED CAPACITIES FOR MEMB 15 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

16 TAPERED PASS BS-4.7 (C) 0.078 201					
20.96 C -0.29 20.21 0.00					

CALCULATED CAPACITIES FOR MEMB 16 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1902.8 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

17 TAPERED PASS BS-4.7 (C) 0.110 201					
17.87 C -0.29 30.95 2.18					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 17 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1642.4 PT= 2166.6 MB= 315.5					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

18 TAPERED PASS BS-4.7 (C) 0.144 207					
23.92 C -0.08 41.01 0.00					

CALCULATED CAPACITIES FOR MEMB 18 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1301.0 PT= 2166.6 MB= 244.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

19 TAPERED PASS BS-4.7 (C) 0.143 207					
23.92 C 0.03 -40.65 0.00					

CALCULATED CAPACITIES FOR MEMB 19 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1207.8 PT= 2166.6 MB= 228.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

20 TAPERED PASS BS-4.7 (C) 0.122 207					
16.57 C -0.09 35.84 0.00					

CALCULATED CAPACITIES FOR MEMB 20 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1488.2 PT= 2166.6 MB= 280.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

21 TAPERED PASS BS-4.7 (C) 0.081 213					
20.97 C 0.34 20.75 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 21 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1902.8 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

22 TAPERED PASS BS-4.8.2.2 0.198 211					
105.49 T -0.06 -48.31 0.00					

CALCULATED CAPACITIES FOR MEMB 22 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

23 TAPERED PASS BS-4.8.2.2 0.199 211					
106.90 T 0.06 48.31 1.02					

CALCULATED CAPACITIES FOR MEMB 23 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

24 TAPERED PASS BS-4.7 (C) 0.259 202					
124.29 C -0.14 -48.34 2.70					

CALCULATED CAPACITIES FOR MEMB 24 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1421.2 PT= 2063.1 MB= 210.2					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

25 TAPERED PASS BS-4.7 (C) 0.259 214					
124.51 C -0.14 -48.35 2.70					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 25 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1421.2 PT= 2063.1 MB= 202.9					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

26 TAPERED PASS BS-4.8.2.2 0.199 211					
107.06 T 0.07 48.26 1.02					

CALCULATED CAPACITIES FOR MEMB 26 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

27 TAPERED PASS BS-4.8.2.2 0.198 211					
105.63 T -0.06 -48.26 0.00					

CALCULATED CAPACITIES FOR MEMB 27 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

28 TAPERED PASS BS-4.7 (C) 0.078 201					
20.43 C 0.28 20.25 0.00					

CALCULATED CAPACITIES FOR MEMB 28 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1902.8 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

29 TAPERED PASS BS-4.7 (C) 0.109 201					
17.33 C 0.26 30.94 2.18					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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30	TAPERED	PASS	BS-4.7 (C)	0.143	207
24.16 C 0.06 40.96 0.00					

CALCULATED CAPACITIES FOR MEMB 30 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1301.0 PT= 2166.6 MB= 244.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

31	TAPERED	PASS	BS-4.7 (C)	0.143	207
24.16 C -0.03 -40.66 0.00					

CALCULATED CAPACITIES FOR MEMB 31 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1207.8 PT= 2166.6 MB= 228.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

32	TAPERED	PASS	BS-4.7 (C)	0.122	207
16.73 C 0.09 35.81 0.00					

CALCULATED CAPACITIES FOR MEMB 32 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1488.2 PT= 2166.6 MB= 280.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

33	TAPERED	PASS	BS-4.7 (C)	0.081	213
21.00 C -0.34 20.74 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 33 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1902.8 PT= 2166.6 MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

34	TAPERED	PASS	BS-4.8.2.2	0.198	211
105.49 T 0.06 -48.31 0.00					

CALCULATED CAPACITIES FOR MEMB 34 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1663.1 PT= 2166.6 MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

35	TAPERED	PASS	BS-4.8.2.2	0.199	211
106.90 T -0.06 48.31 1.02					

CALCULATED CAPACITIES FOR MEMB 35 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1663.1 PT= 2166.6 MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

36	TAPERED	PASS	BS-4.7 (C)	0.259	202
124.32 C 0.14 -48.34 2.70					

CALCULATED CAPACITIES FOR MEMB 36 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY=	56.6	PC=	1421.2 PT= 2063.1 MB= 210.2
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d =	280 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

37	TAPERED	PASS	BS-4.7 (C)	0.262	214
93.79 C -3.69 -37.46 2.70					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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38	TAPERED	PASS	BS-4.8.2.2	0.154	211
		80.47 T	-0.25	36.49	1.02

CALCULATED CAPACITIES FOR MEMB 37 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.6	PC=	1421.2 PT= 2063.1 MB= 202.9
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d =	280 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

39	TAPERED	PASS	BS-4.8.2.2	0.153	211
		79.54 T	0.24	-36.49	0.00

CALCULATED CAPACITIES FOR MEMB 39 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1663.1 PT= 2166.6 MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

40	TAPERED	PASS	BS-4.7 (C)	0.073	201
		19.00 C	-0.75	16.15	0.00

CALCULATED CAPACITIES FOR MEMB 40 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1902.8 PT= 2166.6 MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 41 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1642.4 PT= 2166.6 MB= 315.5
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

42	TAPERED	PASS	BS-4.7 (C)	0.196	207
		32.02 C	0.10	56.14	3.19

CALCULATED CAPACITIES FOR MEMB 42 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1301.0 PT= 2166.6 MB= 244.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

43	TAPERED	PASS	BS-4.7 (C)	0.197	207
		32.05 C	-0.10	-56.14	0.00

CALCULATED CAPACITIES FOR MEMB 43 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1207.8 PT= 2166.6 MB= 228.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

44	TAPERED	PASS	BS-4.7 (C)	0.127	207
		13.00 C	0.14	37.91	0.00

CALCULATED CAPACITIES FOR MEMB 44 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY=	56.7	PC=	1488.2 PT= 2166.6 MB= 280.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d =	380 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm2
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

45	TAPERED	PASS	BS-4.7 (C)	0.077	213
		20.23 C	0.85	16.66	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 45 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1902.8	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

46	TAPERED	PASS	BS-4.8.2.2	0.153	211
79.43 T -0.24 -36.54 0.00					

CALCULATED CAPACITIES FOR MEMB 46 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1663.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

47	TAPERED	PASS	BS-4.8.2.2	0.154	211
80.34 T 0.24 36.54 1.02					

CALCULATED CAPACITIES FOR MEMB 47 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1663.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

48	TAPERED	PASS	BS-4.7 (C)	0.262	202
93.71 C 3.69 -37.46 2.70					

CALCULATED CAPACITIES FOR MEMB 48 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY= 56.6	PC= 1421.2	PT= 2063.1	MB= 210.2
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

52	TAPERED	PASS	BS-4.7 (C)	0.095	213
17.27 C -0.29 -26.11 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 52 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1763.3	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

53	TAPERED	PASS	BS-4.7 (C)	0.108	213
17.41 C 0.32 -30.33 0.00					

CALCULATED CAPACITIES FOR MEMB 53 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1763.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

54	TAPERED	PASS	BS-4.7 (C)	0.108	213
17.44 C -0.32 -30.31 0.00					

CALCULATED CAPACITIES FOR MEMB 54 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1763.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

55	TAPERED	PASS	BS-4.7 (C)	0.095	213
17.23 C 0.29 -26.10 0.00					

CALCULATED CAPACITIES FOR MEMB 55 UNIT - KN,m SECTION CLASS 3					
MCZ=	324.8	MCY= 56.7	PC= 1763.1	PT= 2166.6	MB= 324.8
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2			
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

56	TAPERED	PASS	BS-4.7 (C)	0.136	207
11.83 C 0.15 -41.04 2.52					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 56 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1532.6 PT= 2166.6 MB= 290.0					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

57 TAPERED PASS BS-4.7 (C) 0.132 207					
16.99 C 0.10 -38.96 2.52					

CALCULATED CAPACITIES FOR MEMB 57 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1532.6 PT= 2166.6 MB= 290.0					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

58 TAPERED PASS BS-4.7 (C) 0.131 207					
16.48 C -0.08 -39.02 2.52					

CALCULATED CAPACITIES FOR MEMB 58 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1532.6 PT= 2166.6 MB= 290.0					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

59 TAPERED PASS BS-4.7 (C) 0.135 207					
11.83 C -0.13 -41.04 2.52					

CALCULATED CAPACITIES FOR MEMB 59 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1532.6 PT= 2166.6 MB= 290.0					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

95 TAPERED PASS BS-4.7 (C) 0.302 214					
68.91 C 6.87 37.46 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 95 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1426.5 PT= 2063.1 MB= 243.9					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qw = 207 N/mm2					
d = 330 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

96 TAPERED PASS BS-4.7 (C) 0.241 215					
89.45 C -0.23 49.36 0.00					

CALCULATED CAPACITIES FOR MEMB 96 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1426.5 PT= 2063.1 MB= 243.9					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qw = 207 N/mm2					
d = 330 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

97 TAPERED PASS BS-4.7 (C) 0.241 215					
89.44 C 0.23 49.36 0.00					

CALCULATED CAPACITIES FOR MEMB 97 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1426.5 PT= 2063.1 MB= 243.9					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qw = 207 N/mm2					
d = 330 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

98 TAPERED PASS BS-4.7 (C) 0.302 214					
68.91 C -6.87 37.46 0.00					

CALCULATED CAPACITIES FOR MEMB 98 UNIT - KN,m SECTION CLASS 3					
MCZ= 275.6 MCY= 56.6 PC= 1426.5 PT= 2063.1 MB= 243.9					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qw = 207 N/mm2					
d = 330 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

99 TAPERED PASS BS-4.7 (C) 0.302 202					
68.82 C -6.87 37.46 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 99 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY= 56.6	PC= 1426.5	PT= 2063.1	MB= 243.9
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qv = 207 N/mm2					
d =	330 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

100	TAPERED	PASS	BS-4.7 (C)	0.241	203
		89.27 C	0.23	49.36	0.00

CALCULATED CAPACITIES FOR MEMB 100 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY= 56.6	PC= 1426.5	PT= 2063.1	MB= 243.9
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qv = 207 N/mm2					
d =	330 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

101	TAPERED	PASS	BS-4.7 (C)	0.241	203
		89.30 C	-0.23	49.36	0.00

CALCULATED CAPACITIES FOR MEMB 101 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY= 56.6	PC= 1426.5	PT= 2063.1	MB= 243.9
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qv = 207 N/mm2					
d =	330 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

102	TAPERED	PASS	BS-4.7 (C)	0.302	202
		68.82 C	6.87	37.46	0.00

CALCULATED CAPACITIES FOR MEMB 102 UNIT - KN,m SECTION CLASS 3					
MCZ=	275.6	MCY= 56.6	PC= 1426.5	PT= 2063.1	MB= 243.9
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 409 kN : qv = 207 N/mm2					
d =	330 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

201	TAPERED	PASS	BS-4.7 (C)	0.092	207
		0.55 C	-1.75	-7.99	1.50

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 201 UNIT - KN,m SECTION CLASS 2					
MCZ=	170.4	MCY= 38.8	PC= 1266.8	PT= 0.0	MB= 190.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qv = 207 N/mm2					
d =	280 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

202	TAPERED	PASS	BS-4.7 (C)	0.307	201
		43.82 C	-0.13	-28.13	3.79

CALCULATED CAPACITIES FOR MEMB 202 UNIT - KN,m SECTION CLASS 2					
MCZ=	170.4	MCY= 38.8	PC= 1266.8	PT= 1614.6	MB= 190.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qv = 207 N/mm2					
d =	280 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

203	TAPERED	PASS	BS-4.7 (C)	0.293	201
		37.62 C	0.05	-31.67	3.00

CALCULATED CAPACITIES FOR MEMB 203 UNIT - KN,m SECTION CLASS 2					
MCZ=	170.4	MCY= 38.8	PC= 1266.8	PT= 1614.6	MB= 190.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qv = 207 N/mm2					
d =	280 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

204	TAPERED	PASS	BS-4.7 (C)	0.307	201
		43.96 C	-0.13	-28.12	2.71

CALCULATED CAPACITIES FOR MEMB 204 UNIT - KN,m SECTION CLASS 2					
MCZ=	170.4	MCY= 38.8	PC= 1266.8	PT= 1614.6	MB= 190.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qv = 207 N/mm2					
d =	280 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

205	TAPERED	PASS	BS-4.7 (C)	0.092	207
		0.55 C	1.75	7.99	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 205 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1267.4 PT= 0.0 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

206 TAPERED PASS BS-4.7 (C) 0.083 207					
0.17 C 0.55 -11.69 1.80					

CALCULATED CAPACITIES FOR MEMB 206 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1174.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

207 TAPERED PASS BS-4.3.6 0.206 218					
0.28 T 0.01 8.91 3.25					

CALCULATED CAPACITIES FOR MEMB 207 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 5224.1 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

208 TAPERED PASS BS-4.3.6 0.191 218					
0.13 T -0.09 7.74 2.80					

CALCULATED CAPACITIES FOR MEMB 208 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 839.5 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

209 TAPERED PASS BS-4.3.6 0.206 218					
0.28 T 0.01 8.91 3.25					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 209 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 3734.2 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

210 TAPERED PASS BS-4.7 (C) 0.083 207					
0.18 C -0.56 11.69 0.00					

CALCULATED CAPACITIES FOR MEMB 210 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1174.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

211 TAPERED PASS BS-4.3.6 0.215 201					
0.01 T 0.02 35.20 3.00					

CALCULATED CAPACITIES FOR MEMB 211 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

212 TAPERED PASS BS-4.3.6 0.254 218					
1.66 T -0.01 11.36 0.00					

CALCULATED CAPACITIES FOR MEMB 212 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

213 TAPERED PASS BS-4.3.6 0.228 218					
0.54 C 0.00 10.55 3.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 213 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 282.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

214 TAPERED PASS BS-4.3.6 0.253 218					
2.17 T 0.00 11.38 6.50					

CALCULATED CAPACITIES FOR MEMB 214 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

215 TAPERED PASS BS-4.3.6 0.215 201					
0.01 T -0.03 35.20 0.00					

CALCULATED CAPACITIES FOR MEMB 215 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

216 TAPERED PASS BS-4.3.6 0.275 207					
3.38 T 0.00 44.73 3.20					

CALCULATED CAPACITIES FOR MEMB 216 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 719.1 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

217 TAPERED PASS BS-4.3.6 0.303 218					
1.37 T 0.00 23.80 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 217 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

218 TAPERED PASS BS-4.3.6 0.240 212					
0.44 C 0.00 8.55 3.00					

CALCULATED CAPACITIES FOR MEMB 218 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

220 TAPERED PASS BS-4.3.6 0.275 207					
3.39 T 0.00 44.73 0.00					

CALCULATED CAPACITIES FOR MEMB 220 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 719.1 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

221 TAPERED PASS BS-4.3.6 0.215 207					
0.01 T -0.03 35.22 3.00					

CALCULATED CAPACITIES FOR MEMB 221 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

222 TAPERED PASS BS-4.3.6 0.253 206					
2.35 T 0.01 11.33 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 222 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

223 TAPERED PASS BS-4.3.6 0.228 206					
0.15 C 0.00 10.55 3.00					

CALCULATED CAPACITIES FOR MEMB 223 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 282.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

224 TAPERED PASS BS-4.3.6 0.253 206					
2.35 T -0.01 11.33 6.50					

CALCULATED CAPACITIES FOR MEMB 224 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

225 TAPERED PASS BS-4.3.6 0.215 207					
0.01 T 0.03 35.22 0.00					

CALCULATED CAPACITIES FOR MEMB 225 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

226 TAPERED PASS BS-4.7 (C) 0.085 207					
0.23 C -0.62 -11.68 1.80					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 226 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1174.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

227 TAPERED PASS BS-4.3.6 0.207 206					
0.34 T -0.02 8.90 3.25					

CALCULATED CAPACITIES FOR MEMB 227 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 5224.1 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

228 TAPERED PASS BS-4.3.6 0.191 206					
0.17 T 0.10 7.73 3.00					

CALCULATED CAPACITIES FOR MEMB 228 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

229 TAPERED PASS BS-4.3.6 0.206 206					
0.34 T -0.02 8.90 3.25					

CALCULATED CAPACITIES FOR MEMB 229 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 3457.6 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

230 TAPERED PASS BS-4.7 (C) 0.085 207					
0.23 C 0.62 11.68 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 230 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1174.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

231 TAPERED PASS BS-4.7 (C) 0.090 207					
0.60 C 1.64 -7.97 1.50					

CALCULATED CAPACITIES FOR MEMB 231 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1267.4 PT= 0.0 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

232 TAPERED PASS BS-4.7 (C) 0.308 213					
44.28 C 0.12 -28.12 3.79					

CALCULATED CAPACITIES FOR MEMB 232 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

233 TAPERED PASS BS-4.7 (C) 0.290 213					
36.44 C -0.04 -31.67 3.00					

CALCULATED CAPACITIES FOR MEMB 233 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 282.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

234 TAPERED PASS BS-4.7 (C) 0.308 213					
44.27 C 0.12 -28.12 2.71					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 234 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 248.4 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

235 TAPERED PASS BS-4.7 (C) 0.090 207					
0.60 C -1.64 7.97 0.00					

CALCULATED CAPACITIES FOR MEMB 235 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 1267.4 PT= 0.0 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

236 TAPERED PASS BS-4.3.6 0.193 218					
0.15 T 0.08 7.86 0.27					

CALCULATED CAPACITIES FOR MEMB 236 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 719.1 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

238 TAPERED PASS BS-4.3.6 0.240 212					
0.44 C 0.00 8.55 0.00					

CALCULATED CAPACITIES FOR MEMB 238 UNIT - KN,m SECTION CLASS 2					
MCZ= 170.4 MCY= 38.8 PC= 777.3 PT= 1614.6 MB= 190.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d = 280 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

240 TAPERED PASS BS-4.3.6 0.191 206					
0.17 T -0.10 7.73 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 240 UNIT - KN,m SECTION CLASS 2					
MCZ=	170.4	MCY= 38.8	PC= 777.3	PT= 1614.6	MB= 190.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d =	280 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

241	TAPERED	PASS	BS-4.3.6	0.295	212
0.42 C 0.00 9.80 6.50					

CALCULATED CAPACITIES FOR MEMB 241 UNIT - KN,m SECTION CLASS 2					
MCZ=	170.4	MCY= 38.8	PC= 248.4	PT= 1614.6	MB= 190.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 347 kN : qw = 207 N/mm2					
d =	280 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

301	TAPERED	PASS	BS-4.7 (C)	0.063	214
0.00 -3.87 6.28 1.30					

CALCULATED CAPACITIES FOR MEMB 301 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 2334.0	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

302	TAPERED	PASS	BS-4.8.2.2	0.237	211
2.46 T -10.34 -34.31 3.25					

CALCULATED CAPACITIES FOR MEMB 302 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 10382.0	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

303	TAPERED	PASS	BS-4.8.2.2	0.187	214
3.17 T -5.56 33.45 3.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 303 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 46181.7	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

304	TAPERED	PASS	BS-4.8.2.2	0.237	211
2.46 T -10.34 -34.31 3.25					

CALCULATED CAPACITIES FOR MEMB 304 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC=205426.3	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

305	TAPERED	PASS	BS-4.7 (C)	0.063	214
0.00 3.87 -6.28 0.00					

CALCULATED CAPACITIES FOR MEMB 305 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 2334.1	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

306	TAPERED	PASS	BS-4.7 (C)	0.063	202
0.00 -3.87 -6.28 1.30					

CALCULATED CAPACITIES FOR MEMB 306 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 2334.1	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

307	TAPERED	PASS	BS-4.8.2.2	0.237	211
2.45 T -10.34 34.31 3.25					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 307 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 10382.5	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

308	TAPERED	PASS	BS-4.8.2.2	0.187	202
3.16 T -5.56 -33.45 3.00					

CALCULATED CAPACITIES FOR MEMB 308 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 46183.5	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

309	TAPERED	PASS	BS-4.8.2.2	0.237	211
2.45 T -10.34 34.31 3.25					

CALCULATED CAPACITIES FOR MEMB 309 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC=205434.5	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

310	TAPERED	PASS	BS-4.7 (C)	0.063	202
0.00 3.87 6.28 0.00					

CALCULATED CAPACITIES FOR MEMB 310 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 2334.1	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

311	TAPERED	PASS	BS-4.7 (C)	0.048	214
0.00 -3.28 3.93 1.30					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 311 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 2334.1	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

312	TAPERED	PASS	BS-4.7 (C)	0.168	214
3.35 C -7.86 22.30 3.25					

CALCULATED CAPACITIES FOR MEMB 312 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 1045.9	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

313	TAPERED	PASS	BS-4.7 (C)	0.135	214
3.53 C -5.18 20.92 3.00					

CALCULATED CAPACITIES FOR MEMB 313 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 1154.2	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

314	TAPERED	PASS	BS-4.7 (C)	0.168	214
3.35 C -7.86 22.31 3.25					

CALCULATED CAPACITIES FOR MEMB 314 UNIT - KN,m SECTION CLASS 3					
MCZ=	257.7	MCY= 100.2	PC= 1045.9	PT= 2537.8	MB= 257.7
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2					
d =	226 mm	t = 6 mm	a = 0 mm	pyf = 344 N/mm2	
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

315	TAPERED	PASS	BS-4.7 (C)	0.048	214
0.00 3.28 -3.93 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

CALCULATED CAPACITIES FOR MEMB 315 UNIT - kN,m SECTION CLASS 3									
MCZ=	257.7	MCY= 100.2	PC= 2334.1	PT= 2537.8	MB= 257.7				
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00									
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2									
d = 226 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2		= 207 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS									

316	TAPERED	PASS	BS-4.7 (C)	0.048	202				
0.00 -3.28 -3.93 1.30									

CALCULATED CAPACITIES FOR MEMB 316 UNIT - kN,m SECTION CLASS 3									
MCZ=	257.7	MCY= 100.2	PC= 2334.1	PT= 2537.8	MB= 257.7				
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00									
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2									
d = 226 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2		= 207 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS									

317	TAPERED	PASS	BS-4.7 (C)	0.168	202				
3.33 C -7.86 -22.31 3.25									

CALCULATED CAPACITIES FOR MEMB 317 UNIT - kN,m SECTION CLASS 3									
MCZ=	257.7	MCY= 100.2	PC= 1045.9	PT= 2537.8	MB= 257.7				
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00									
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2									
d = 226 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2		= 207 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS									

318	TAPERED	PASS	BS-4.7 (C)	0.135	202				
3.52 C -5.19 -20.92 3.00									

CALCULATED CAPACITIES FOR MEMB 318 UNIT - kN,m SECTION CLASS 3									
MCZ=	257.7	MCY= 100.2	PC= 1154.2	PT= 2537.8	MB= 257.7				
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00									
Shear Buckling check is required : Vb = 280 kN : qw = 207 N/mm2									
d = 226 mm :	t = 6 mm	: a = 0 mm : pyf = 344 N/mm2		= 207 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS									

319	TAPERED	PASS	BS-4.7 (C)	0.168	202				
3.33 C -7.86 -22.31 3.25									

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)									
MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION				
=====									
501 ST	UA70X70X6	PASS	BS-4.6 (T)	0.005	207				
=====									
MATERIAL DATA									
Grade of steel = S 275									
Modulus of elasticity = 205 kN/mm2									
Design Strength (py) = 344 N/mm2									
=====									
SECTION PROPERTIES (units - cm)									
Member Length = 349.70									
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20									
=====									
Moment of inertia : 15.391 y-y axis									
Plastic modulus : 60.246									
Elastic modulus : 21.909									
Effective modulus : 12.171									
Shear Area : 2.520									
=====									
DESIGN DATA (units - kN,m)									
Section Class : SLENDER									
Tension Capacity : 282.9 y-y axis									
Shear Capacity : 52.2									
=====									
BUCKLING CALCULATIONS (units - kN,m)									
(axis nomenclature as per design code)									
LTB Moment Capacity (kNm) and LTB Length (m): 2.69, 3.497									
LTB Coefficients & Associated Moments (kNm):									
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00									
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00									
=====									
CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):									
=====									
CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY		
BS-4.6 (T)	0.005	207	1.3	-	-	-	-		
Torsion and deflections have not been considered in the design.									
=====									

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)									
MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION				
=====									
502 ST	UA70X70X6	PASS	BS-4.6 (T)	0.001	207				
=====									
MATERIAL DATA									
Grade of steel = S 275									
Modulus of elasticity = 205 kN/mm2									
Design Strength (py) = 344 N/mm2									
=====									
SECTION PROPERTIES (units - cm)									
Member Length = 363.58									
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20									
=====									
Moment of inertia : 15.391 z-z axis									
Plastic modulus : 60.246 y-y axis									
Elastic modulus : 21.909									
Effective modulus : 12.171									
Shear Area : 2.520									
=====									
DESIGN DATA (units - kN,m)									
Section Class : BS5950-1/2000									
Squash Load : 282.90									
Axial force/Squash load : 0.001									
=====									
Compression Capacity : 46.3 z-z axis									
Tension Capacity : 282.9 y-y axis									
Shear Capacity : 52.2									
=====									
BUCKLING CALCULATIONS (units - kN,m)									
(axis nomenclature as per design code)									
Slenderness : 172.501 a-a axis									
Radius of gyration (cm) : 1.370 b-b axis									
Effective Length : 2.363									
=====									
LTB Moment Capacity (kNm) and LTB Length (m): 2.63, 3.636									
LTB Coefficients & Associated Moments (kNm):									
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00									
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00									
=====									
CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):									
=====									
CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY		
BS-4.6 (T)	0.001	207	0.4	-	-	-	-		
BS-4.7 (C)	0.001	218	0.0	-	-	-	-		
=====									

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====	=====	=====	=====	=====	=====

503 ST UA70X70X6	PASS	BS-4.7	(C)	0.001	220
	0.02 C	0.00		0.00	0.00
=====	=====	=====	=====	=====	=====

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 437.92
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.000

	Z-Z axis	Y-Y axis
Compression Capacity	: 33.1	71.9
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 207.770	133.911	133.911
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.846	2.846	2.846

LTB Moment Capacity (kNm) and LTB Length (m): 2.37, 4.379
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE RATIO LOAD FX VY VZ MZ MY
Torsion and deflections have not been considered in the design.

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

504 ST UA70X70X6	PASS	BS-4.7 (C)	0.080	207	0.00
	2.49 C	0.00	0.00		
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 451.85
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.009

	z-z axis	y-y axis
Compression Capacity	: 31.2	68.2
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 214.382	138.172	138.172
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.937	2.937	2.937

LTB Moment Capacity (kNm) and LTB Length (m): 2.33, 4.519
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.001	212	0.4	-	-	-	-
BS-4.7 (C)	0.080	207	2.5	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

505 ST UA70X70X6	PASS	BS-4.7 (C)	0.083	207
	2.40 C	0.00	0.00	4.71

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 471.31
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.008

	z-z axis	y-y axis
Compression Capacity	: 28.9	63.5
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 223.614	144.122	144.122
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 3.064	3.064	3.064

LTB Moment Capacity (kNm) and LTB Length (m): 2.27, 4.713
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.001	212	0.4	-	-	-	-
BS-4.7 (C)	0.083	207	2.4	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====	=====	=====	=====	=====	=====

506 ST UA70X70X6 PASS BS-4.7 (C) 0.001 207
0.02 C 0.00 0.00 4.58

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 457.97
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.000

	Z-Z axis	Y-Y axis
Compression Capacity	: 30.4	66.7
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 217.283	140.042	140.042
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.977	2.977	2.977

LTB Moment Capacity (kNm) and LTB Length (m): 2.31, 4.580
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6	(T)	0.001	207	0.4	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====	=====	=====	=====	=====	=====

507 ST UA70X70X6 PASS BS-4.6 (T) 0.001 207
0.42 T 0.00 0.00 0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 329.16
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.001

	Z-Z axis	Y-Y axis
Compression Capacity	: 55.2	112.8
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 156.172	100.655	100.655
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.140	2.140	2.140

LTB Moment Capacity (kNm) and LTB Length (m): 2.77, 3.292
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6	(T)	0.001	207	0.4	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
508 ST UA70X70X6	PASS	1.26 T	BS-4.6 (T) 0.00	0.004 0.00	207 0.00

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 344 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 313.76
 Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

Moment of inertia : 15.391 y-y axis 60.246
 Plastic modulus : 10.133 21.909
 Elastic modulus : 5.629 12.171
 Effective modulus : 5.629 12.171
 Shear Area : 2.520

DESIGN DATA (units - kN,m) BS5950-1/2000
 Section Class : SLENDER

Tension Capacity : 282.9 y-y axis 282.9
 Shear Capacity : 52.2 52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB Moment Capacity (kNm) and LTB Length (m): 2.83, 3.138
 LTB Coefficients & Associated Moments (kNm):
 mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
 Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe RATIO LOAD FX VY VZ MZ MY
 BS-4.6 (T) 0.004 207 1.3 - - -
 Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
509 ST UA70X70X6	PASS	0.36 T	BS-4.6 (T) 0.00	0.001 0.00	207 3.64

MATERIAL DATA

Grade of steel = S 275
 Modulus of elasticity = 205 kN/mm²
 Design Strength (py) = 344 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 363.58
 Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

Moment of inertia : 15.391 z-z axis 60.246 y-y axis
 Plastic modulus : 10.133 21.909
 Elastic modulus : 5.629 12.171
 Effective modulus : 5.629 12.171
 Shear Area : 2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
 Section Class : SLENDER
 Squash Load : 282.9
 Axial Force/Squash load : 0.001

Compression Capacity : 46.3 z-z axis 97.2 y-y axis
 Tension Capacity : 282.9 282.9
 Shear Capacity : 52.2 52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

Slenderness : 172.501 a-a axis 111.179 b-b axis 111.179
 Radius of gyration (cm) : 1.370 2.126 2.126
 Effective Length : 2.363 2.363 2.363

LTB Moment Capacity (kNm) and LTB Length (m): 2.63, 3.636
 LTB Coefficients & Associated Moments (kNm):
 mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
 Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe RATIO LOAD FX VY VZ MZ MY
 BS-4.6 (T) 0.001 207 0.4 - - -
 BS-4.7 (C) 0.001 218 0.1 - - -

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====	=====	=====	=====	=====	=====

510 ST UA70X70X6	PASS	BS-4.6	(T)	0.005	207
	1.30 T	0.00		0.00	3.50

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 349.70
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000
Section Class : SLENDER

	z-z axis	y-y axis
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)
(axis nomenclature as per design code)

LTB Moment Capacity (kNm) and LTB Length (m): 2.69, 3.497
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.005	207	1.3	-	-	-	-

Torsion and deflections have not been considered in the design.

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

511 ST UA70X70X6	PASS	BS-4.7 (C)	0.080	207	0.00
	2.49 C	0.00	0.00		
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 451.85
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.009

	Z-Z axis	Y-Y axis
Compression Capacity	: 31.2	68.2
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 214.382	138.172	138.172
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.937	2.937	2.937

LTB Moment Capacity (kNm) and LTB Length (m): 2.33, 4.519
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.001	212	0.4	-	-	-	-
BS-4.7 (C)	0.080	207	2.5	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====	=====	=====	=====	=====	=====

512 ST UA70X70X6 PASS BS-4.7 (C) 0.001 222
0.02 C 0.00 0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 437.92
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)
BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.000

	Z-Z axis	Y-Y axis
Compression Capacity	: 33.1	71.9
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)
(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 207.770	133.911	133.911
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.846	2.846	2.846

LTB Moment Capacity (kNm) and LTB Length (m): 2.37, 4.379
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe RATIO LOAD FX VY VZ MZ MY
Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====	=====	=====	=====	=====	=====

513 ST UA70X70X6 PASS BS-4.7 (C) 0.001 207
0.02 C 0.00 0.00 4.58

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 457.97
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)
BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.000

	Z-Z axis	Y-Y axis
Compression Capacity	: 30.4	66.7
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)
(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 217.283	140.042	140.042
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.977	2.977	2.977

LTB Moment Capacity (kNm) and LTB Length (m): 2.31, 4.580
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe RATIO LOAD FX VY VZ MZ MY
Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

514 ST UA70X70X6	PASS	BS-4.7 (C)	0.083	207
	2.39 C	0.00	0.00	4.71

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 471.31
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.008

	z-z axis	y-y axis
Compression Capacity	: 28.9	63.5
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 223.614	144.122	144.122
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 3.064	3.064	3.064

LTB Moment Capacity (kNm) and LTB Length (m): 2.27, 4.713
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.001	212	0.4	-	-	-	-
BS-4.7 (C)	0.083	207	2.4	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
515 ST UA70X70X6	PASS	0.42 T	BS-4.6 (T)	0.001	207
			0.00	0.00	0.00

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
516 ST UA70X70X6	PASS	1.26 T	BS-4.6 (T)	0.004	207
			0.00	0.00	3.14

MATERIAL DATA

Grade of steel = S 275

Modulus of elasticity = 205 kN/mm²

Design Strength (py) = 344 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 329.16

Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000

Section Class : SLENDER

Squash Load : 282.90

Axial force/Squash load : 0.001

	Z-Z axis	Y-Y axis
Compression Capacity	: 55.2	112.8
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 156.172	100.655	100.655
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.140	2.140	2.140

LTB Moment Capacity (kNm) and LTB Length (m): 2.77, 3.292

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00

Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.001	207	0.4	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
516 ST UA70X70X6	PASS	1.26 T	BS-4.6 (T)	0.004	207
			0.00	0.00	3.14

MATERIAL DATA

Grade of steel = S 275

Modulus of elasticity = 205 kN/mm²

Design Strength (py) = 344 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 313.76

Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000

Section Class : SLENDER

	Z-Z axis	Y-Y axis
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

LTB Moment Capacity (kNm) and LTB Length (m): 2.83, 3.138

LTB Coefficients & Associated Moments (kNm):

mLT = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00

Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.004	207	1.3	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

522 ST UA70X70X6	PASS	BS-4.7 (C)	0.313	221	
	9.86 C	0.00	0.00	0.00	
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 449.65
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.035

	Z-Z axis	Y-Y axis
Compression Capacity	: 31.5	68.8
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 213.340	137.500	137.500
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.923	2.923	2.923

LTB Moment Capacity (kNm) and LTB Length (m): 2.33, 4.497
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.009	228	2.7	-	-	-	-
BS-4.7 (C)	0.313	221	9.9	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

523 ST UA70X70X6	PASS	BS-4.7 (C)	0.224	201
9.73 C	0.00	0.00	0.00	0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 376.60
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.034

	Z-Z axis	Y-Y axis
Compression Capacity	: 43.5	92.0
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 178.677	115.160	115.160
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.448	2.448	2.448

LTB Moment Capacity (kNm) and LTB Length (m): 2.58, 3.766
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mX = 0.00 : mY = 0.00 : mYX = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.009	218	2.6	-	-	-	-
BS-4.7 (C)	0.224	201	9.7	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

524 ST UA70X70X6	PASS	BS-4.7 (C)	0.274	219	
	9.69 C	0.00	0.00	0.00	
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 422.12
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.034

	z-z axis	y-y axis
Compression Capacity	: 35.4	76.5
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 200.278	129.082	129.082
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.744	2.744	2.744

LTB Moment Capacity (kNm) and LTB Length (m): 2.42, 4.221
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.009	218	2.5	-	-	-	-
BS-4.7 (C)	0.274	219	9.7	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

525 ST UA70X70X6	PASS	BS-4.7 (C)	0.251	219	
	9.48 C	0.00	0.00	0.00	
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 407.22
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.034

	z-z axis	y-y axis
Compression Capacity	: 37.7	81.1
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 193.205	124.523	124.523
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.647	2.647	2.647

LTB Moment Capacity (kNm) and LTB Length (m): 2.47, 4.072
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.009	218	2.5	-	-	-	-
BS-4.7 (C)	0.251	219	9.5	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

530 ST UA70X70X6	PASS	BS-4.7 (C)	0.137	207	0.00
4.51 C	0.00	0.00	0.00	0.00	0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 437.92
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.016

	Z-Z axis	Y-Y axis
Compression Capacity	: 33.1	71.9
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 207.770	133.911	133.911
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.846	2.846	2.846

LTB Moment Capacity (kNm) and LTB Length (m): 2.37, 4.379
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mX = 0.00 : mY = 0.00 : mYX = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.004	212	1.1	-	-	-	-
BS-4.7 (C)	0.137	207	4.5	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

531 ST UA70X70X6	PASS	BS-4.7 (C)	0.144	207	
	4.37 C	0.00	0.00	4.58	
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 457.97
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.015

	z-z axis	y-y axis
Compression Capacity	: 30.4	66.7
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 217.283	140.042	140.042
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.977	2.977	2.977

LTB Moment Capacity (kNm) and LTB Length (m): 2.31, 4.580
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.004	212	1.1	-	-	-	-
BS-4.7 (C)	0.144	207	4.4	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

532 ST UA70X70X6	PASS	BS-4.7 (C)	0.136	207	0.00
4.48 C	0.00	0.00	0.00	0.00	0.00
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 437.92
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.016

	Z-Z axis	Y-Y axis
Compression Capacity	: 33.1	71.9
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 207.770	133.911	133.911
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.846	2.846	2.846

LTB Moment Capacity (kNm) and LTB Length (m): 2.37, 4.379
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.004	212	1.1	-	-	-	-
BS-4.7 (C)	0.136	207	4.5	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

533 ST UA70X70X6	PASS	BS-4.7 (C)	0.142	207	
	4.32 C	0.00	0.00	4.58	
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 457.97
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	z-z axis	y-y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.015

	z-z axis	y-y axis
Compression Capacity	: 30.4	66.7
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	v-v axis	a-a axis	b-b axis
Slenderness	: 217.283	140.042	140.042
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.977	2.977	2.977

LTB Moment Capacity (kNm) and LTB Length (m): 2.31, 4.580
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.004	218	1.1	-	-	-	-
BS-4.7 (C)	0.142	207	4.3	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

538 ST UA70X70X6	PASS	BS-4.7 (C)	0.258	213	
	10.04 C	0.00	0.00	4.00	
=====					

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 400.28
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.035

	Z-Z axis	Y-Y axis
Compression Capacity	: 38.9	83.4
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 189.914	122.402	122.402
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.602	2.602	2.602

LTB Moment Capacity (kNm) and LTB Length (m): 2.50, 4.003
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mX = 0.00 : mY = 0.00 : mYX = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.010	206	2.9	-	-	-	-
BS-4.7 (C)	0.258	213	10.0	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

539 ST UA70X70X6	PASS	BS-4.7 (C)	0.268	222
	10.11 C	0.00	0.00	4.07

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 407.37
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.036

	Z-Z axis	Y-Y axis
Compression Capacity	: 37.7	81.1
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 193.276	124.569	124.569
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.648	2.648	2.648

LTB Moment Capacity (kNm) and LTB Length (m): 2.47, 4.074

LTB Coefficients & Associated Moments (kNm):

mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.010	206	2.9	-	-	-	-
BS-4.7 (C)	0.268	222	10.1	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

540 ST UA70X70X6	PASS	BS-4.7 (C)	0.268	220
	10.11 C	0.00	0.00	0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm2
Design Strength (py) = 344 N/mm2

SECTION PROPERTIES (units - cm)

Member Length = 407.37
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of inertia	: 15.391	60.246
Plastic modulus	: 10.133	21.909
Elastic modulus	: 5.629	12.171
Effective modulus	: 5.629	12.171
Shear Area	: 2.520	2.520

DESIGN DATA (units - kN,m) BS5950-1/2000

Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.036

	Z-Z axis	Y-Y axis
Compression Capacity	: 37.7	81.1
Tension Capacity	: 282.9	282.9
Shear Capacity	: 52.2	52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	V-V axis	a-a axis	b-b axis
Slenderness	: 193.276	124.569	124.569
Radius of gyration (cm)	: 1.370	2.126	2.126
Effective Length	: 2.648	2.648	2.648

LTB Moment Capacity (kNm) and LTB Length (m): 2.47, 4.074
LTB Coefficients & Associated Moments (kNm):
mLt = 1.00 : mx = 0.00 : my = 0.00 : myx = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSe	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.010	206	2.9	-	-	-	-
BS-4.7 (C)	0.268	220	10.1	-	-	-	-

Torsion and deflections have not been considered in the design.

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

541 ST UA70X70X6 PASS BS-4.7 (C) 0.258 213
10.04 C 0.00 0.00 0.00

MATERIAL DATA

Grade of steel = S 275
Modulus of elasticity = 205 kN/mm²
Design Strength (py) = 344 N/mm²

SECTION PROPERTIES (units - cm)

Member Length = 400.28
Gross Area = 8.20 Net Area = 8.20 Eff. Area = 8.20

	Z-Z axis	Y-Y axis
Moment of Inertia	: 15.391	: 60.246
Plastic modulus	: 10.133	: 21.909
Elastic modulus	: 5.629	: 12.171
Effective modulus	: 5.629	: 12.171
Shear Area	: 2.520	: 2.520

DESIGN DATA (units - kN,m)

BS5950-1/2000
Section Class : SLENDER
Squash Load : 282.90
Axial force/Squash load : 0.035

	Z-Z axis	Y-Y axis
Compression Capacity	: 38.9	: 83.4
Tension Capacity	: 282.9	: 282.9
Shear Capacity	: 52.2	: 52.2

BUCKLING CALCULATIONS (units - kN,m)

(axis nomenclature as per design code)

	a-a axis	b-b axis
Slenderness	: 189.914	: 122.402
Radius of gyration (cm)	: 1.370	: 2.126
Effective Length	: 2.602	: 2.602

LTB Moment Capacity (kNm) and LTB Length (m): 2.50, 4.003
LTB Coefficients & Associated Moments (kNm):
mLT = 1.00 : mX = 0.00 : mY = 0.00 : myX = 0.00
Mlt = 0.00 : Mx = 0.00 : My = 0.00 : My = 0.00

CRITICAL LOADS FOR EACH CLAUSE CHECK (units- kN,m):

CLAUSE	RATIO	LOAD	FX	VY	VZ	MZ	MY
BS-4.6 (T)	0.010	206	2.9	-	-	-	-
BS-4.7 (C)	0.258	213	10.0	-	-	-	-

Torsion and deflections have not been considered in the design.

542	TAPERED	PASS	BS-4.7 (C)	0.229	222
		62.99 C	-5.43	6.42	0.50

CALCULATED CAPACITIES FOR MEMB 542 UNIT - kN,m SECTION CLASS 4					
MCZ=	177.6	MCY=	35.1 PC= 1550.7 PT= 1691.9	MB=	177.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 0.00, mY = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm ²					
d =	284 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm ²
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

543	TAPERED	PASS	BS-4.7 (C)	0.241	221
		66.36 C	-5.72	-6.70	0.50

CALCULATED CAPACITIES FOR MEMB 543 UNIT - kN,m SECTION CLASS 4					
MCZ=	177.6	MCY=	35.1 PC= 1550.7 PT= 1691.9	MB=	177.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 0.00, mY = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm ²					
d =	284 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm ²
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

544	TAPERED	PASS	BS-4.7 (C)	0.068	213
		85.58 C	-0.01	-2.80	0.50

CALCULATED CAPACITIES FOR MEMB 544 UNIT - kN,m SECTION CLASS 4					
MCZ=	177.6	MCY=	35.1 PC= 1550.7 PT= 1691.9	MB=	177.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 0.00, mY = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm ²					
d =	284 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm ²
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

545	TAPERED	PASS	BS-4.7 (C)	0.072	201
		90.24 C	-0.01	3.05	0.50

CALCULATED CAPACITIES FOR MEMB 545 UNIT - kN,m SECTION CLASS 4					
MCZ=	177.6	MCY=	35.1 PC= 1550.7 PT= 1691.9	MB=	177.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 0.00, mY = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm ²					
d =	284 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm ²
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

546	TAPERED	PASS	BS-4.7 (C)	0.067	213
		85.66 C	0.01	-2.71	0.50

CALCULATED CAPACITIES FOR MEMB 546 UNIT - kN,m SECTION CLASS 4					
MCZ=	177.6	MCY=	35.1 PC= 1550.7 PT= 1691.9	MB=	177.6
BUCKLING CO-EFFICIENTS mLT = 1.00, mX = 0.00, mY = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm ²					
d =	284 mm	t =	6 mm	a =	0 mm : pyf = 344 N/mm ²
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

547	TAPERED	PASS	BS-4.7 (C)	0.073	201
		90.22 C	0.01	3.16	0.50

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 547 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

548 TAPERED PASS 62.99 C 5.44 BS-4.7 (C) 0.229 220 6.42 0.50					

CALCULATED CAPACITIES FOR MEMB 548 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

549 TAPERED PASS 66.37 C 5.77 BS-4.7 (C) 0.242 219 -0.70 0.50					

CALCULATED CAPACITIES FOR MEMB 549 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

550 TAPERED PASS 86.23 C -7.66 BS-4.7 (C) 0.285 207 2.71 0.20					

CALCULATED CAPACITIES FOR MEMB 550 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

551 TAPERED PASS 99.55 C -0.16 BS-4.7 (C) 0.089 208 4.48 0.20					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 551 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

552 TAPERED PASS 99.52 C 0.16 BS-4.7 (C) 0.089 208 4.50 0.20					

CALCULATED CAPACITIES FOR MEMB 552 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

553 TAPERED PASS 86.24 C 7.66 BS-4.7 (C) 0.285 207 2.71 0.20					

CALCULATED CAPACITIES FOR MEMB 553 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

554 TAPERED PASS 83.98 C -7.62 BS-4.7 (C) 0.282 207 -2.59 0.20					

CALCULATED CAPACITIES FOR MEMB 554 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

555 TAPERED PASS 97.28 C -0.14 BS-4.7 (C) 0.087 208 -4.46 0.20					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 555 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

556 TAPERED PASS BS-4.7 (C) 0.087 208					
97.28 C 0.14 -4.48 0.20					

CALCULATED CAPACITIES FOR MEMB 556 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

557 TAPERED PASS BS-4.7 (C) 0.281 207					
83.98 C 7.59 -2.59 0.20					

CALCULATED CAPACITIES FOR MEMB 557 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

558 TAPERED PASS BS-4.7 (C) 0.085 203					
36.68 C -0.01 -11.12 0.44					

CALCULATED CAPACITIES FOR MEMB 558 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

559 TAPERED PASS BS-4.7 (C) 0.110 203					
46.89 C 0.02 -14.36 0.44					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 559 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

560 TAPERED PASS BS-4.7 (C) 0.110 203					
46.85 C -0.02 -14.36 0.44					

CALCULATED CAPACITIES FOR MEMB 560 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

561 TAPERED PASS BS-4.7 (C) 0.085 203					
36.68 C 0.01 -11.12 0.44					

CALCULATED CAPACITIES FOR MEMB 561 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

562 TAPERED PASS BS-4.8.2.2 0.138 207					
25.04 T 5.03 -12.26 0.66					

CALCULATED CAPACITIES FOR MEMB 562 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

564 TAPERED PASS BS-4.8.2.2 0.101 212					
56.70 T 0.17 -23.46 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 564 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

566 TAPERED PASS BS-4.8.2.2 0.101 212					
56.71 T -0.17 -23.46 0.00					

CALCULATED CAPACITIES FOR MEMB 566 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

567 TAPERED PASS BS-4.8.2.2 0.138 207					
25.04 T -5.03 -12.26 0.66					

CALCULATED CAPACITIES FOR MEMB 567 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

568 TAPERED PASS BS-4.7 (C) 0.085 215					
36.82 C -0.01 -11.13 0.44					

CALCULATED CAPACITIES FOR MEMB 568 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

569 TAPERED PASS BS-4.7 (C) 0.110 215					
46.94 C 0.02 -14.35 0.44					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
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CALCULATED CAPACITIES FOR MEMB 569 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

570 TAPERED PASS BS-4.7 (C) 0.110 215					
46.93 C -0.02 -14.35 0.44					

CALCULATED CAPACITIES FOR MEMB 570 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

571 TAPERED PASS BS-4.7 (C) 0.085 215					
36.81 C 0.01 -11.13 0.44					

CALCULATED CAPACITIES FOR MEMB 571 UNIT - KN,m SECTION CLASS 4					
MCZ= 177.6 MCY= 35.1 PC= 1550.7 PT= 1691.9 MB= 177.6					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 352 kN : qw = 207 N/mm2					
d = 284 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

572 TAPERED PASS BS-4.8.2.2 0.136 207					
25.32 T -5.02 -11.78 0.66					

CALCULATED CAPACITIES FOR MEMB 572 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myX = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

573 TAPERED PASS BS-4.8.2.2 0.101 212					
56.65 T -0.16 -23.37 0.00					

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					

CALCULATED CAPACITIES FOR MEMB 573 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

574	TAPERED	PASS	BS-4.8.2.2	0.101	212
		56.65 T	0.16	-23.37	0.00

CALCULATED CAPACITIES FOR MEMB 574 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

575	TAPERED	PASS	BS-4.8.2.2	0.136	207
		25.32 T	5.02	-11.78	0.66

CALCULATED CAPACITIES FOR MEMB 575 UNIT - KN,m SECTION CLASS 3					
MCZ= 324.8 MCY= 56.7 PC= 1663.1 PT= 2166.6 MB= 324.8					
BUCKLING CO-EFFICIENTS mLT = 1.00, mx = 0.00, my = 0.00, myx = 0.00					
Shear Buckling check is required : Vb = 471 kN : qw = 207 N/mm2					
d = 380 mm : t = 6 mm : a = 0 mm : pyf = 344 N/mm2					
BS-4.4.3.2 status = PASS : BS-4.4.3.3 status = PASS					

***** END OF TABULATED RESULT OF DESIGN *****

610. STEEL TAKE OFF ALL

STEEL TAKE OFF

STEEL TAKE-OFF

PROFILE	LENGTH(METE)	WEIGHT (KN)

ISECT: TAPERED	1	21.60
ISECT: TAPERED	2	87.01
ISECT: TAPERED	95	21.22
ISECT: TAPERED	201	164.60
ISECT: TAPERED	301	86.40
ST UA70X70X6	114.13	7.300
ISECT: TAPERED	542	9.08

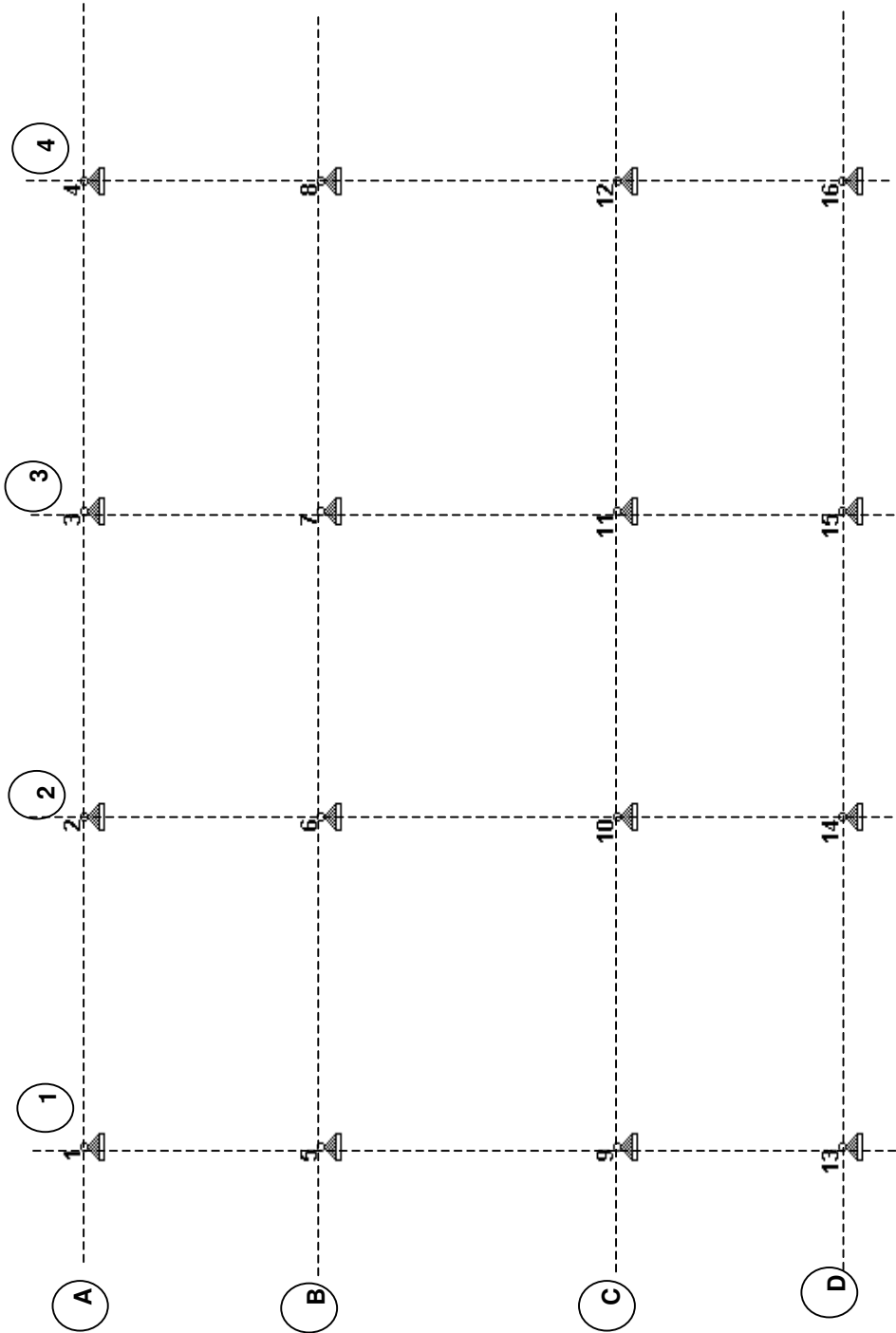
TOTAL =		183.023

***** END OF DATA FROM INTERNAL STORAGE *****

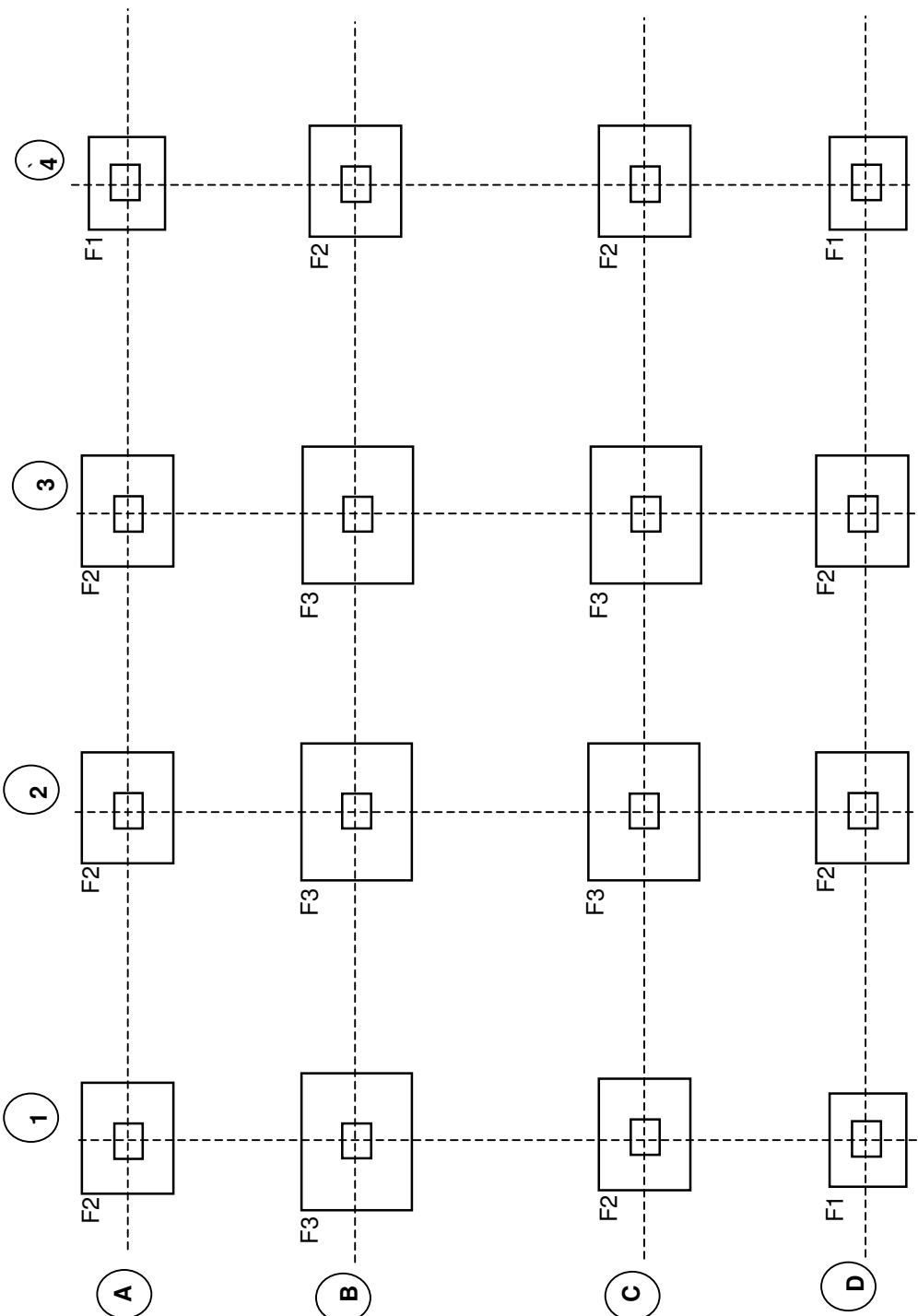
611. FINISH

***** END OF THE STAAD.Pro RUN *****

5 Foundation Design



Grid Marking With Node Details (STAAD) @ Foundation Level



RC Foundation Layout Plan

Footing Design Summary

(A) Isolated Footing

Ref. node on STAAD	Location on drawing	Footing ID on Drg.	Length I_x (M)	Width I_y (M)	Depth of footing D (mm)	Bottom reinforcement (Parallel to)			Top reinforcement (Parallel to)		
						Length (x dir.)		Width (y dir.)		Length (x dir.)	
						Bar dia	Nos.	Bar dia	Nos.	Bar dia	Nos.
						Req./Pro.	Req./Pro.	Req./Pro.	Req./Pro.	Req./Pro.	Req./Pro.
4 , 13, 16	A-4 , D-1 D-4	F1	2.6	2.6	600	12 / 12	18 / 21	12 / 12	18 / 21	0 / 10	0 / 18
1,2,3, 8,9,12 14,15	A-1,A-2,A-3, B-4,C-1,C-4 D-2, D-3	F2	3.2	3.2	600	12 / 12	22 / 26	12 / 12	22 / 26	0 / 10	0 / 22
5, 6 , 7 10 , 11	B-1, B-2 B-3 , C-2, C-3	F3	4.0	4.0	600	16 / 16	29 / 32	16 / 16	29 / 32	0 / 10	0 / 27

Reinforced Concrete Footing Design Based On BS8110

Footing Location on Drg.

A-1

Ref. node on staad

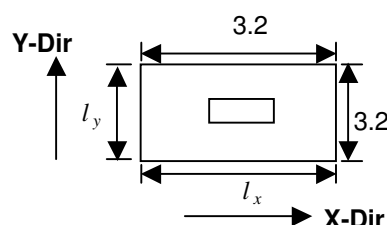
1

Material properties

Characteristic strength of concrete f_{cu}	40 N/mm ²	Net Bearing capacity of soil	75 kN/m ²
Characteristic strength of steel f_y	460 N/mm ²	(Note - Reduced bearing capacity of soil due to effect of water table)	
Concrete density	24 kN/mm ³		
Steel density	78 kN/mm ³		

Length of column	c_x	0.55 m
Width of column	c_y	0.55 m
Length of footing	l_x	3.2 m
Width of footing	l_y	3.2 m
Depth of footing	D	600 mm
Conc. Ten. Cover	d_t	75 mm
Conc. Comp. Cover	d_c	75 mm
Tension bar dia.(x- dir)	Φ_x	12 mm
Tension bar dia.(y- dir)	Φ_y	12 mm
Comp. bar dia.(x- dir)	Φ_{cx}	10 mm
Comp. bar dia.(y- dir)	Φ_{cy}	10 mm
Effect. Depth(x- dir)	d_x	519 mm
Effect. Depth(y- dir)	d_y	507 mm
Area of footing	A_B	10.24 m ²

Self weight of footing 36.86 kN



Reactions

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
101	558.3	58.1	58.1	200	803.1	83.8	83.8
102	504.6	52.9	52.9	201	688.5	72.6	72.6
103	459.0	48.4	48.4	202	653.0	69.2	69.2
104	466.5	49.2	49.2	203	472.6	51.6	51.6
105	445.6	47.1	47.1	204	668.4	70.7	70.7
106	449.7	47.5	47.5	205	629.5	66.9	66.9
107	455.8	48.1	48.1	206	449.1	49.3	49.3
108	462.4	48.8	48.8	207	683.7	72.2	72.2
109	429.2	45.5	45.5	208	647.4	68.6	68.6
110	429.2	45.5	45.5	209	467.0	51.0	51.0
111	435.2	46.1	46.1	210	643.8	68.3	68.3
112	436.7	46.2	46.2	211	600.8	64.1	64.1
113	421.7	44.8	44.8	212	420.4	46.5	46.5
114	419.8	44.6	44.6	213	652.8	69.2	69.2
115	293.7	32.3	32.3	214	611.3	65.1	65.1
116	383.0	41.0	41.0	215	430.9	47.5	47.5
117	519.0	54.3	54.3	216	632.5	67.2	67.2
118	608.3	63.0	63.0	217	587.7	62.8	62.8
119	270.5	30.0	30.0	218	407.3	45.2	45.2
120	339.3	36.7	36.7	219	474.3	51.7	51.7

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
121	562.7	58.6	58.6	220	608.2	64.8	64.8
122	631.5	65.3	65.3	221	818.5	85.3	85.3
123	248.6	27.9	27.9	222	952.4	98.4	98.4
124	337.9	36.6	36.6	223	438.5	48.2	48.2
125	473.9	49.9	49.9	224	541.8	58.3	58.3
126	563.2	58.6	58.6	225	884.9	91.8	91.8
127	225.4	25.6	25.6	226	988.2	101.9	101.9
128	294.2	32.3	32.3	227	207.4	25.7	25.7
129	517.6	54.1	54.1	228	341.3	38.7	38.7
130	586.4	60.9	60.9	229	551.6	59.3	59.3
131	412.5	43.9	43.9	230	685.6	72.4	72.4
132	481.5	50.6	50.6	231	171.6	22.2	22.2
133	581.5	60.4	60.4	232	274.9	32.3	32.3
134	650.5	67.1	67.1	233	618.1	65.8	65.8
135	395.3	42.2	42.2	234	721.3	75.8	75.8
136	448.5	47.4	47.4	235	546.6	58.8	58.8
137	614.4	63.6	63.6	236	543.3	58.5	58.5
138	667.6	68.8	68.8				
L/C - Load combination cases as per specification					P= Fy + self weight of footing		

P _{max,net}	68.8	kN/m ²	(for SLS)
P _{max,net}	101.9	kN/m ²	(for ULS)

Length of cantilever in x- Dir	1.33 m
Length of cantilever in y- Dir	1.33 m
Moment (x - Dir)	90.13 kN-m/m
Moment (y - Dir)	90.13 kN-m/m

About x- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} \quad 0.008 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.991 d$$

Hence, $z = 0.95 d$

$$A_s = \frac{M}{(0.95 \times f_y \times z)}$$

$$= 418 \text{ mm}^2/\text{m}$$

Min area of steel (0.13 % of bD) = 780 mm²/m

Area of bottom reinforcement req 780 mm²/m

Area of bottom reinforcement pro. 905 mm²/m

T - 12 @ 122 c/c

No. of bar
26

Area of top reinforcement req = 0 mm²/m

Area of top reinforcement pro. = 524 mm²/m

T - 10 @ 145 c/c

No. of bar
22

About y- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} = 0.009 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.99d$$

Hence, $z = 0.95d$

$$A_s = \frac{M}{(0.95 \times f_y \times z)}$$

$$= 428 \text{ mm}^2/\text{m}$$

$$\text{Min area of steel (0.13 \% of } bD) = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement req} = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement pro.} = 905 \text{ mm}^2/\text{m}$$

$$\boxed{T - 12 @ 122} \text{ c/c}$$

No. of bar
26

$$\text{Area of top reinforcement req} = 0 \text{ mm}^2/\text{m}$$

$$\text{Area of top reinforcement pro.} = 524 \text{ mm}^2/\text{m}$$

$$\boxed{T - 10 @ 145} \text{ c/c}$$

No. of bar
22

Check for shear

About x- dir

$$\text{Shear force at face} = 135.53 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.26 \text{ (N/mm}^2\text{)} \text{ O.K}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	519	82.64	0.16	0.17	0.41	0.82	Not reqd
2d	1038	29.7548	0.06	0.17	0.41	0.41	Not reqd

About y- dir

$$\text{Shear force at face} = 135.53 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.27 \text{ (N/mm}^2\text{)} \text{ O.K}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	507	83.86	0.17	0.18	0.358	0.72	Not reqd
2d	1014	32.2	0.06	0.18	0.358	0.36	Not reqd

Check for punching shear

P

803.1 kN

p

78.43 kN/m²

Punching surface perimeter

4600 mm

Net P

706.71 kN

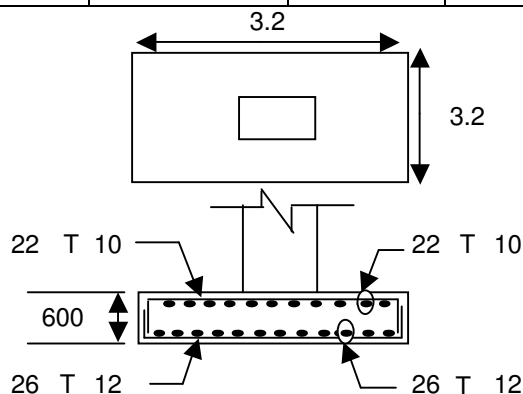
ζ_v

0.3 N/mm²

ζ_{v,design}

5 N/mm²

> ζ_v ∴ Safe



Reinforced Concrete Footing Design Based On BS8110

Footing Location on Drg.

B-1

Ref. node on staad

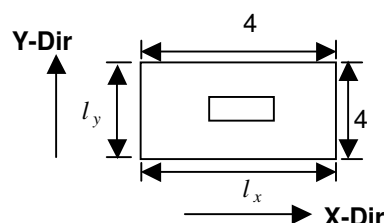
5

Material properties

Characteristic strength of concrete f_{cu}	40 N/mm ²	Net Bearing capacity of soil	75 kN/m ²
Characteristic strength of steel f_y	460 N/mm ²	(Note - Reduced bearing capacity of soil due to effect of water table)	
Concrete density	24 kN/mm ³		
Steel density	78 kN/mm ³		

Length of column	C_x	0.55 m
Width of column	C_y	0.55 m
Length of footing	l_x	4 m
Width of footing	l_y	4 m
Depth of footing	D	600 mm
Conc. Ten. Cover	d_t	75 mm
Conc. Comp. Cover	d_c	75 mm
Tension bar dia.(x- dir)	Φ_x	16 mm
Tension bar dia.(y- dir)	Φ_y	16 mm
Comp. bar dia.(x- dir)	Φ_{cx}	10 mm
Comp. bar dia.(y- dir)	Φ_{cy}	10 mm
Effect. Depth(x- dir)	d_x	517 mm
Effect. Depth(y- dir)	d_y	501 mm
Area of footing	A_B	16 m ²

Self weight of footing 57.6 kN



Reactions

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
101	1077.3	70.9	70.9	200	1569.5	103.5	103.5
102	924.1	61.4	61.4	201	1256.1	83.9	83.9
103	837.4	55.9	55.9	202	1036.4	70.2	70.2
104	740.3	49.9	49.9	203	728.1	50.9	50.9
105	817.8	54.7	54.7	204	1226.7	82.1	82.1
106	715.8	48.3	48.3	205	1002.2	68.0	68.0
107	881.6	58.7	58.7	206	693.8	48.8	48.8
108	795.5	53.3	53.3	207	1322.4	88.1	88.1
109	827.0	55.3	55.3	208	1113.7	75.0	75.0
110	727.4	49.1	49.1	209	805.4	55.7	55.7
111	886.7	59.0	59.0	210	1240.6	82.9	82.9
112	801.9	53.7	53.7	211	1018.3	69.1	69.1
113	851.9	56.8	56.8	212	710.0	49.8	49.8
114	758.5	51.0	51.0	213	1330.0	88.5	88.5
115	648.1	44.1	44.1	214	1122.7	75.6	75.6
116	605.8	41.5	41.5	215	814.3	56.3	56.3
117	935.9	62.1	62.1	216	1277.9	85.3	85.3
118	893.6	59.5	59.5	217	1061.8	71.8	71.8
119	796.0	53.4	53.4	218	753.5	52.5	52.5
120	883.9	58.8	58.8	219	1166.4	78.3	78.3

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
121	657.8	44.7	44.7	220	1103.1	74.3	74.3
122	745.7	50.2	50.2	221	1606.1	105.8	105.8
123	571.0	39.3	39.3	222	1542.8	101.8	101.8
124	528.8	36.7	36.7	223	1394.2	92.5	92.5
125	858.8	57.3	57.3	224	1526.1	100.8	100.8
126	816.6	54.6	54.6	225	1183.1	79.3	79.3
127	718.9	48.5	48.5	226	1315.0	87.6	87.6
128	806.8	54.0	54.0	227	575.0	41.3	41.3
129	580.7	39.9	39.9	228	511.6	37.4	37.4
130	668.7	45.4	45.4	229	1014.6	68.8	68.8
131	909.1	60.4	60.4	230	951.3	64.9	64.9
132	876.4	58.4	58.4	231	802.7	55.6	55.6
133	1124.9	73.9	73.9	232	934.6	63.8	63.8
134	1092.3	71.9	71.9	233	591.6	42.4	42.4
135	1018.5	67.3	67.3	234	723.6	50.6	50.6
136	1086.5	71.5	71.5	235	1062.3	71.8	71.8
137	914.9	60.8	60.8	236	1084.2	73.2	73.2
138	982.8	65.0	65.0				
L/C - Load combination cases as per specification					P= Fy + self weight of footing		

$P_{max,net}$ 73.9 kN/m² (for SLS)
 $P_{max,net}$ 105.8 kN/m² (for ULS)

Length of cantilever in x- Dir 1.73 m
 Length of cantilever in y- Dir 1.73 m
 Moment (x - Dir) 158.29 kN-m/m
 Moment (y - Dir) 158.29 kN-m/m

About x- dir

$K = M/(b \times d^2 \times f_{cu})$ 0.015 < $K' = 0.156$
 since $K < K'$ compression reinforcement not required

$z = d\{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$
 $= 0.983 d$
 Hence, $z = 0.95 d$

$A_s = M/(0.95 \times f_y \times z)$
 $= 737 \text{ mm}^2/\text{m}$

Min area of steel (0.13 % of bD) = 780 mm²/m
 Area of bottom reinforcement req 780 mm²/m
 Area of bottom reinforcement pro. 1608 mm²/m

T - 16 @ 124 c/c

No. of bar
32

Area of top reinforcement req = 0 mm²/m
 Area of top reinforcement pro. = 524 mm²/m

T - 10 @ 148 c/c

No. of bar
27

About y- dir

$$K = M/(b \times d^2 \times f_{cu}) = 0.016 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d\{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.982 d$$

Hence, $z = 0.95 d$

$$A_s = M/(0.95 \times f_y \times z)$$

$$= 761 \text{ mm}^2/\text{m}$$

$$\text{Min area of steel (0.13 \% of } bD) = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement req} = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement pro.} = 1608 \text{ mm}^2/\text{m}$$

$$\boxed{T - 16 @ 124} \text{ c/c}$$

No. of bar
32

$$\text{Area of top reinforcement req} = 0 \text{ mm}^2/\text{m}$$

$$\text{Area of top reinforcement pro.} = 524 \text{ mm}^2/\text{m}$$

$$\boxed{T - 10 @ 148} \text{ c/c}$$

No. of bar
27

Check for shear

About x- dir

$$\text{Shear force at face} = 183 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.35 \text{ (N/mm}^2\text{)} \text{ O.K}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	517	128.31	0.25	0.31	0.5	1	Not reqd
2d	1034	73.62288	0.14	0.31	0.5	0.5	Not reqd

About y- dir

$$\text{Shear force at face} = 183 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.37 \text{ (N/mm}^2\text{)} \text{ O.K}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	501	130	0.26	0.32	0.428	0.86	Not reqd
2d	1002	77.01	0.15	0.32	0.428	0.43	Not reqd

Check for punching shear

P

1569.5 kN

p

98.09 kN/m²

Punching surface perimeter

4600 mm

Net P

1449.08 kN

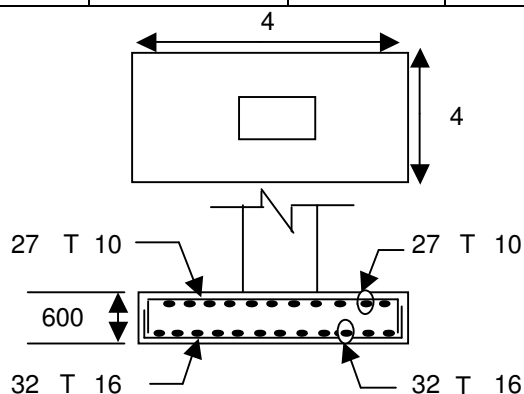
ζ_v

0.61 N/mm²

ζ_{v,design}

5 N/mm²

> ζ_v ∴ Safe



Reinforced Concrete Footing Design Based On BS8110

Footing Location on Drg.

B-3

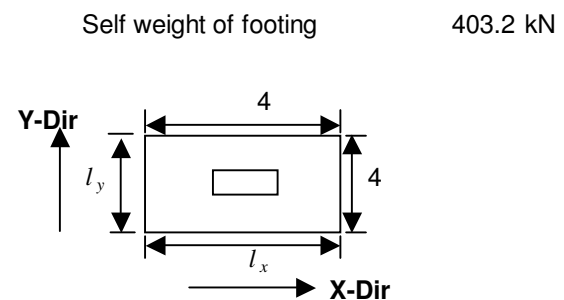
Ref. node on staad

7

Material properties

Characteristic strength of concrete f_{cu}	40 N/mm ²	Net Bearing capacity of soil	150 kN/m ²
Characteristic strength of steel f_y	460 N/mm ²	at	1.2 m
Concrete density	24 kN/mm ³		Below G.L
Steel density	78 kN/mm ³		

Length of column	C_x	0.55 m
Width of column	C_y	0.55 m
Length of footing	l_x	4 m
Width of footing	l_y	4 m
Depth of footing	D	600 mm
Conc. Ten. Cover	d_t	75 mm
Conc. Comp. Cover	d_c	75 mm
Tension bar dia.(x- dir)	Φ_x	16 mm
Tension bar dia.(y- dir)	Φ_y	16 mm
Comp. bar dia.(x- dir)	Φ_{cx}	10 mm
Comp. bar dia.(y- dir)	Φ_{cy}	10 mm
Effect. Depth(x- dir)	d_x	517 mm
Effect. Depth(y- dir)	d_y	501 mm
Area of footing	A_B	16 m ²



Reactions

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
101	1800.9	137.8	137.8	201	2637.9	202.7	202.7
102	1509.3	119.5	119.5	202	2125.0	170.6	170.6
103	1416.7	113.7	113.7	203	1662.7	141.7	141.7
104	1187.7	99.4	99.4	204	1175.6	111.3	111.3
105	1400.2	112.7	112.7	205	2100.3	169.1	169.1
106	1167.1	98.2	98.2	206	1634.0	139.9	139.9
107	1457.8	116.3	116.3	207	1146.9	109.5	109.5
108	1239.1	102.6	102.6	208	2186.7	174.5	174.5
109	1410.7	113.4	113.4	209	1734.7	146.2	146.2
110	1180.2	99.0	99.0	210	1247.6	115.8	115.8
111	1465.1	116.8	116.8	211	2116.0	170.1	170.1
112	1248.3	103.2	103.2	212	1652.3	141.1	141.1
113	1434.6	114.9	114.9	213	1165.2	110.6	110.6
114	1210.1	100.8	100.8	214	2197.7	175.2	175.2
115	1231.2	102.2	102.2	215	1747.6	147.0	147.0
116	1176.7	98.8	98.8	216	1260.5	116.6	116.6
117	1258.7	103.9	103.9	217	2151.9	172.3	172.3
118	1204.2	100.5	100.5	218	1694.1	143.7	143.7
119	1302.7	106.6	106.6	219	1207.0	113.2	113.2
120	1311.1	107.1	107.1	220	2268.7	179.6	179.6

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
121	1124.4	95.5	95.5	221	2187.0	174.5	174.5
122	1132.8	96.0	96.0	222	2310.7	182.2	182.2
123	1109.4	94.5	94.5	223	2229.0	177.1	177.1
124	1055.0	91.1	91.1	224	2378.8	186.5	186.5
125	1136.9	96.3	96.3	225	2391.4	187.3	187.3
126	1082.5	92.9	92.9	226	2106.4	169.5	169.5
127	1180.9	99.0	99.0	227	2119.0	170.2	170.2
128	1189.3	99.5	99.5	228	1225.4	114.4	114.4
129	1002.6	87.9	87.9	229	1143.7	109.3	109.3
130	1011.0	88.4	88.4	230	1267.4	117.0	117.0
131	1665.8	129.3	129.3	231	1185.7	111.9	111.9
132	1623.7	126.7	126.7	232	1335.4	121.3	121.3
133	1686.4	130.6	130.6	233	1348.0	122.1	122.1
134	1644.4	128.0	128.0	234	1063.1	104.2	104.2
135	1718.7	132.6	132.6	235	1075.7	105.0	105.0
136	1725.2	133.0	133.0	236	1799.8	150.3	150.3
137	1585.0	124.3	124.3				
138	1591.5	124.7	124.7				
L/C - Load combination cases as per specification					P= Fy + self weight of footing		

$$P_{\max, \text{net}} = 137.8 \text{ kN/m}^2 \text{ (for SLS)}$$

$$P_{\max, \text{net}} = 202.7 \text{ kN/m}^2 \text{ (for ULS)}$$

$$\begin{aligned} \text{Length of cantilever in x- Dir} &= 1.73 \text{ m} \\ \text{Length of cantilever in y- Dir} &= 1.73 \text{ m} \\ \text{Moment (x - Dir)} &= 303.29 \text{ kN-m/m} \\ \text{Moment (y - Dir)} &= 303.29 \text{ kN-m/m} \end{aligned}$$

About x- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} = 0.028 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$\begin{aligned} z &= d\{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d \\ &= 0.968 d \\ \text{Hence, } z &= 0.95 d \end{aligned}$$

$$\begin{aligned} A_s &= \frac{M}{(0.95 \times f_y \times z)} \\ &= 1413 \text{ mm}^2/\text{m} \end{aligned}$$

$$\begin{aligned} \text{Min area of steel (0.13 \% of bD)} &= 780 \text{ mm}^2/\text{m} \\ \text{Area of bottom reinforcement req} &= 1413 \text{ mm}^2/\text{m} \\ \text{Area of bottom reinforcement pro.} &= 1608 \text{ mm}^2/\text{m} \end{aligned}$$

T - 16 @ 124 c/c

No. of bar
32

$$\begin{aligned} \text{Area of top reinforcement req} &= 0 \text{ mm}^2/\text{m} \\ \text{Area of top reinforcement pro.} &= 524 \text{ mm}^2/\text{m} \end{aligned}$$

T - 10 @ 148 c/c

No. of bar
27

About y- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} = 0.03 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.965 d$$

$$\text{Hence, } z = 0.95 d$$

$$A_s = \frac{M}{(0.95 \times f_y \times z)} = 1458 \text{ mm}^2/\text{m}$$

$$\text{Min area of steel (0.13 \% of } bD) = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement req} = 1458 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement pro.} = 1608 \text{ mm}^2/\text{m}$$

$$\boxed{T - 16 @ 124} \text{ c/c}$$

No. of bar
32

$$\text{Area of top reinforcement req} = 0 \text{ mm}^2/\text{m}$$

$$\text{Area of top reinforcement pro.} = 524 \text{ mm}^2/\text{m}$$

$$\boxed{T - 10 @ 148} \text{ c/c}$$

No. of bar
27

Check for shear

About x- dir

$$\text{Shear force at face} = 350.62 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.68 \text{ (N/mm}^2\text{)} \text{ O.K}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	517	245.84	0.48	0.31	0.5	1	Not reqd
2d	1034	141.06	0.27	0.31	0.5	0.5	Not reqd

About y- dir

$$\text{Shear force at face} = 350.62 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.7 \text{ (N/mm}^2\text{)} \text{ O.K}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	501	249.08	0.5	0.32	0.428	0.86	Not reqd
2d	1002	147.54	0.29	0.32	0.428	0.43	Not reqd

Check for punching shear

P

$$2637.9 \text{ kN}$$

p

$$164.87 \text{ kN/m}^2$$

Punching surface perimeter

$$4600 \text{ mm}$$

Net P

$$2435.62 \text{ kN}$$

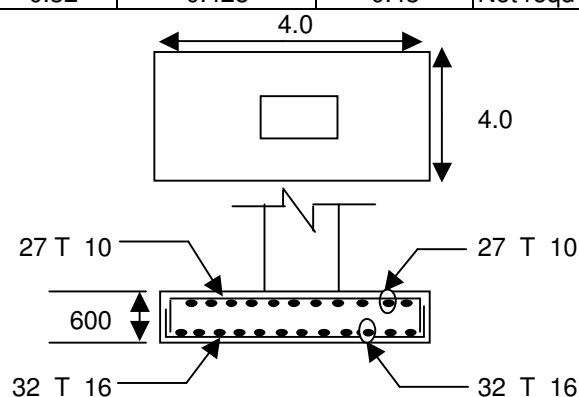
ζ_v

$$1.02 \text{ N/mm}^2$$

$\zeta_{v, \text{design}}$

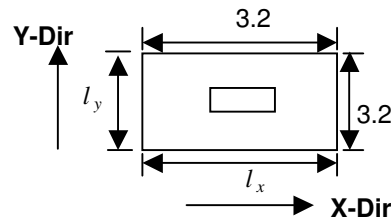
$$5 \text{ N/mm}^2$$

$$> \zeta_v \therefore \text{Safe}$$



Reinforced Concrete Footing Design Based On BS8110

Footing Location on Drg.		C-1	Ref. node on staad	9
Material properties				
Characteristic strength of concrete f_{cu}		40 N/mm ²	Net Bearing capacity of soil	150 kN/m ²
Characteristic strength of steel f_y		460 N/mm ²	at	1.2 m
Concrete density		24 kN/mm ³		Below G.L
Steel density		78 kN/mm ³		
Length of column	c_x	0.55 m		
Width of column	c_y	0.55 m		
Length of footing	l_x	3.2 m		
Width of footing	l_y	3.2 m		
Depth of footing	D	600 mm	Self weight of footing	258.05 kN
Conc. Ten. Cover	d_t	75 mm		
Conc. Comp. Cover	d_c	75 mm		
Tension bar dia.(x- dir)	Φ_x	12 mm		
Tension bar dia.(y- dir)	Φ_y	12 mm		
Comp. bar dia.(x- dir)	Φ_{cx}	12 mm		
Comp. bar dia.(y- dir)	Φ_{cy}	12 mm		
Effect. Depth(x- dir)	d_x	519 mm		
Effect. Depth(y- dir)	d_y	507 mm		
Area of footing	A_B	10.24 m ²		



Reactions

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
101	1120.7	134.6	134.6	200	1628.7	196.9	196.9
102	971.2	120.1	120.1	201	1385.1	173.1	173.1
103	923.4	115.4	115.4	202	1197.6	154.8	154.8
104	855.5	108.7	108.7	203	868.9	122.7	122.7
105	887.0	111.8	111.8	204	1330.5	167.7	167.7
106	809.9	104.3	104.3	205	1133.9	148.5	148.5
107	917.0	114.8	114.8	206	805.1	116.4	116.4
108	847.5	108.0	108.0	207	1375.5	172.1	172.1
109	860.1	109.2	109.2	208	1186.4	153.7	153.7
110	776.3	101.0	101.0	209	857.7	121.6	121.6
111	870.2	110.2	110.2	210	1290.2	163.8	163.8
112	788.9	102.2	102.2	211	1086.9	143.9	143.9
113	849.9	108.2	108.2	212	758.2	111.8	111.8
114	763.5	99.8	99.8	213	1305.3	165.3	165.3
115	655.8	89.2	89.2	214	1104.5	145.7	145.7
116	705.0	94.1	94.1	215	775.8	113.6	113.6
117	938.6	116.9	116.9	216	1274.8	162.3	162.3
118	987.9	121.7	121.7	217	1068.9	142.2	142.2
119	698.0	93.4	93.4	218	740.2	110.1	110.1
120	784.5	101.8	101.8	219	1160.5	151.1	151.1

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
121	859.2	109.1	109.1	220	1234.4	158.4	158.4
122	945.6	117.6	117.6	221	1592.7	193.3	193.3
123	573.6	81.2	81.2	222	1666.5	200.6	200.6
124	622.8	86.0	86.0	223	1225.6	157.5	157.5
125	856.5	108.8	108.8	224	1355.3	170.2	170.2
126	905.7	113.7	113.7	225	1471.8	181.5	181.5
127	615.9	85.3	85.3	226	1601.4	194.2	194.2
128	702.3	93.8	93.8	227	560.6	92.6	92.6
129	777.0	101.1	101.1	228	634.5	99.8	99.8
130	863.4	109.5	109.5	229	992.8	134.8	134.8
131	920.9	115.1	115.1	230	1066.6	142.0	142.0
132	958.9	118.8	118.8	231	625.7	98.9	98.9
133	1133.0	135.8	135.8	232	755.3	111.6	111.6
134	1171.0	139.6	139.6	233	871.9	123.0	123.0
135	952.1	118.2	118.2	234	1001.5	135.6	135.6
136	1018.9	124.7	124.7	235	1106.2	145.8	145.8
137	1073.0	130.0	130.0	236			
138	1139.8	136.5	136.5				
L/C - Load combination cases as per specification					P= Fy + self weight of footing		

P _{max,net}	139.6	kN/m ²	(for SLS)
P _{max,net}	200.6	kN/m ²	(for ULS)

Length of cantilever in x- Dir	1.33 m
Length of cantilever in y- Dir	1.33 m
Moment (x - Dir)	177.38 kN-m/m
Moment (y - Dir)	177.38 kN-m/m

About x- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} \quad 0.016 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.982 d$$

Hence, $z = 0.95 d$

$$A_s = \frac{M}{(0.95 \times f_y \times z)}$$

$$= 823 \text{ mm}^2/\text{m}$$

Min area of steel (0.13 % of bD) = 780 mm²/m

Area of bottom reinforcement req 823 mm²/m

Area of bottom reinforcement pro. 905 mm²/m

T - 12 @ 122 c/c

No. of bar
26

Area of top reinforcement req = 0 mm²/m

Area of top reinforcement pro. = 524 mm²/m

T - 10 @ 145 c/c

No. of bar
22

About y- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} = 0.017 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.981 d$$

Hence, $z = 0.95 d$

$$A_s = \frac{M}{(0.95 \times f_y \times z)}$$

$$= 843 \text{ mm}^2/\text{m}$$

Min area of steel (0.13 % of bD) = 780 mm²/m

Area of bottom reinforcement req 843 mm²/m

Area of bottom reinforcement pro. 905 mm²/m

T - 12 @ 122 c/c

No. of bar
26

Area of top reinforcement req = 0 mm²/m

Area of top reinforcement pro. = 524 mm²/m

T - 10 @ 145 c/c

No. of bar
22

Check for shear

About x- dir

Shear force at face = 266.73 kN/m

Shear stress at face = 0.51 (N/mm²) O.K

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	519	162.65	0.31	0.17	0.41	0.82	Not reqd
2d	1038	58.56	0.11	0.17	0.41	0.41	Not reqd

About y- dir

Shear force at face = 266.73 kN/m

Shear stress at face = 0.53 (N/mm²) O.K

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	507	165.05	0.33	0.18	0.358	0.72	Not reqd
2d	1014	63.37	0.12	0.18	0.358	0.36	Not reqd

Check for punching shear

P

1628.7 kN

p

159.05 kN/m²

Punching surface perimeter

4600 mm

Net P

1433.14 kN

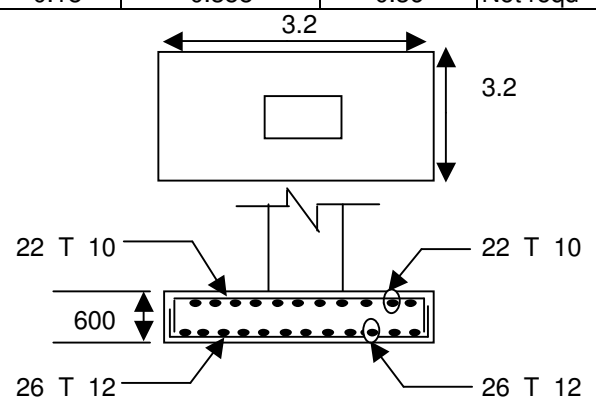
ζ_v

0.6 N/mm²

ζ_{v,design}

5 N/mm²

> ζ_v ∴ Safe



Reinforced Concrete Footing Design Based On BS8110

Footing Location on Drg.

D-1

Ref. node on staad

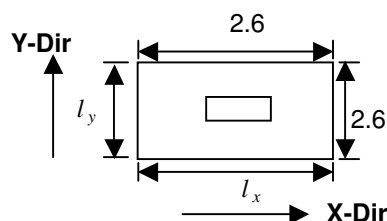
13

Material properties

Characteristic strength of concrete f_{cu}	40 N/mm ²	Net Bearing capacity of soil	150 kN/m ²
Characteristic strength of steel f_y	460 N/mm ²	at	1.2 m
Concrete density	24 kN/mm ³		Below G.L
Steel density	78 kN/mm ³		

Length of column	c_x	0.55 m
Width of column	c_y	0.55 m
Length of footing	l_x	2.6 m
Width of footing	l_y	2.6 m
Depth of footing	D	600 mm
Conc. Ten. Cover	d_t	75 mm
Conc. Comp. Cover	d_c	75 mm
Tension bar dia.(x- dir)	Φ_x	12 mm
Tension bar dia.(y- dir)	Φ_y	12 mm
Comp. bar dia.(x- dir)	Φ_{cx}	10 mm
Comp. bar dia.(y- dir)	Φ_{cy}	10 mm
Effect. Depth(x- dir)	d_x	519 mm
Effect. Depth(y- dir)	d_y	507 mm
Area of footing	A_B	6.76 m ²

Self weight of footing 170.35 kN



Reactions

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
101	631.0	118.5	118.5	201	906.5	171.9	171.9
102	573.0	110.0	110.0	202	737.0	146.8	146.8
103	491.3	97.9	97.9	203	697.5	141.0	141.0
104	498.2	98.9	98.9	204	491.5	110.5	110.5
105	478.4	96.0	96.0	205	717.7	144.0	144.0
106	482.2	96.5	96.5	206	675.0	137.7	137.7
107	514.0	101.2	101.2	207	469.0	107.2	107.2
108	526.5	103.1	103.1	208	770.9	151.8	151.8
109	487.6	97.3	97.3	209	737.2	146.9	146.9
110	493.6	98.2	98.2	210	531.1	116.4	116.4
111	518.9	102.0	102.0	211	731.4	146.0	146.0
112	532.8	104.0	104.0	212	691.1	140.0	140.0
113	505.2	99.9	99.9	213	485.0	109.6	109.6
114	515.7	101.5	101.5	214	778.4	153.0	153.0
115	450.9	91.9	91.9	215	745.9	148.1	148.1
116	354.6	77.7	77.7	216	539.9	117.7	117.7
117	675.5	125.1	125.1	217	757.9	149.9	149.9
118	579.2	110.9	110.9	218	721.9	144.6	144.6
119	638.3	119.6	119.6	219	515.9	114.1	114.1
120	706.9	129.8	129.8	220	708.1	142.5	142.5

L/C	Serviceability			L/C	Design		
	Fy(kN)	P/A	p		Fy(kN)	P/A	p
121	323.2	73.0	73.0	221	563.6	121.2	121.2
122	391.8	83.2	83.2	222	1051.1	193.3	193.3
123	399.4	84.3	84.3	223	906.6	171.9	171.9
124	303.1	70.0	70.0	224	996.7	185.2	185.2
125	624.0	117.5	117.5	225	1099.6	200.5	200.5
126	527.7	103.3	103.3	226	515.2	114.0	114.0
127	586.8	112.0	112.0	227	618.1	129.2	129.2
128	655.4	122.2	122.2	228	410.6	98.5	98.5
129	271.7	65.4	65.4	229	266.2	77.2	77.2
130	340.3	75.5	75.5	230	753.6	149.3	149.3
131	555.0	107.3	107.3	231	609.2	127.9	127.9
132	480.6	96.3	96.3	232	699.2	141.2	141.2
133	723.4	132.2	132.2	233	802.1	156.5	156.5
134	649.0	121.2	121.2	234	217.7	70.0	70.0
135	693.7	127.8	127.8	235	320.6	85.2	85.2
136	746.7	135.7	135.7	236	619.6	129.5	129.5
137	457.3	92.9	92.9				
138	510.3	100.7	100.7				
L/C - Load combination cases as per specification					P= Fy + self weight of footing		

$P_{max,net}$	135.7	kN/m ²	(for SLS)
$P_{max,net}$	200.5	kN/m ²	(for ULS)

Length of cantilever in x- Dir	1.03 m
Length of cantilever in y- Dir	1.03 m
Moment (x - Dir)	106.33 kN-m/m
Moment (y - Dir)	106.33 kN-m/m

About x- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} \quad 0.01 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.989 d$$

Hence, $z = 0.95 d$

$$A_s = \frac{M}{(0.95 \times f_y \times z)}$$

$$= 493 \text{ mm}^2/\text{m}$$

$$\text{Min area of steel (0.13 \% of bD)} = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement req} = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement pro.} = 905 \text{ mm}^2/\text{m}$$

$$\boxed{T - 12 @ 122} \text{ c/c}$$

No. of bar
21

$$\text{Area of top reinforcement req} = 0 \text{ mm}^2/\text{m}$$

$$\text{Area of top reinforcement pro.} = 524 \text{ mm}^2/\text{m}$$

$$\boxed{T - 10 @ 144} \text{ c/c}$$

No. of bar
18

About y- dir

$$K = \frac{M}{(b \times d^2 \times f_{cu})} = 0.01 < K' = 0.156$$

since $K < K'$ compression reinforcement not required

$$z = d \{0.5 + \sqrt{0.25 - K/0.9}\} \leq 0.95d$$

$$= 0.989 d$$

Hence, $z = 0.95 d$

$$A_s = \frac{M}{(0.95 \times f_y \times z)} = 505 \text{ mm}^2/\text{m}$$

$$\text{Min area of steel (0.13 \% of } bD) = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement req} = 780 \text{ mm}^2/\text{m}$$

$$\text{Area of bottom reinforcement pro.} = 905 \text{ mm}^2/\text{m}$$

$$\boxed{T - 12 @ 122} \text{ c/c}$$

No. of bar
21

$$\text{Area of top reinforcement req} = 0 \text{ mm}^2/\text{m}$$

$$\text{Area of top reinforcement pro.} = 524 \text{ mm}^2/\text{m}$$

$$\boxed{T - 10 @ 144} \text{ c/c}$$

No. of bar
18

Check for shear

About x- dir

$$\text{Shear force at face} = 206.47 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.4 \text{ (N/mm}^2\text{)} \text{ O.K.}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	519	102.44	0.2	0.17	0.41	0.82	Not reqd
2d	1038	0	0	0.17	0.41	0.41	Not reqd

About y- dir

$$\text{Shear force at face} = 206.47 \text{ kN/m}$$

$$\text{Shear stress at face} = 0.41 \text{ (N/mm}^2\text{)} \text{ O.K.}$$

Distance from support (mm)	av (mm)	V (kN)	v (N/mm ²)	100As/ (bv.d)	v _c (N/mm ²)	Enhanced v _c (N/mm ²)	Remarks
d	507	104.84	0.21	0.18	0.358	0.72	Not reqd
2d	1014	3.21	0.01	0.18	0.358	0.36	Not reqd

Check for punching shear

P

906.5 kN

p

134.1 kN/m²

Punching surface perimeter

4600 mm

Net P

741.66 kN

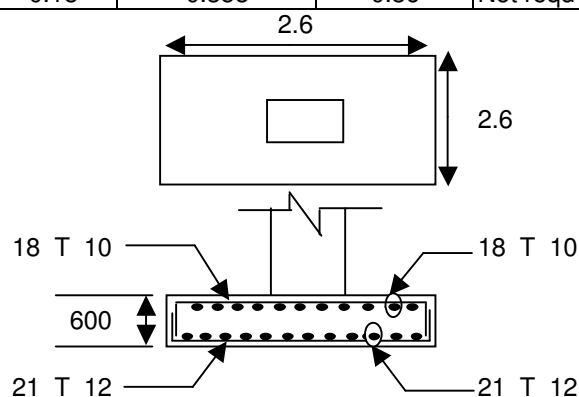
ζ_v

0.31 N/mm²

ζ_{v,design}

5 N/mm²

> ζ_v ∴ Safe



6 Slab Design

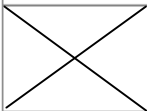
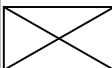
Slab Design Summary

(A) Ground Slab Layout Plan

Level	Slab Id no	Length l_x (M)	Width l_y (M)	Thickness of Slab (mm)	Bottom reinforcement				Top reinforcement			
					Parallel to Length (x dir.)		Parallel to Width (y dir.)		Parallel to Length (x dir.)		Parallel to Width (y dir.)	
					Req.	Provided	Bar dia + Spacing (mm)(c/c)	Provided	Req.	Provided	Bar dia + Spacing (mm)(c/c)	Provided
Level - 01	S1	4.70	5.31	225	T12 @ 265	T12 @ 200	T10 @ 221	T10 @ 200	T12 @ 198	T12 @ 150	T10 @ 167	T10 @ 125
Level - 01	S2	4.63	5.31	225	T12 @ 300	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 259	T12 @ 150	T10 @ 210	T10 @ 150
Level - 01	S3	1.50	6.50	225	T12 @ 386	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 386	T12 @ 150	T10 @ 269	T10 @ 150
Level - 01	S4	1.65	6.50	225	T12 @ 386	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 386	T12 @ 150	T10 @ 269	T10 @ 150
Level - 01	S5	1.49	2.33	225	T12 @ 387	T12 @ 150	T10 @ 269	T10 @ 200	T12 @ 387	T12 @ 125	T12 @ 387	T10 @ 150
Level - 01	S6	1.38	2.16	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 387	T12 @ 150	T12 @ 387	T10 @ 150
Level - 01	S7	2.16	2.65	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 387	T12 @ 150	T10 @ 269	T10 @ 150
Level - 01	S8	2.16	3.85	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 387	T12 @ 150	T10 @ 269	T10 @ 150
Level - 01	S9	6.50	6.65	225	T12 @ 192	T12 @ 150	T12 @ 201	T12 @ 150	T12 @ 144	T12 @ 125	T12 @ 151	T12 @ 125
Level - 01	S10	6.00	6.65	225	T12 @ 250	T12 @ 200	T12 @ 275	T12 @ 200	T12 @ 189	T12 @ 125	T12 @ 207	T12 @ 150
Level - 01	S11	2.65	6.65	225	T12 @ 386	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 386	T12 @ 125	T12 @ 386	T12 @ 150
Level - 01	S12	3.85	6.65	225	T12 @ 366	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 277	T12 @ 125	T12 @ 387	T12 @ 150
Level - 01	S13	2.60	5.04	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 387	T12 @ 150	T10 @ 269	T10 @ 125
Level - 01	S14	3.90	5.04	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 387	T12 @ 150	T10 @ 269	T10 @ 125
Level - 01	S15	4.30	5.04	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 384	T12 @ 150	T10 @ 269	T10 @ 150
Level - 01	S16	1.70	5.04	225	T12 @ 387	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 386	T12 @ 150	T10 @ 269	T10 @ 150
Level - 01	S17	4.68	5.04	225	T12 @ 375	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 282	T12 @ 150	T10 @ 221	T10 @ 150
Level - 01	S18	1.83	5.04	225	T12 @ 386	T12 @ 200	T10 @ 269	T10 @ 200	T12 @ 386	T12 @ 150	T10 @ 269	T10 @ 150

(B) Rc Roof Layout plan

Roof	RS-1	5.31	6.50	175	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 200	T10 @ 263	T10 @ 150	T10 @ 345	T10 @ 150
Roof	RS-2	5.31	6.00	175	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 200	T10 @ 331	T10 @ 150	T10 @ 345	T10 @ 150
Roof	RS-3	6.50	6.65	175	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 200	T10 @ 273	T10 @ 150	T10 @ 286	T10 @ 150
Roof	RS-4	6.00	6.65	175	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 150	T10 @ 345	T10 @ 150
Roof	RS-5	5.04	6.50	175	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 200	T10 @ 275	T10 @ 150	T10 @ 345	T10 @ 150
Roof	RS-6	5.04	6.00	175	T10 @ 345	T10 @ 200	T10 @ 345	T10 @ 200	T10 @ 340	T10 @ 150	T10 @ 345	T10 @ 150

	S-1	S-2	S-3	 	S-4
			S-3		
S-5	S-9	S-10	S-6	S-7	S-8
S-11			S-12		
S-13	S-14	S-15	S-16	S-17	 S-18

Ground Slab Layout Plan

Design of Slab Panel - S1

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

4.7

Long span of the slab, l_y

5.31

Span ratio, l_y/l_x

1.13

TWO WAY SLAB

Edge Condition

Two Adjacent Edges Discont.



Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	15 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	33.94 kN/m²

Deflection check	
f_s , N/mm ²	231
M.F.	1.52
Span/ d_{eff}	23
d_{eff} Req'd	134
$D_{req'd}$, mm	175
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	43.55	184	0.04	174.80	12	570	198	150	755
Short span at bottom	32.61	184	0.03	174.80	12	427	265	200	566
Long span at top	33.74	173	0.03	164.35	10	470	167	125	629
Long span at bottom	25.50	173	0.03	164.35	10	355	221	200	393

End Condition : Two adjacent edges discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.058	43.55 kNm
Along short span at bottom	β_{sx} (+)	0.043	32.61 kNm
Along long span at top	β_{sy} (-)	0.045	33.74 kNm
Along long span at bottom	β_{sy} (+)	0.034	25.50 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.04 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	570 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	198 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	427 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	265 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	470 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	167 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	628 mm²	
Provided % of steel	=	0.37 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	355 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 222 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.23 % **Provided area of steel is OK**

Provide 10T @ 125 c/c at top (Negative Reinf.)

Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 4.7 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.410 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 232 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 102 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.449	71.61	0.39	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.296	47.21	0.26	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.400	63.81	0.37	0.37	0.63	1.25	0	No. shear reinf.

Design of Slab Panel - S2

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x	4.625
Long span of the slab, l_y	5.31
Span ratio, l_y/l_x	1.15

TWO WAY SLAB
Edge Condition
Loading

One Short Edge Discontinuous

Deflection check	
f_s , N/mm ²	177
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Req'd	89
$D_{req'd}$, mm	130
$D_{prov.}$ mm	225
Safe in deflection	

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	15 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	33.94 kN/m²

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	33.35	184	0.03	174.80	12	437	259	150	755
Short span at bottom	25.01	184	0.02	174.80	12	327	300	200	566
Long span at top	26.87	173	0.03	164.35	10	374	210	150	524
Long span at bottom	20.33	173	0.02	164.35	10	293	269	200	393

End Condition : One short edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.046	33.35 kNm
Along short span at bottom	β_{sx} (+)	0.034	25.01 kNm
Along long span at top	β_{sy} (-)	0.037	26.87 kNm
Along long span at bottom	β_{sy} (+)	0.028	20.33 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	437 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	259 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	327 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	346 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	374 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	210 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.31 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	283 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 278 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.23 % **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 4.625 m
Actul l/d ratio = 26
Provided % of steel at bottom along short span = 0.410 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 178 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 89 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.404	63.48	0.35	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.360	56.51	0.33	0.31	0.59	1.18	0	No. shear reinf.

Design of Slab Panel - S3

Material Properties

Grade of the concrete, f_{cu}	=	40	N/mm ²
Grade of the steel, f_y	=	460	N/mm ²
Concrete Density, γ_c	=	24	kN/m ³

Data

Thickness of slab, D	=	225	mm
Width of slab, b	=	1000	mm
Bottom cover	=	35	mm
Short span, l_x	=	1.5	m
Long span, l_y	=	6.5	m
Span ratio, l_y/l_x	=	4.33	Oneway

Edge condition

Both ends continuous ▼

Loading

Self weight of the slab	=	5.4	kN/m ²
Floor finish	=	1.2	kN/m ²
Partitions	=	0	kN/m ²
M&E services	=	0.5	kN/m ²
Total Dead load, Gk	=	7.1	kN/m ²
Total Live load, Qk	=	10	kN/m ²
Design load on slab, $n = 1.4*G_k + 1.6*Q_k$	=	25.94	kN/m ²

Deflection check

f_s , N/mm ²	159
M.F.	2.00
Span/ d_{eff}	26
$d_{eff, reqd}$	29
D_{reqd} , mm	70
$D_{provided}$, mm	225
Safe in deflection	

(-ve) moment at support, $M_t = n * l^2/10$	=	5.84	kN.m
(+ve) moment at span, $M_b = n * l^2/10$	=	5.84	kN.m
Minimum Area of steel (0.13% of Gross c/s area)	=	293	mm ²

Location	K = $M/f_{cu}bd$	d_{eff} , mm	Dia of ϕ	Required		Provided	
				As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	0.01	184	12	293	386	150	755
Short span at bottom	0.01	184	12	293	386	200	566
Distribution, 0.13*b*D			10	293	269	200	393

Area of steel at support

Available effective depth, d_{eff}	=	184	mm
At support, $M_t / f_{cu}bd^2$ (K)	=	0.01	
Z	=	174.8	mm
Required area of the steel, A_s	=	76	mm ²
Minimum Area of steel	=	293	mm ²
Diameter of bar	=	12	mm
Required spacing of reinf.	=	386	mm
Provided spacing of reinf.	=	150	mm
Provided area of the steel	=	754	mm ²
Provided % of steel	=	0.41	%

OK

Provide 12T @ 150 c/c at top (Negative Reinf.)

Area of steel at span

Available effective depth, d_{eff}	=	184 mm	
At support, $M_b / f_{cu} b d^2$	=	0.01	
Z	=	174.80	
Required area of the steel, A_s	=	76 mm ²	
Minimum Area of steel	=	293 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	386 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	566 mm ²	
Provided % of steel	=	0.31 %	OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Distribution steel, 0.13% of Gross C/A	=	293 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	269 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.20 %	OK
Provide 10T @ 200 c/c			

Check for Deflection

BS8110 -Section 3.4.6.5, Table 3.10

Length of shorter span	=	1.5 m	
Actual l/d ratio	=	26	
Provided % of steel at bottom along short span	=	0.310 %	
$f_s = (2 \cdot f_y \cdot \text{Req. steel}) / (3 \cdot \text{Provided steel})$	=	41 N/mm ²	
M.F. for tension reinforcement,	=	2.00	
Required effective depth	=	29 mm	
Available effective depth	=	184 mm	Safe in Deflection

Check for shear

Shear force, V_{sx}	=	19.455 kN
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Distance from support, m	av (m)	Vsx (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_{cs} (N/mm ²)	Asv/sv	Remarks
d	0.184	14.68	0.08	0.67	1.34	0.00	No. shear reinf.
2d	0.368	9.91	0.05	0.67	0.67	0.00	No. shear reinf.
3d	0.552	5.14	0.03	0.67	0.67	0.00	No. shear reinf.

Design of Slab Panel - S4

Material Properties

Grade of the concrete, f_{cu}	=	40	N/mm ²
Grade of the steel, f_y	=	460	N/mm ²
Concrete Density, γ_c	=	24	kN/m ³

Data

Thickness of slab, D	=	225	mm
Width of slab, b	=	1000	mm
Bottom cover	=	35	mm
Short span, l_x	=	1.65	m
Long span, l_y	=	6.5	m
Span ratio, l_y/l_x	=	3.94	Oneway

Edge condition

One end continuous ▼

Loading

Self weight of the slab	=	5.4	kN/m ²
Floor finish	=	1.2	kN/m ²
Partitions	=	0	kN/m ²
M&E services	=	0.5	kN/m ²
Total Dead load, Gk	=	7.1	kN/m ²
Total Live load, Qk	=	10	kN/m ²
Design load on slab, $n = 1.4*G_k + 1.6*Q_k$	=	25.94	kN/m ²

Deflection check

f_s , N/mm ²	159
M.F.	2.00
Span/ d_{eff}	23
$d_{eff, reqd}$	36
D_{reqd} , mm	77
$D_{provided}$, mm	225
Safe in deflection	

(-ve) moment at support, $M_t = n * l^2/10$	=	7.06	kN.m
(+ve) moment at span, $M_b = n * l^2/10$	=	7.06	kN.m
Minimum Area of steel (0.13% of Gross c/s area)	=	293	mm ²

Location	K = $M/f_{cu}bd$	d_{eff} , mm	Dia of ϕ	Required		Provided	
				As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	0.01	184	12	293	386	150	755
Short span at bottom	0.01	184	12	293	386	200	566
Distribution, 0.13*b*D			10	293	269	200	393

Area of steel at support

Available effective depth, d_{eff}	=	184	mm
At support, $M_t / f_{cu}bd^2$ (K)	=	0.01	
Z	=	174.8	mm
Required area of the steel, A_s	=	92	mm ²
Minimum Area of steel	=	293	mm ²
Diameter of bar	=	12	mm
Required spacing of reinf.	=	386	mm
Provided spacing of reinf.	=	150	mm
Provided area of the steel	=	754	mm ²
Provided % of steel	=	0.41	%

OK

Provide 12T @ 150 c/c at top (Negative Reinf.)

Area of steel at span

Available effective depth, d_{eff}	=	184 mm	
At support, $M_b / f_{cu} b d^2$	=	0.01	
Z	=	174.80	
Required area of the steel, A_s	=	92 mm ²	
Minimum Area of steel	=	293 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	386 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	566 mm ²	
Provided % of steel	=	0.31 %	OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Distribution steel, 0.13% of Gross C/A	=	293 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	269 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.20 %	OK
Provide 10T @ 200 c/c			

Check for Deflection

BS8110 -Section 3.4.6.5, Table 3.10

Length of shorter span	=	1.65 m	
Actual l/d ratio	=	23	
Provided % of steel at bottom along short span	=	0.310 %	
$f_s = (2 \cdot f_y \cdot \text{Req. steel}) / (3 \cdot \text{Provided steel})$	=	50 N/mm ²	
M.F. for tension reinforcement,	=	2.00	
Required effective depth	=	36 mm	
Available effective depth	=	184 mm	Safe in Deflection

Check for shear

Shear force, V_{sx}	=	21.4005 kN
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Distance from support, m	a_v (m)	V_{sx} (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_{cs} (N/mm ²)	A_{sv}/s_v	Remarks
d	0.184	16.63	0.09	0.67	1.34	0.00	No. shear reinf.
2d	0.368	11.85	0.06	0.67	0.67	0.00	No. shear reinf.
3d	0.552	7.08	0.04	0.67	0.67	0.00	No. shear reinf.

Design of Slab Panel - S5

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

1.485

Long span of the slab, l_y

2.325

Span ratio, l_y/l_x

1.57

TWO WAY SLAB

Edge Condition

Two Adjacent Edges Discont.

Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check

f_s , N/mm ²	119
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Req'd	33
$D_{req'd}$, mm	74
$D_{prov.}$, mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	4.60	184	0.01	174.80	12	293	387	125	905
Short span at bottom	3.47	184	0.01	174.80	12	293	387	150	755
Long span at top	2.58	172	0.01	163.40	12	293	387	150	755
Long span at bottom	1.95	173	0.01	164.35	10	293	269	200	393

End Condition : Two adjacent edges discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.080	4.60 kNm
Along short span at bottom	β_{sx} (+)	0.061	3.47 kNm
Along long span at top	β_{sy} (-)	0.045	2.58 kNm
Along long span at bottom	β_{sy} (+)	0.034	1.95 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	60 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	1878 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	905 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 125 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	45 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	2490 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 150 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	172 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	163.40 mm	
Required area of the steel, A_s	=	36 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	3130 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm²	
Provided % of steel	=	0.44 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	163.40 mm	
Required area of the steel, A_s	=	27 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 2876 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.23 % **Provided area of steel is OK**

Provide 12T @ 150 c/c at top (Negative Reinf.)

Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 1.485 m
Actual l/d ratio = 23
Provided % of steel at bottom along short span = 0.500 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 18 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 32 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.548	21.10	0.12	0.50	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.358	13.79	0.08	0.50	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.400	15.41	0.09	0.44	0.67	1.33	0	No. shear reinf.

Design of Slab Panel - S6

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x	1.375
Long span of the slab, l_y	2.16
Span ratio, l_y/l_x	1.57

TWO WAY SLAB
Edge Condition
Loading

Interior Panel

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check	
f_s , N/mm ²	158
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Reqd	27
D_{reqd} , mm	68
D_{prov} , mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	2.69	184	0.01	174.80	12	293	387	150	755
Short span at bottom	2.02	184	0.01	174.80	12	293	387	200	566
Long span at top	1.57	172	0.01	163.40	12	293	387	150	755
Long span at bottom	1.18	173	0.01	164.35	10	293	269	200	393

End Condition : Interior panel

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.055	2.69 kNm
Along short span at bottom	β_{sx} (+)	0.041	2.02 kNm
Along long span at top	β_{sy} (-)	0.032	1.57 kNm
Along long span at bottom	β_{sy} (+)	0.024	1.18 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	35 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	3212 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	26 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	4277 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	172 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	163.40 mm	
Required area of the steel, A_s	=	22 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	5144 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.44 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ²	
Lever arm z	=	163.40 mm	
Required area of the steel, A_s	=	17 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 4753 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.23 % **Provided area of steel is OK**

Provide 12T @ 150 c/c at top (Negative Reinf.)

Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 1.375 m
Actul l/d ratio = 26
Provided % of steel at bottom along short span = 0.410 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 14 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 26 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.459	16.35	0.09	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.330	11.77	0.07	0.44	0.67	1.33	0	No. shear reinf.

Design of Slab Panel - S7

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

2.16

Long span of the slab, l_y

2.65

Span ratio, l_y/l_x

1.23

TWO WAY SLAB

Edge Condition

Interior Panel

Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check	
f_s , N/mm ²	158
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Req'd	42
$D_{req'd}$, mm	83
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	5.22	184	0.01	174.80	12	293	387	150	755
Short span at bottom	3.98	184	0.01	174.80	12	293	387	200	566
Long span at top	3.88	173	0.01	164.35	10	293	269	150	524
Long span at bottom	2.91	173	0.01	164.35	10	293	269	200	393

End Condition : Interior panel

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.043	5.22 kNm
Along short span at bottom	β_{sx} (+)	0.033	3.98 kNm
Along long span at top	β_{sy} (-)	0.032	3.88 kNm
Along long span at bottom	β_{sy} (+)	0.024	2.91 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	68 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	1655 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	52 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	2171 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	54 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	1454 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.31 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	41 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **1939 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.23 %** **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **2.16 m**
Actul l/d ratio = **26**
Provided % of steel at bottom along short span = **0.410 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **28 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **42 mm**
Available effective depth = **184 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.395	22.15	0.13	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.330	18.49	0.11	0.31	0.59	1.18	0	No. shear reinf.

Design of Slab Panel - S8

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

2.16

Long span of the slab, l_y

3.85

Span ratio, l_y/l_x

1.78

TWO WAY SLAB

Edge Condition

One Short Edge Discontinuous

Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check

f_s , N/mm ²	158
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Req'd	42
$D_{req'd}$, mm	83
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	7.69	184	0.01	174.80	12	293	387	150	755
Short span at bottom	5.74	184	0.01	174.80	12	293	387	200	566
Long span at top	4.48	173	0.01	164.35	10	293	269	150	524
Long span at bottom	3.39	173	0.01	164.35	10	293	269	200	393

End Condition : One short edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.064	7.69 kNm
Along short span at bottom	β_{sx} (+)	0.047	5.74 kNm
Along long span at top	β_{sy} (-)	0.037	4.48 kNm
Along long span at bottom	β_{sy} (+)	0.028	3.39 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	101 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	1123 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	75 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	1506 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	62 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	1259 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.31 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	47 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **1664 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.23 %** **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **2.16 m**
Actul l/d ratio = **26**
Provided % of steel at bottom along short span = **0.410 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **41 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **42 mm**
Available effective depth = **184 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.503	28.16	0.16	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.360	20.17	0.12	0.31	0.59	1.18	0	No. shear reinf.

Design of Slab Panel - S9

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x	6.5
Long span of the slab, l_y	6.65
Span ratio, l_y/l_x	1.02

TWO WAY SLAB

Edge Condition	One Short Edge Discontinuous
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Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	15 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	33.94 kN/m²

Deflection check	
f_s , N/mm ²	228
M.F.	1.38
Span/ d_{eff}	26
d_{eff} Req'd	182
$D_{req'd}$, mm	223
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	57.58	184	0.05	173.14	12	761	149	125	905
Short span at bottom	42.91	184	0.04	174.80	12	562	201	150	755
Long span at top	53.06	172	0.05	161.84	12	750	151	125	905
Long span at bottom	40.16	172	0.04	163.40	12	562	201	150	755

End Condition : One short edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.040	57.58 kNm
Along short span at bottom	β_{sx} (+)	0.030	42.91 kNm
Along long span at top	β_{sy} (-)	0.037	53.06 kNm
Along long span at bottom	β_{sy} (+)	0.028	40.16 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b * d^2 * f_{cu}$	=	0.05 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	173.14 mm	
Required area of the steel, A_s	=	761 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	149 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	905 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 125 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b * d^2 * f_{cu}$	=	0.04 N/mm ²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	562 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	202 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 150 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	172 mm	
$K = m_{sy} (Top) / b * d^2 * f_{cu}$	=	0.05 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	161.84 mm	
Required area of the steel, A_s	=	750 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	151 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	905 mm ²	
Provided % of steel	=	0.53 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b * d^2 * f_{cu}$	=	0.04 N/mm ²	
Lever arm z	=	163.40 mm	
Required area of the steel, A_s	=	562 mm ²	
Diameter of bar	=	12 mm	

Required spacing of reinf. = 202 mm
Provided spacing of reinf. = 150 mm
Provided area of the steel = 754 mm²
Provided % of steel = 0.44 % **Provided area of steel is OK**
Provide 12T @ 125 c/c at top (Negative Reinf.)
Provide 12T @ 150 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 6.5 m
Actul l/d ratio = 26
Provided % of steel at bottom along short span = 0.500 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 228 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 125 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.367	80.95	0.44	0.50	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.50	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.360	79.42	0.46	0.53	0.71	1.41	0	No. shear reinf.

Design of Slab Panel - S10

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

6

Long span of the slab, l_y

6.65

Span ratio, l_y/l_x

1.11

TWO WAY SLAB

Edge Condition

Interior Panel

Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	15 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	33.94 kN/m²

Deflection check

f_s , N/mm ²	246
M.F.	1.47
Span/ d_{eff}	26
d_{eff} Req'd	158
$D_{req'd}$, mm	199
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	45.72	184	0.04	174.80	12	599	189	125	905
Short span at bottom	34.62	184	0.03	174.80	12	453	250	200	566
Long span at top	39.10	172	0.04	163.40	12	548	207	150	755
Long span at bottom	29.33	172	0.03	163.40	12	411	275	200	566

End Condition : Interior panel

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.037	45.72 kNm
Along short span at bottom	β_{sx} (+)	0.028	34.62 kNm
Along long span at top	β_{sy} (-)	0.032	39.10 kNm
Along long span at bottom	β_{sy} (+)	0.024	29.33 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b * d^2 * f_{cu}$	=	0.04 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	599 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	189 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	905 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 125 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b * d^2 * f_{cu}$	=	0.03 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	453 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	250 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	172 mm	
$K = m_{sy} (Top) / b * d^2 * f_{cu}$	=	0.04 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	163.40 mm	
Required area of the steel, A_s	=	548 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	207 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.44 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b * d^2 * f_{cu}$	=	0.03 N/mm²	
Lever arm z	=	163.40 mm	
Required area of the steel, A_s	=	411 mm ²	
Diameter of bar	=	12 mm	

Required spacing of reinf. = 276 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 566 mm²
Provided % of steel = 0.33 % **Provided area of steel is OK**

Provide 12T @ 150 c/c at top (Negative Reinf.)

Provide 12T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 6 m
Actul l/d ratio = 26
Provided % of steel at bottom along short span = 0.500 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 246 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 115 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.363	73.82	0.41	0.50	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.50	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.330	67.20	0.39	0.44	0.67	1.33	0	No. shear reinf.

Area of steel at span

Available effective depth, d_{eff}	=	184 mm	
At support, $M_b / f_{cu} b d^2$	=	0.02	
Z	=	174.80	
Required area of the steel, A_s	=	260 mm ²	
Minimum Area of steel	=	293 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	386 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	566 mm ²	
Provided % of steel	=	0.31 %	OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Distribution steel, 0.13% of Gross C/A	=	293 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	269 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.20 %	OK
Provide 10T @ 200 c/c			

Check for Deflection

BS8110 -Section 3.4.6.5, Table 3.10

Length of shorter span	=	2.65 m	
Actual l/d ratio	=	26	
Provided % of steel at bottom along short span	=	0.310 %	
$f_s = (2 \cdot f_y \cdot \text{Req. steel}) / (3 \cdot \text{Provided steel})$	=	141 N/mm ²	
M.F. for tension reinforcement,	=	2.00	
Required effective depth	=	51 mm	
Available effective depth	=	184 mm	Safe in Deflection

Check for shear

Shear force, V_{sx}	=	44.9705 kN
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Distance from support, m	av (m)	Vsx (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_{cs} N/mm ²	Asv/sv	Remarks
d	0.184	38.73	0.21	0.71	1.43	0.00	No. shear reinf.
2d	0.368	32.48	0.18	0.71	0.71	0.00	No. shear reinf.
3d	0.552	26.24	0.14	0.71	0.71	0.00	No. shear reinf.

Design of Slab Panel - S12

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

3.85

Long span of the slab, l_y

6.65

Span ratio, l_y/l_x

1.73

TWO WAY SLAB
Edge Condition
Loading

One long Edge Distcont.



Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check	
f_s , N/mm ²	167
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Reqd	84
D_{reqd} , mm	125
D_{prov} , mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	31.22	184	0.03	174.80	12	409	277	125	905
Short span at bottom	23.60	184	0.02	174.80	12	309	366	200	566
Long span at top	14.23	172	0.02	163.40	12	293	387	150	755
Long span at bottom	10.77	173	0.01	164.35	10	293	269	200	393

End Condition : One long edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.081	31.22 kNm
Along short span at bottom	β_{sx} (+)	0.061	23.60 kNm
Along long span at top	β_{sy} (-)	0.037	14.23 kNm
Along long span at bottom	β_{sy} (+)	0.028	10.77 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	409 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	277 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	905 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 125 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	309 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	367 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.50 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	172 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	163.40 mm	
Required area of the steel, A_s	=	199 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	568 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm²	
Provided % of steel	=	0.44 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	163.40 mm	
Required area of the steel, A_s	=	151 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 521 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.23 % **Provided area of steel is OK**
Provide 12T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 3.85 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.500 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 168 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 84 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.546	54.56	0.3	0.50	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.358	35.77	0.2	0.50	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.360	35.95	0.21	0.44	0.67	1.33	0	No. shear reinf.

Design of Slab Panel - S11

Material Properties

Grade of the concrete, f_{cu}	=	40	N/mm ²
Grade of the steel, f_y	=	460	N/mm ²
Concrete Density, γ_c	=	24	kN/m ³

Data

Thickness of slab, D	=	225	mm
Width of slab, b	=	1000	mm
Bottom cover	=	35	mm
Short span, l_x	=	2.65	m
Long span, l_y	=	6.65	m
Span ratio, l_y/l_x	=	2.51	Oneway

Edge condition

Both ends continuous ▼

Loading

Self weight of the slab	=	5.4	kN/m ²
Floor finish	=	1.2	kN/m ²
Partitions	=	0	kN/m ²
M&E services	=	0.5	kN/m ²
Total Dead load, Gk	=	7.1	kN/m ²
Total Live load, Qk	=	15	kN/m ²
Design load on slab, $n = 1.4*G_k + 1.6*Q_k$	=	33.94	kN/m ²

Deflection check

f_s , N/mm ²	159
M.F.	2.00
Span/ d_{eff}	26
$d_{eff, reqd}$	51
D_{reqd} , mm	92
$D_{provided}$, mm	225
Safe in deflection	

(-ve) moment at support, $M_t = n * l^2/12$	=	19.86	kN.m
(+ve) moment at span, $M_b = n * l^2/12$	=	19.86	kN.m
Minimum Area of steel (0.13% of Gross c/s area)	=	293	mm ²

Location	K = $M/f_{cu}bd$	d_{eff} , mm	Dia of ϕ	Required		Provided	
				As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	0.02	184	12	293	386	125	905
Short span at bottom	0.01	184	12	293	386	200	566
Distribution, 0.13*b*D			10	293	269	200	393

Area of steel at support

Available effective depth, d_{eff}	=	184	mm
At support, $M_t / f_{cu}bd^2$ (K)	=	0.02	
Z	=	174.8	mm
Required area of the steel, A_s	=	260	mm ²
Minimum Area of steel	=	293	mm ²
Diameter of bar	=	12	mm
Required spacing of reinf.	=	386	mm
Provided spacing of reinf.	=	125	mm
Provided area of the steel	=	905	mm ²
Provided % of steel	=	0.50	%

OK

Provide 12T @ 125 c/c at top (Negative Reinf.)

Design of Slab Panel - S13

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x	2.6
Long span of the slab, l_y	5.04
Span ratio, l_y/l_x	1.94

TWO WAY SLAB
Edge Condition
Loading

Two Adjacent Edges Discont.

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check	
f_s , N/mm ²	158
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Req'd	57
$D_{req'd}$, mm	98
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	16.05	184	0.02	174.80	12	293	387	150	755
Short span at bottom	12.06	184	0.01	174.80	12	293	387	200	566
Long span at top	7.90	173	0.01	164.35	10	293	269	125	629
Long span at bottom	5.97	173	0.01	164.35	10	293	269	200	393

End Condition : Two adjacent edges discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.092	16.05 kNm
Along short span at bottom	β_{sx} (+)	0.069	12.06 kNm
Along long span at top	β_{sy} (-)	0.045	7.90 kNm
Along long span at bottom	β_{sy} (+)	0.034	5.97 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	210 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	538 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	158 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	717 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	110 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	714 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	628 mm²	
Provided % of steel	=	0.37 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	83 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **945 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.23 %** **Provided area of steel is OK**

Provide 10T @ 125 c/c at top (Negative Reinf.)

Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **2.6 m**
Actual l/d ratio = **23**
Provided % of steel at bottom along short span = **0.410 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **86 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **57 mm**
Available effective depth = **184 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.593	39.97	0.22	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.395	26.65	0.15	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.400	26.98	0.16	0.37	0.63	1.25	0	No. shear reinf.

Design of Slab Panel - S14

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

3.9

Long span of the slab, l_y

5.04

Span ratio, l_y/l_x

1.29

TWO WAY SLAB

Edge Condition

One Short Edge Discontinuous

Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check

f_s , N/mm ²	158
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Req'd	75
$D_{req'd}$, mm	116
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	20.40	184	0.02	174.80	12	293	387	150	755
Short span at bottom	15.30	184	0.02	174.80	12	293	387	200	566
Long span at top	14.60	173	0.02	164.35	10	293	269	125	629
Long span at bottom	11.05	173	0.01	164.35	10	293	269	200	393

End Condition : One short edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.052	20.40 kNm
Along short span at bottom	β_{sx} (+)	0.039	15.30 kNm
Along long span at top	β_{sy} (-)	0.037	14.60 kNm
Along long span at bottom	β_{sy} (+)	0.028	11.05 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	267 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	423 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	200 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	565 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	203 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	386 mm	
Provided spacing of reinf.	=	125 mm	
Provided area of the steel	=	628 mm ²	
Provided % of steel	=	0.37 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.01 N/mm ²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	154 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **511 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.23 %** **Provided area of steel is OK**

Provide 10T @ 125 c/c at top (Negative Reinf.)

Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **3.9 m**
Actual l/d ratio = **26**
Provided % of steel at bottom along short span = **0.410 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **109 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **75 mm**
Available effective depth = **184 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.438	44.36	0.25	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.360	36.42	0.21	0.37	0.63	1.25	0	No. shear reinf.

Design of Slab Panel - S15

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x

4.3

Long span of the slab, l_y

5.04

Span ratio, l_y/l_x

1.17

TWO WAY SLAB

Edge Condition

One Short Edge Discontinuous

Loading

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check	
f_s , N/mm ²	158
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Req'd	83
$D_{req'd}$, mm	124
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	22.49	184	0.02	174.80	12	294	384	150	755
Short span at bottom	16.87	184	0.02	174.80	12	293	387	200	566
Long span at top	17.75	173	0.02	164.35	10	293	269	150	524
Long span at bottom	13.43	173	0.02	164.35	10	293	269	200	393

End Condition : One short edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.047	22.49 kNm
Along short span at bottom	β_{sx} (+)	0.035	16.87 kNm
Along long span at top	β_{sy} (-)	0.037	17.75 kNm
Along long span at bottom	β_{sy} (+)	0.028	13.43 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	294 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	384 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	221 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	513 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	247 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	318 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.31 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	187 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **421 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.23 %** **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **4.3 m**
Actual l/d ratio = **26**
Provided % of steel at bottom along short span = **0.410 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **120 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **83 mm**
Available effective depth = **184 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	#N/A	#N/A	#N/A	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	#N/A	#N/A	#N/A	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.360	40.16	0.23	0.31	0.59	1.18	0	No. shear reinf.

Design of Slab Panel - S16

Material Properties

Grade of the concrete, f_{cu}	=	40	N/mm ²
Grade of the steel, f_y	=	460	N/mm ²
Concrete Density, γ_c	=	24	kN/m ³

Data

Thickness of slab, D	=	225	mm
Width of slab, b	=	1000	mm
Bottom cover	=	35	mm
Short span, l_x	=	1.7	m
Long span, l_y	=	5.04	m
Span ratio, l_y/l_x	=	2.96	Oneway

Edge condition

Both ends continuous ▼

Loading

Self weight of the slab	=	5.4	kN/m ²
Floor finish	=	1.2	kN/m ²
Partitions	=	0	kN/m ²
M&E services	=	0.5	kN/m ²
Total Dead load, Gk	=	7.1	kN/m ²
Total Live load, Qk	=	10	kN/m ²
Design load on slab, $n = 1.4*G_k + 1.6*Q_k$	=	25.94	kN/m ²

Deflection check

f_s , N/mm ²	159
M.F.	2.00
Span/ d_{eff}	26
$d_{eff, reqd}$	33
D_{reqd} , mm	74
$D_{provided}$, mm	225
Safe in deflection	

(-ve) moment at support, $M_t = n * l^2/10$	=	7.50	kN.m
(+ve) moment at span, $M_b = n * l^2/10$	=	7.50	kN.m
Minimum Area of steel (0.13% of Gross c/s area)	=	293	mm ²

Location	K = $M/f_{cu}bd$	d_{eff} , mm	Dia of ϕ	Required		Provided	
				As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	0.01	184	12	293	386	150	755
Short span at bottom	0.01	184	12	293	386	200	566
Distribution, 0.13*b*D			10	293	269	200	393

Area of steel at support

Available effective depth, d_{eff}	=	184	mm
At support, $M_t / f_{cu}bd^2$ (K)	=	0.01	
Z	=	174.8	mm
Required area of the steel, A_s	=	98	mm ²
Minimum Area of steel	=	293	mm ²
Diameter of bar	=	12	mm
Required spacing of reinf.	=	386	mm
Provided spacing of reinf.	=	150	mm
Provided area of the steel	=	754	mm ²
Provided % of steel	=	0.41	%

OK

Provide 12T @ 150 c/c at top (Negative Reinf.)

Area of steel at span

Available effective depth, d_{eff}	=	184 mm	
At support, $Mb / f_{cu}bd^2$	=	0.01	
Z	=	174.80	
Required area of the steel, A_s	=	98 mm ²	
Minimum Area of steel	=	293 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	386 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	566 mm ²	
Provided % of steel	=	0.31 %	OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Distribution steel, 0.13% of Gross C/A	=	293 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	269 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.20 %	OK
Provide 10T @ 200 c/c			

Check for Deflection

BS8110 -Section 3.4.6.5, Table 3.10

Length of shorter span	=	1.7 m	
Actual l/d ratio	=	26	
Provided % of steel at bottom along short span	=	0.310 %	
$f_s = (2*f_y*Req. steel)/(3*Provided steel)$	=	53 N/mm ²	
M.F. for tension reinforcement,	=	2.00	
Required effective depth	=	33 mm	
Available effective depth	=	184 mm	Safe in Deflection

Check for shear

Shear force, V_{sx}	=	22.049 kN
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Distance from support, m	av (m)	Vsx (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_{cs} N/mm ²	Asv/sv	Remarks
d	0.184	17.28	0.09	0.67	1.34	0.00	No. shear reinf.
2d	0.368	12.50	0.07	0.67	0.67	0.00	No. shear reinf.
3d	0.552	7.73	0.04	0.67	0.67	0.00	No. shear reinf.

Design of Slab Panel - S17

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	225 mm
Clear cover	=	35 mm

Short span of the slab, l_x	4.675
Long span of the slab, l_y	5.04
Span ratio, l_y/l_x	1.08

TWO WAY SLAB
Edge Condition
Loading

Two Adjacent Edges Discont.

Self weight of slab	=	5.4 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	7.1 kN/m²
Live load, Qk	=	10 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	25.94 kN/m²

Deflection check	
f_s , N/mm ²	164
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Req'd	102
$D_{req'd}$, mm	143
$D_{prov.}$ mm	225
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	30.63	184	0.03	174.80	12	401	282	150	755
Short span at bottom	23.07	184	0.02	174.80	12	302	375	200	566
Long span at top	25.52	173	0.03	164.35	10	355	221	150	524
Long span at bottom	19.28	173	0.02	164.35	10	293	269	200	393

End Condition : Two adjacent edges discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.054	30.63 kNm
Along short span at bottom	β_{sx} (+)	0.041	23.07 kNm
Along long span at top	β_{sy} (-)	0.045	25.52 kNm
Along long span at bottom	β_{sy} (+)	0.034	19.28 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	184 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	174.80 mm	
Required area of the steel, A_s	=	401 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	282 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	754 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	174.8 mm	
Required area of the steel, A_s	=	302 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	375 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	565 mm ²	
Provided % of steel	=	0.41 %	Provided area of steel is OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	173 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	164.35 mm	
Required area of the steel, A_s	=	355 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	221 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.31 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	164.35 mm	
Required area of the steel, A_s	=	268 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 293 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.23 % **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 4.675 m
Actul l/d ratio = 23
Provided % of steel at bottom along short span = 0.410 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 164 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 102 mm
Available effective depth = 184 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	#N/A	#N/A	#N/A	0.41	0.64	1.28	0	No. shear reinf.
Along shorter direction at discontinuous edge	#N/A	#N/A	#N/A	0.41	0.64	1.28	0	No. shear reinf.
Along longer direction at continuous edge	0.400	48.51	0.28	0.31	0.59	1.18	0	No. shear reinf.

Design of Slab Panel - S18

Material Properties

Grade of the concrete, f_{cu}	=	40	N/mm ²
Grade of the steel, f_y	=	460	N/mm ²
Concrete Density, γ_c	=	24	kN/m ³

Data

Thickness of slab, D	=	225	mm
Width of slab, b	=	1000	mm
Bottom cover	=	35	mm
Short span, l_x	=	1.825	m
Long span, l_y	=	5.04	m
Span ratio, l_y/l_x	=	2.76	Oneway

Edge condition

One end continuous

Loading

Self weight of the slab	=	5.4	kN/m ²
Floor finish	=	1.2	kN/m ²
Partitions	=	0	kN/m ²
M&E services	=	0.5	kN/m ²
Total Dead load, Gk	=	7.1	kN/m ²
Total Live load, Qk	=	10	kN/m ²
Design load on slab, $n = 1.4*G_k + 1.6*Q_k$	=	25.94	kN/m ²

Deflection check

f_s , N/mm ²	159
M.F.	2.00
Span/ d_{eff}	23
$d_{eff, reqd}$	40
D_{reqd} , mm	81
$D_{provided}$, mm	225
Safe in deflection	

(-ve) moment at support, $M_t = n * l^2/10$	=	8.64	kN.m
(+ve) moment at span, $M_b = n * l^2/10$	=	8.64	kN.m
Minimum Area of steel (0.13% of Gross c/s area)	=	293	mm ²

Location	K = $M/f_{cu}bd$	d_{eff} , mm	Dia of ϕ	Required		Provided	
				As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	0.01	184	12	293	386	150	755
Short span at bottom	0.01	184	12	293	386	200	566
Distribution, 0.13*b*D			10	293	269	200	393

Area of steel at support

Available effective depth, d_{eff}	=	184	mm
At support, $M_t / f_{cu}bd^2$ (K)	=	0.01	
Z	=	174.8	mm
Required area of the steel, A_s	=	113	mm ²
Minimum Area of steel	=	293	mm ²
Diameter of bar	=	12	mm
Required spacing of reinf.	=	386	mm
Provided spacing of reinf.	=	150	mm
Provided area of the steel	=	754	mm ²
Provided % of steel	=	0.41	%

OK

Provide 12T @ 150 c/c at top (Negative Reinf.)

Area of steel at span

Available effective depth, d_{eff}	=	184 mm	
At support, $M_b / f_{cu} b d^2$	=	0.01	
Z	=	174.80	
Required area of the steel, A_s	=	113 mm ²	
Minimum Area of steel	=	293 mm ²	
Diameter of bar	=	12 mm	
Required spacing of reinf.	=	386 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	566 mm ²	
Provided % of steel	=	0.31 %	OK
Provide 12T @ 200 c/c at bottom (Positive Reinf.)			

Distribution steel, 0.13% of Gross C/A	=	293 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	269 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.20 %	OK
Provide 10T @ 200 c/c			

Check for Deflection

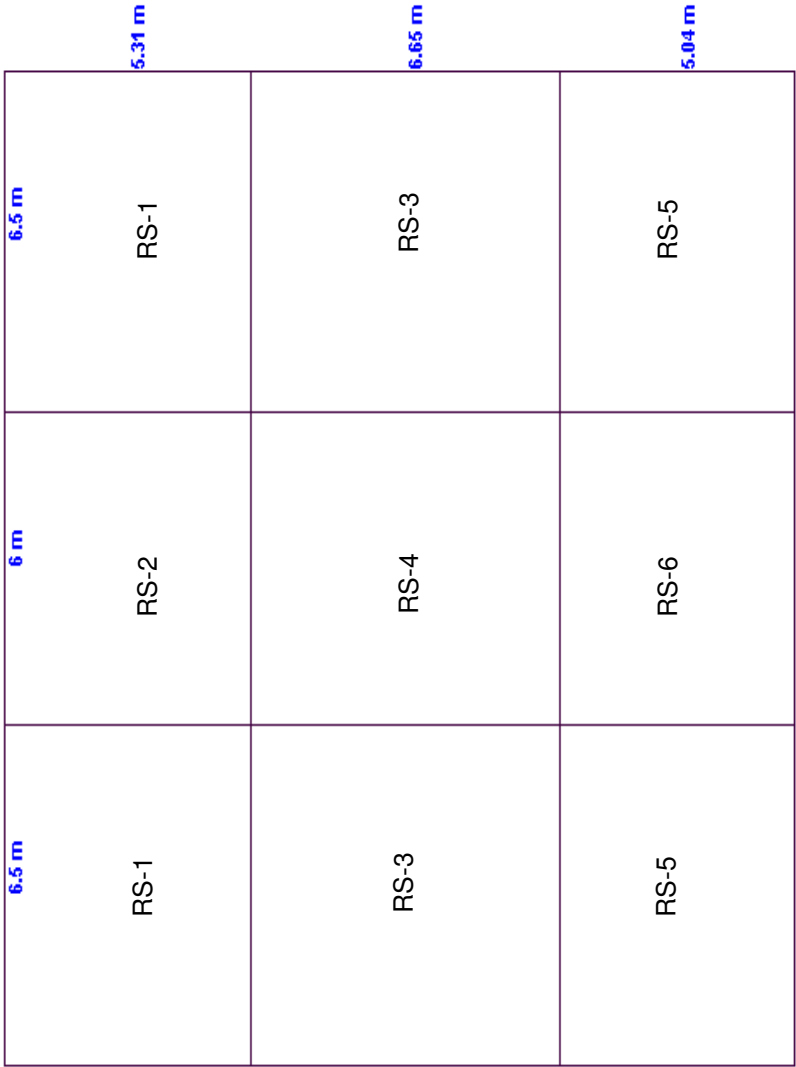
BS8110 -Section 3.4.6.5, Table 3.10

Length of shorter span	=	1.825 m	
Actual l/d ratio	=	23	
Provided % of steel at bottom along short span	=	0.310 %	
$f_s = (2 \cdot f_y \cdot \text{Req. steel}) / (3 \cdot \text{Provided steel})$	=	61 N/mm ²	
M.F. for tension reinforcement,	=	2.00	
Required effective depth	=	40 mm	
Available effective depth	=	184 mm	Safe in Deflection

Check for shear

Shear force, V_{sx}	=	23.67025 kN
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Distance from support, m	av (m)	Vsx (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_{cs} (N/mm ²)	Asv/sv	Remarks
d	0.184	18.90	0.10	0.67	1.34	0.00	No. shear reinf.
2d	0.368	14.12	0.08	0.67	0.67	0.00	No. shear reinf.
3d	0.552	9.35	0.05	0.67	0.67	0.00	No. shear reinf.



Roof Slab Layout Plan

Design of Slab Panel - RS1

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	175 mm
Clear cover	=	35 mm

Short span of the slab, l_x

5.31

Long span of the slab, l_y

6.5

Span ratio, l_y/l_x

1.22

TWO WAY SLAB

Edge Condition

Two Adjacent Edges Discont.

Loading

Self weight of slab	=	4.2 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	5.9 kN/m²
Live load, Qk	=	0.6 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	9.22 kN/m²

Deflection check	
f_s , N/mm ²	178
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Req'd	116
$D_{req'd}$, mm	156
$D_{prov.}$, mm	175
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	16.76	135	0.03	128.25	10	299	263	150	524
Short span at bottom	12.47	135	0.02	128.25	10	228	345	200	393
Long span at top	11.70	125	0.02	118.75	10	228	345	150	524
Long span at bottom	8.84	125	0.02	118.75	10	228	345	200	393

End Condition : Two adjacent edges discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.064	16.76 kNm
Along short span at bottom	β_{sx} (+)	0.048	12.47 kNm
Along long span at top	β_{sy} (-)	0.045	11.70 kNm
Along long span at bottom	β_{sy} (+)	0.034	8.84 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	135 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	299 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	263 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	222 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	353 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	125 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	118.75 mm	
Required area of the steel, A_s	=	225 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	348 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.42 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	118.75 mm	
Required area of the steel, A_s	=	170 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 462 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.32 % **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 5.31 m
Actual l/d ratio = 23
Provided % of steel at bottom along short span = 0.390 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 174 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 115 mm
Available effective depth = 135 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.477	23.36	0.18	0.39	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.315	15.41	0.12	0.39	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.400	19.58	0.16	0.42	0.71	1.42	0	No. shear reinf.

Design of Slab Panel - RS2

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	175 mm
Clear cover	=	35 mm

Short span of the slab, l_x

5.31

Long span of the slab, l_y

6

Span ratio, l_y/l_x

1.13

TWO WAY SLAB

Edge Condition

One long Edge Discont.

Loading

Self weight of slab	=	4.2 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	5.9 kN/m²
Live load, Qk	=	0.6 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	9.22 kN/m²

Deflection check	
f_s , N/mm ²	178
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Reqd	116
D_{reqd} , mm	156
D_{prov} , mm	175
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	13.29	135	0.02	128.25	10	237	331	150	524
Short span at bottom	9.83	135	0.02	128.25	10	228	345	200	393
Long span at top	9.62	125	0.02	118.75	10	228	345	150	524
Long span at bottom	7.28	125	0.02	118.75	10	228	345	200	393

End Condition : One long edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.051	13.29 kNm
Along short span at bottom	β_{sx} (+)	0.038	9.83 kNm
Along long span at top	β_{sy} (-)	0.037	9.62 kNm
Along long span at bottom	β_{sy} (+)	0.028	7.28 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	135 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	237 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	331 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ²	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	175 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	448 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	125 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	118.75 mm	
Required area of the steel, A_s	=	185 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	424 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.42 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm ²	
Lever arm z	=	118.75 mm	
Required area of the steel, A_s	=	140 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 560 mm
 Provided spacing of reinf. = 200 mm
 Provided area of the steel = 393 mm²
 Provided % of steel = 0.32 % **Provided area of steel is OK**

Provide 10T @ 150 c/c at top (Negative Reinf.)

Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 5.31 m
 Actul l/d ratio = 23
 Provided % of steel at bottom along short span = 0.390 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 137 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
 Required effective depth = 115 mm
 Available effective depth = 135 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.412	20.17	0.15	0.39	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.276	13.51	0.11	0.39	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.360	17.62	0.14	0.42	0.71	1.42	0	No. shear reinf.

Design of Slab Panel - RS3

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	175 mm
Clear cover	=	35 mm

Short span of the slab, l_x

6.5

Long span of the slab, l_y

6.65

Span ratio, l_y/l_x

1.02

TWO WAY SLAB

Edge Condition

One long Edge Discont.

Loading

Self weight of slab	=	4.2 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	5.9 kN/m²
Live load, Qk	=	0.6 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	9.22 kN/m²

Deflection check

f_s , N/mm ²	178
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Reqd	125
D_{reqd} , mm	165
$D_{prov.}$ mm	175
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	16.10	135	0.03	128.25	10	287	273	150	524
Short span at bottom	12.23	135	0.02	128.25	10	228	345	200	393
Long span at top	14.42	125	0.03	118.75	10	278	283	150	524
Long span at bottom	10.91	125	0.02	118.75	10	228	345	200	393

End Condition : One long edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.041	16.10 kNm
Along short span at bottom	β_{sx} (+)	0.031	12.23 kNm
Along long span at top	β_{sy} (-)	0.037	14.42 kNm
Along long span at bottom	β_{sy} (+)	0.028	10.91 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	135 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	287 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	273 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	218 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	360 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	125 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	118.75 mm	
Required area of the steel, A_s	=	278 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	283 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.42 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	118.75 mm	
Required area of the steel, A_s	=	210 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 374 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.32 % **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 6.5 m
Actul l/d ratio = 26
Provided % of steel at bottom along short span = 0.390 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = 170 N/mm²
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = 2.00
Required effective depth = 125 mm
Available effective depth = 135 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.369	22.13	0.17	0.39	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.247	14.80	0.11	0.39	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.360	21.57	0.17	0.42	0.71	1.42	0	No. shear reinf.

Design of Slab Panel - RS4

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	175 mm
Clear cover	=	35 mm

Short span of the slab, l_x

6

Long span of the slab, l_y

6.65

Span ratio, l_y/l_x

1.11

TWO WAY SLAB

Edge Condition

Interior Panel

Loading

Self weight of slab	=	4.2 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	5.9 kN/m²
Live load, Qk	=	0.6 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	9.22 kN/m²

Deflection check	
f_s , N/mm ²	178
M.F.	2.00
Span/ d_{eff}	26
d_{eff} Req'd	116
$D_{req'd}$, mm	156
$D_{prov.}$ mm	175
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	12.42	135	0.02	128.25	10	228	345	150	524
Short span at bottom	9.41	135	0.02	128.25	10	228	345	200	393
Long span at top	10.63	125	0.02	118.75	10	228	345	150	524
Long span at bottom	7.97	125	0.02	118.75	10	228	345	200	393

End Condition : Interior panel

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.037	12.42 kNm
Along short span at bottom	β_{sx} (+)	0.028	9.41 kNm
Along long span at top	β_{sy} (-)	0.032	10.63 kNm
Along long span at bottom	β_{sy} (+)	0.024	7.97 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	135 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	222 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	354 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	168 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	468 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	125 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	118.75 mm	
Required area of the steel, A_s	=	205 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	383 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.42 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	118.75 mm	
Required area of the steel, A_s	=	154 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = 512 mm
Provided spacing of reinf. = 200 mm
Provided area of the steel = 393 mm²
Provided % of steel = 0.32 % **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = 6 m
Actual l/d ratio = 26
Provided % of steel at bottom along short span = 0.390 %
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel}) = 131 \text{ N/mm}^2$
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))] = 2.00$
Required effective depth = 115 mm
Available effective depth = 135 mm **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.363	20.05	0.15	0.39	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.000	0.00	0	0.39	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.330	18.26	0.15	0.42	0.71	1.42	0	No. shear reinf.

Design of Slab Panel - RS5

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	175 mm
Clear cover	=	35 mm

Short span of the slab, l_x

5.04

Long span of the slab, l_y

6.5

Span ratio, l_y/l_x

1.29

TWO WAY SLAB

Edge Condition

Two Adjacent Edges Discont.

Loading

Self weight of slab	=	4.2 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	5.9 kN/m²
Live load, Qk	=	0.6 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	9.22 kN/m²

Deflection check	
f_s , N/mm ²	178
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Req'd	110
$D_{req'd}$, mm	150
$D_{prov.}$, mm	175
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	16.02	135	0.03	128.25	10	286	275	150	524
Short span at bottom	11.85	135	0.02	128.25	10	228	345	200	393
Long span at top	10.54	125	0.02	118.75	10	228	345	150	524
Long span at bottom	7.97	125	0.02	118.75	10	228	345	200	393

End Condition : Two adjacent edges discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.068	16.02 kNm
Along short span at bottom	β_{sx} (+)	0.051	11.85 kNm
Along long span at top	β_{sy} (-)	0.045	10.54 kNm
Along long span at bottom	β_{sy} (+)	0.034	7.97 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	135 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.03 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	286 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	275 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	211 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	372 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	125 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	118.75 mm	
Required area of the steel, A_s	=	203 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	387 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.42 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	118.75 mm	
Required area of the steel, A_s	=	154 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **512 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.32 %** **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **5.04 m**
Actual l/d ratio = **23**
Provided % of steel at bottom along short span = **0.390 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **165 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **110 mm**
Available effective depth = **135 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.497	23.09	0.18	0.39	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.328	15.24	0.12	0.39	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.400	18.59	0.15	0.42	0.71	1.42	0	No. shear reinf.

Design of Slab Panel - RS6

Material Properties

Characteristic strength of concrete, f_{cu}	=	40 N/mm ²
Characteristic strength of steel, f_y	=	460 N/mm ²
Concrete Density, γ_c	=	24 kN/m ³

Data

Thickness of slab	=	175 mm
Clear cover	=	35 mm

Short span of the slab, l_x

5.04

Long span of the slab, l_y

6

Span ratio, l_y/l_x

1.19

TWO WAY SLAB

Edge Condition

One long Edge Discont.

Loading

Self weight of slab	=	4.2 kN/m ²
Floor finish	=	1.2 kN/m ²
Partitions	=	0 kN/m ²
M&E services	=	0.5 kN/m ²
Total Dead load, Gk	=	5.9 kN/m²
Live load, Qk	=	0.6 kN/m²
Design load, N= 1.4*Gk+1.6*Qk	=	9.22 kN/m²

Deflection check	
f_s , N/mm ²	178
M.F.	2.00
Span/ d_{eff}	23
d_{eff} Req'd	110
D_{reqd} , mm	150
D_{prov} , mm	175
Safe in deflection	

Location	B.M., kN.m	d_{eff} , mm	K = $M/f_{cu}b d^2$	Lever Arm z	Dia of bar ϕ	Required Reinforcement		Provided Reinforcement	
						As, mm ²	Spacing	Spacing	As, mm ²
Short span at top	12.96	135	0.02	128.25	10	231	340	150	524
Short span at bottom	9.71	135	0.02	128.25	10	228	345	200	393
Long span at top	8.67	125	0.02	118.75	10	228	345	150	524
Long span at bottom	6.56	125	0.02	118.75	10	228	345	200	393

End Condition : One long edge discontinuous

Coefficients from BS 8110-1, Table 3.14

Along short span at top	β_{sx} (-)	0.055	12.96 kNm
Along short span at bottom	β_{sx} (+)	0.041	9.71 kNm
Along long span at top	β_{sy} (-)	0.037	8.67 kNm
Along long span at bottom	β_{sy} (+)	0.028	6.56 kNm

Reinforcement along the Short Span

Available effective depth, d_{eff}	=	135 mm	
At Top, $K = m_{sx} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	231 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	340 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 150 c/c at top (Negative Reinf.)			

At bottom, $K = m_{sx} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	128.25 mm	
Required area of the steel, A_s	=	173 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	454 mm	
Provided spacing of reinf.	=	200 mm	
Provided area of the steel	=	393 mm ²	
Provided % of steel	=	0.39 %	Provided area of steel is OK
Provide 10T @ 200 c/c at bottom (Positive Reinf.)			

Reinforcement along the Long Span

Available effective depth, d_{eff}	=	125 mm	
$K = m_{sy} (Top) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm² OK	
K'	=	0.156 (If redistribution not exceed 10%)	
Lever Arm z	=	118.75 mm	
Required area of the steel, A_s	=	167 mm ²	
Diameter of bar	=	10 mm	
Required spacing of reinf.	=	470 mm	
Provided spacing of reinf.	=	150 mm	
Provided area of the steel	=	524 mm²	
Provided % of steel	=	0.42 %	Provided area of steel is OK
$K = m_{sy} (Bot) / b \cdot d^2 \cdot f_{cu}$	=	0.02 N/mm²	
Lever arm z	=	118.75 mm	
Required area of the steel, A_s	=	126 mm ²	
Diameter of bar	=	10 mm	

Required spacing of reinf. = **622 mm**
Provided spacing of reinf. = **200 mm**
Provided area of the steel = **393 mm²**
Provided % of steel = **0.32 %** **Provided area of steel is OK**
Provide 10T @ 150 c/c at top (Negative Reinf.)
Provide 10T @ 200 c/c at bottom (Positive Reinf.)

Check for Deflection

BS8110- Section 3.4.6.5, Table 3.10

Length of shorter span = **5.04 m**
Actual l/d ratio = **23**
Provided % of steel at bottom along short span = **0.390 %**
 $f_s = (2*f_y * \text{Req. steel}) / (3 * \text{Provided steel})$ = **135 N/mm²**
 $0.55 + [(477 - f_s) / (120 * (0.9 + M/bd^2))]$ = **2.00**
Required effective depth = **110 mm**
Available effective depth = **135 mm** **Safe in Deflection**

Check for shear

Shear coefficients from Table 3.15

v_c from Table 3.8

Location	Coeff.	S.For ce V (kN)	v N/mm ²	% of steel	v_c , N/mm ²	Enhanced v_c , N/mm ²	A_{sv}/s_v	Remarks
Along shorter direction at continuous edge	0.436	20.27	0.16	0.39	0.68	1.36	0	No. shear reinf.
Along shorter direction at discontinuous edge	0.288	13.39	0.1	0.39	0.68	1.36	0	No. shear reinf.
Along longer direction at continuous edge	0.360	16.73	0.13	0.42	0.71	1.42	0	No. shear reinf.

7 Base Plate Design

Base Plate Design Type-I

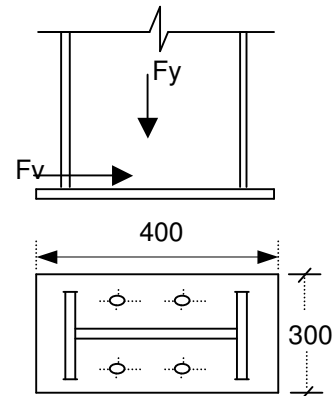
Load case No. 214 214 1.4DL+1.4WL5 [Ref. Node on Staad = 14]

Applied Load

		Force (kN)	Nature
Axial force (P)	=	126.439	Compression
	=	54.223	Tension
Horizontal Shear (V)	=	18.038	Compression
Horizontal Shear (V)	=	25.76	Tension

Column Section

Depth ,(D)	=	300	mm
Width ,(B)	=	200	mm
Flange thickness ,(tf)	=	10	mm
Web thickness ,(tw)	=	6	mm



Bolt

Bolt grade	=	8.8	
Bolt size	=	M 16	
No. of bolt required (n)	=	4	nos.
Tension capacity , T _b	=	88	kN
Shear capacity , v _b	=	59	kN
Shear in each bolt	=	$\sqrt{(18.038^2 + 25.76^2)} / 4$	
	=	7.87	kN
		< Shear capacity	Hence OK
Tension in each bolt	=	54.223/4	
	=	13.56	kN
		< Tension capacity	Hence OK

Combined check for tension and shear

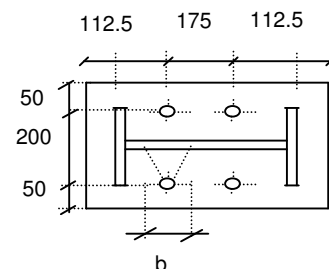
(Actual tension /Capacity in tension)+(Actual Shear /Capacity in Shear) < 1.4

$$\frac{13.56}{88} + \frac{7.87}{59} < 1.4$$

0.29 < 1.4, Hence Ok

Base Plate

Width of Plate ,(B _P)	=	300	mm
Length of Plate ,(L _P)	=	400	mm
Thicknees of Plate ,(T _P)	=	20	mm
p _{yp}	=	345	N/mm ²
f _{cu}	=	30	N/mm ²
0.6 f _{cu}	=	18	N/mm ²
Tension in one bolt	=	13.56	kN
Edge distance (e)	=	47	mm
Resisting width by 30 deg	=	16 + 2x47 x Tan(30)	
disperssion	=	70.28	mm
Moment due to tension on bolt	=	13.56 x 47	
	=	637.32	kNmm



$$\begin{aligned}\text{Thickness of base plate reqrd } (t_p) &= \sqrt{6xM_{\max}/p_y \times b} \\ &= \sqrt{6 \times 637.3210^3 / (.66 \times 345 \times 70.28)} \\ &= 15.46 \quad \text{mm}\end{aligned}$$

Design by the effective area method

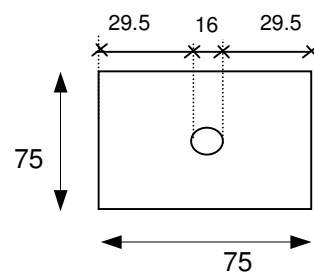
$$\begin{aligned}\text{Area of flange} &= 2000 \text{ mm}^2 \\ \text{x-sectional area} &= 5680 \text{ mm}^2 \\ \text{Column perimeter} &= 1388 \text{ mm} \\ \text{Area of base Plate} &= 120000 \text{ mm}^2 \\ \text{Area Required} &= 7024.39 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Bearing Area} &= \text{Hatched Area} \\ 4c^2 + (\text{column perimeter}) \times c + \text{column area} &= \text{Hatched Area} \\ 4c^2 + 1388 \times c + 5680 &= 7024.39 \\ c &= 1 \quad \text{mm}\end{aligned}$$

$$\begin{aligned}\text{Thickness of Base Plate } (t_p) &= c[3w/p_{yp}]^{0.5} \\ \text{Reqd.} &= 0.4 \quad \text{mm} \\ \text{Provide thickness of base plate} &= 20 \quad \text{mm} \\ &> \text{Reqrd, Hence Ok !}\end{aligned}$$

Design of anchor plate

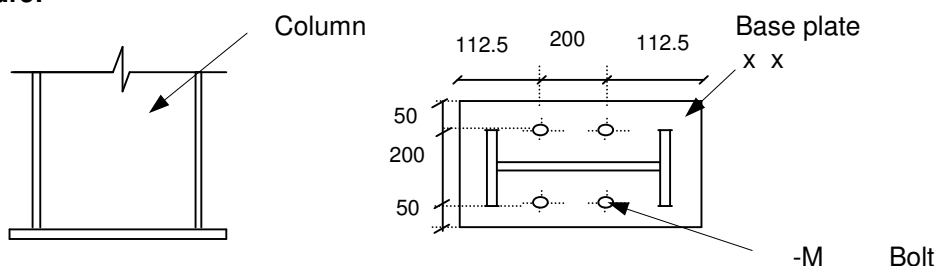
$$\begin{aligned}\text{Maxm tension force on each bolt} &= 13.56 \quad \text{kN} \\ \text{Size of anchor plate} &= 75 \times 75 \\ \text{Pressure on anchor plate} &= 13.56 / (75 \times 75) \\ \text{due to bolt} &= 2410.67 \text{ kN/mm}^2 \\ \text{Edge dist from , e} &= 29.5 \quad \text{mm} \\ \text{outer face of bolt}\end{aligned}$$



$$\begin{aligned}\text{Bending moment at} &= 2410.67 \times 29.5^2 / 2 \times 75 \\ \text{outer face of bolt} &= 78.68 \quad \text{kNm} \\ \text{Bending stress} &= 0.66 p_y \\ &= 227.7 \quad \text{N/mm}^2\end{aligned}$$

$$\begin{aligned}\text{Thickness reqrd of anchor} &= \sqrt{6 \times 78.68 / (75 \times 227.7)} \\ \text{plate} &= 5.26 \quad \text{mm} \\ \text{Provide thick of anchor} &= 10 \quad \text{mm} \quad \text{Hence OK} \\ \text{plate}\end{aligned}$$

Figure:



Base Plate Design Type-2 [Roof level]

[Ref. Node on Staad = 96]

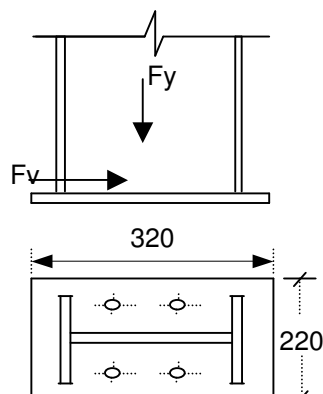
Load case No. 207 1.2DL+1.2LL+1.2WL3

Applied Load

		Force (kN)	Nature
Axial force (P)	=	116.52	Compression
	=	30.58	Tension
Horizontal Shear (V)	=	21.812	Compression
Horizontal Shear (V)	=	7.087	Tension

Column Section

Depth ,(D)	=	300	mm
Width ,(B)	=	200	mm
Flange thickness ,(tf)	=	8	mm
Web thickness ,(tw)	=	6	mm



Bolt

Bolt grade		8.8	
Bolt size		M 16	
No. of bolt required (n)	=	4	nos.
Tension capacity , Tb	=	88	kN
Shear capacity , vb	=	59	kN
Shear in each bolt	=	$\sqrt{(21.812^2 + 7.087^2)} / 4$	
	=	5.74	kN
		< Shear capacity	Hence OK
Tension in each bolt	=	30.58/4	
	=	7.65	kN
		< Tension capacity	Hence OK

Combined check for tension and shear

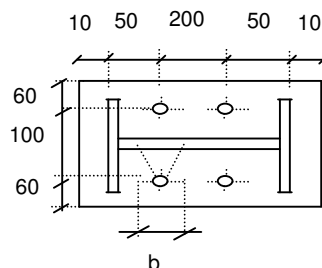
(Actual tension /Capacity in tension)+(Actual Shear /Capacity in Shear) < 1.4

$$\frac{7.65}{88} + \frac{5.74}{59} < 1.4$$

0.19 < 1.4, Hence Ok

Base Plate

Width of Plate ,(Bp)	=	220	mm
Length of Plate , (Lp)	=	320	mm
Thicknees of Plate ,(Tp)	=	20	mm
Pyp	=	345	N/mm2
fcu	=	30	N/mm2
0.6 fcu	=	18	N/mm2
Tension in one bolt	=	7.65	kN
Edge distance (e)	=	47	mm
Resisting width by 30 deg	=	$16 + 2 \times 47 \tan(30)$	
disperssion	=	70.28	mm



Moment due to tension on bolt	=	7.65 x 47	
	=	359.55	kNmm

$$\begin{aligned}\text{Thickness of base plate reqrd (t}_p) &= \sqrt{6M_{\max}/p_y \times b} \\ &= \sqrt{6 \times 359.5510^3 / (.66 \times 345 \times 70.28)} \\ &= 11.62 \quad \text{mm}\end{aligned}$$

Design by the effective area method

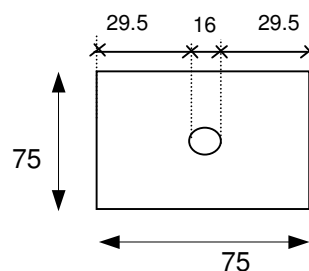
$$\begin{aligned}\text{Area of flange} &= 1600 \text{ mm}^2 \\ \text{x-sectional area} &= 4904 \text{ mm}^2 \\ \text{Column perimeter} &= 1388 \text{ mm} \\ \text{Area of base Plate} &= 70400 \text{ mm}^2 \\ \text{Area Required} &= 6473.34 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}\text{Bearing Area} &= \text{Hatched Area} \\ 4c^2 + (\text{column perimeter}) \times c + \text{column area} &= \text{Hatched Area} \\ 4c^2 + 1388 \times c + 4904 &= 6473.34 \\ c &= 1.2 \quad \text{mm}\end{aligned}$$

$$\begin{aligned}\text{Thickness of Base Plate (t}_p) &= c[3w/p_{yp}]^{0.5} \\ \text{Reqd.} &= 0.5 \quad \text{mm} \\ \text{Provide thickness of base plate} &= 20 \quad \text{mm} \\ &> \text{Reqrd, Hence Ok !}\end{aligned}$$

Design of anchor plate

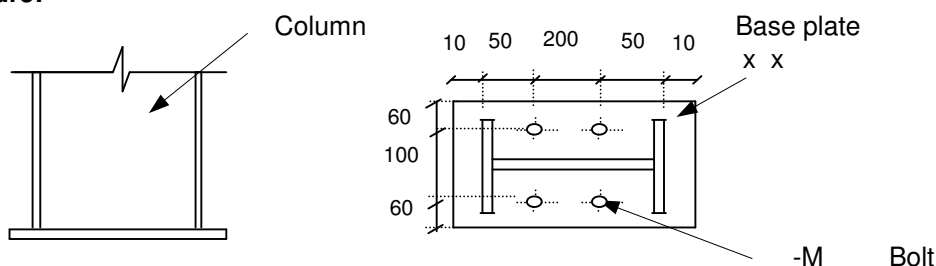
$$\begin{aligned}\text{Maxm tension force on each bolt} &= 7.65 \quad \text{kN} \\ \text{Size of anchor plate} &= 75 \times 75 \\ \text{Pressure on anchor plate} &= 7.65/(75 \times 75) \\ \text{due to bolt} &= 1360 \quad \text{kN/mm}^2 \\ \text{Edge dist from , e} &= 29.5 \quad \text{mm} \\ \text{outer face of bolt}\end{aligned}$$



$$\begin{aligned}\text{Bending moment at} &= 1360 \times 29.5^2 / 2 \times 75 \\ \text{outer face of bolt} &= 44.39 \quad \text{kNm} \\ \text{Bending stress} &= 0.66 p_y \\ &= 227.7 \quad \text{N/mm}^2\end{aligned}$$

$$\begin{aligned}\text{Thickness reqrd of anchor} &= \sqrt{6 \times 44.39 / (75 \times 227.7)} \\ \text{plate} &= 3.95 \quad \text{mm} \\ \text{Provide thick of anchor} &= 10 \quad \text{mm} \quad \text{Hence OK} \\ \text{plate}\end{aligned}$$

Figure:



Base Plate Design Type-3 [Roof level]

[Ref. Node on Staad = 109]

Load case No. 207 1.2DL+1.2LL+1.2WL3

Applied Load

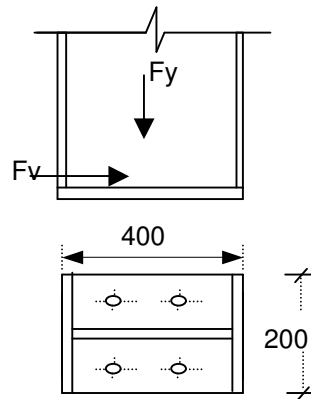
		Force (kN)	Nature
Axial force (P)	=	119.699	Compression
	=	66.961	Tension

Column Section

Depth ,(D)	=	400	mm
Width ,(B)	=	200	mm
Flange thickness ,(tf)	=	10	mm
Web thickness ,(tw)	=	10	mm

Bolt

Bolt grade		8.8	
Bolt size		M 16	
No. of bolt required (n)	=	4	nos.
Tension capacity , T_b	=	88	kN
Shear capacity , v_b	=	59	kN



Tension in each bolt	=	66.961/4	
	=	16.75	kN
		< Tension capacity	Hence OK

Combined check for tension and shear

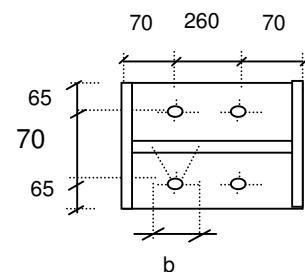
(Actual tension /Capacity in tension)+(Actual Shear /Capacity in Shear) < 1.4

$$\frac{16.75}{88} + \frac{0.2}{0.2} < 1.4$$

Hence Ok

Base Plate

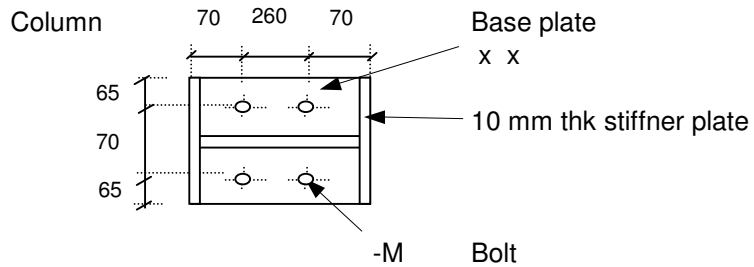
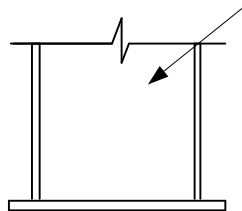
Width of Plate ,(B _P)	=	200	mm
Length of Plate ,(L _P)	=	400	mm
Thicknees of Plate ,(T _P)	=	20	mm
p_{yp}	=	345	N/mm ²
f_{cu}	=	30	N/mm ²
0.6 f_{cu}	=	18	N/mm ²
Tension in one bolt	=	16.75	kN
Edge distance (e)	=	30	mm
Resisting width by 30 deg dispersion	=	$16 + 2 \times 30 \tan(30)$	mm
	=	50.65	mm



Moment due to tension on bolt	=	16.75 x 30	
	=	502.5	kNmm

$$\begin{aligned}
 \text{Thickness of base plate } (t_p) &= \sqrt{\frac{6M_{\max}}{p_y \times b}} \\
 \text{reqrd} &= \sqrt{\frac{6 \times 502.510^3}{.66 \times 345 \times 50.65}} \\
 &= 16.17 \text{ mm} \\
 \text{Provide thickness of base plate} &= 20 \text{ mm} \\
 &> \text{Reqrd, Hence Ok !}
 \end{aligned}$$

Figure:



Base Plate Design Type-4 [Roof level]

[Ref. Node on Staad = 116]

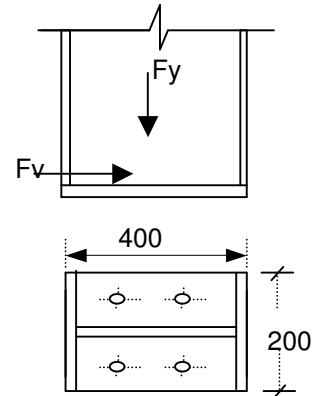
Load case No. 215 1DL+1.4WL5

Applied Load

		Force (kN)	Nature
Axial force (P)	=	45.735	Compression
	=	45.782	Tension

Column Section

Depth ,(D)	=	400	mm
Width ,(B)	=	200	mm
Flange thickness ,(tf)	=	10	mm
Web thickness ,(tw)	=	10	mm



Bolt

Bolt grade		8.8	
Bolt size		M 16	
No. of bolt required (n)	=	4	nos.
Tension capacity , T _b	=	88	kN
Shear capacity , v _b	=	59	kN

Tension in each bolt	=	45.782/4	
	=	11.4455	kN
		< Tension capacity	Hence OK

Combined check for tension and shear

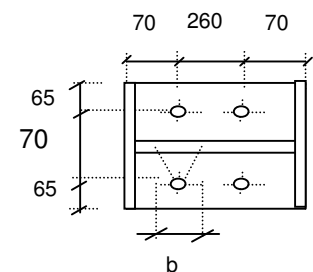
(Actual tension /Capacity in tension)+(Actual Shear /Capacity in Shear) < 1.4

$$\frac{11.4455}{88} + \frac{0.14}{59} < 1.4$$

Hence Ok

Base Plate

Width of Plate ,(B _P)	=	200	mm
Length of Plate , (L _P)	=	400	mm
Thicknees of Plate ,(T _P)	=	20	mm
p _{yp}	=	345	N/mm ²
f _{cu}	=	30	N/mm ²
0.6 f _{cu}	=	18	N/mm ²
Tension in one bolt	=	11.4455	kN
Edge distance (e)	=	30	mm
Resisting width by 30 deg dispersion	=	16 + 2x30Tan(30)	mm
	=	50.65	mm



Moment due to tension on bolt	=	11.4455 x 30	
	=	343.365	kNmm

$$\begin{aligned}
 \text{Thickness of base plate } (t_p) &= \sqrt{6M_{\max} / p_y \times b} \\
 \text{reqrd} &= \sqrt{6 \times 343.36510^3 / (.66 \times 345 \times 50.65)} \\
 &= 13.37 \quad \text{mm} \\
 \text{Provide thickness of base plate} &= 20 \quad \text{mm} \\
 &> \text{Reqrd, Hence Ok !}
 \end{aligned}$$

Design of anchor plate

$$\begin{aligned}
 \text{Maxm tension force on each bolt} &= 11.4455 \quad \text{kN} \\
 \text{Size of anchor plate} &= 75 \times 75 \\
 \text{Pressure on anchor plate} &= 11.4455 / (75 \times 75) \\
 \text{due to bolt} &= 2034.76 \quad \text{kN/mm}^2 \\
 \text{Edge dist from , e} &= 29.5 \quad \text{mm} \\
 \text{outer face of bolt} &= 29.5 \quad \text{mm} \\
 \text{Bending moment at} &= 2034.76 \times 29.5^2 / 2 \times 75 \\
 \text{outer face of bolt} &= 66.41 \quad \text{kNm} \\
 \text{Bending stress} &= 0.66 p_y \\
 &= 227.7 \quad \text{N/mm}^2 \\
 \text{Thickness reqrd of anchor} &= \sqrt{6 \times 66.41 / (75 \times 227.7)} \\
 \text{plate} &= 4.84 \quad \text{mm} \\
 \text{Provide thick of anchor} &= 10 \quad \text{mm} \quad \text{Hence OK} \\
 \text{plate} &
 \end{aligned}$$

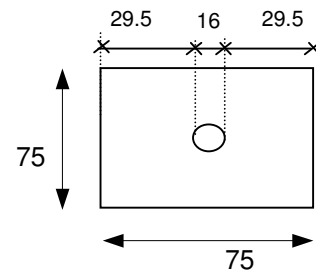
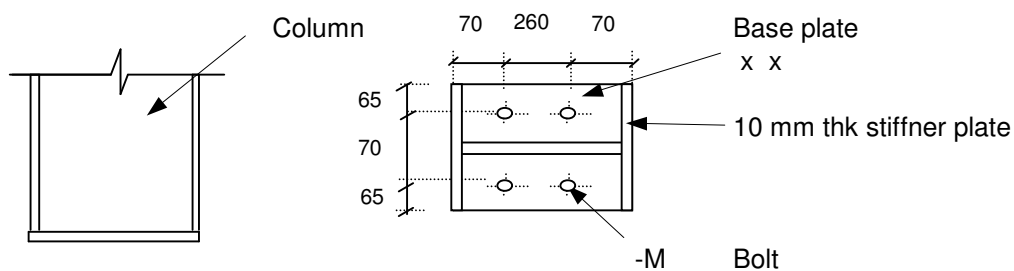


Figure:



8 Connection Design

Moment connection of Beam to Rafter

For Beams

SB2

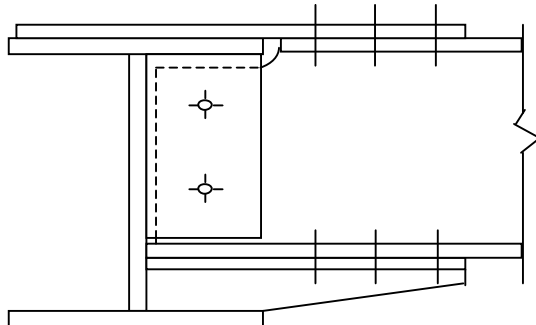
[Ref. Node on Staad 46]

Data :	Load case	207 1.2DL+1.2LL+1.2WL3	
Section	Beam	SB2 Rafter	MF1
Properties of section	h	300 mm	400 mm
	b	150 mm	200 mm
	t _w	6 mm	6 mm
	t _f	10 mm	10 mm
	r mm	0 mm	0 mm
Forces :	Axial force N	3.389 kN	
	Vert shear force T	34.624 kN	
	Hori shear force H	0 kN	
	Majr axis moment(M)	44.724 kN-m	

Geometrical data :

Bolt grade	8.8	
Shear stgth of Bolt p_s	375	N/mm ²
Tension stgth p_t	560	N/mm ²
Dia of bolt	M 20	
Net Area of Bolt A_{net}	245	mm ²
Steel grade	S 345	
Bearing stgth p_{bs}	460	N/mm ²
Ult tensile stgth(U_s)	410	N/mm ²

Calculation :



Lever Arm	=	300	mm	
Bolt Tension due to M	=	Moment / Lev arm		
(T)	=	44.724x10 ⁶ / 300		
	=	149.1	kN	
Total Tension in top plate	=	N/2 + T		
	=	3.389/2 + 149.1	=	150.8 kN
Therefore total shear in Bolt	=	150.8	kN	
Dia of Bolt for shear	=	M 20		$A_{net} = 245$
Shear capacity of one bolt	=	$A_{net} \times p_s / 1000$		
	=	245x375 / 1000		
	=	91.9	kN	
No of bolts reqd for shear	=	V / Shear capacity of Bolt		
	=	150.7945/91.9		
	=	1.60		
Provide no of bolts	=	3	No's	ok
(At Top & same at Botm)				

Shear in each bolt = 50.26 kN

Flange Plate

Trial plate thk (top & botm) = 15 mm
Bearing strength of plate = $(22 \times 15 \times 460 / 1000)$
= 151.8 kN ok

Tensile strenght of plate = Net c/s plt area x 345/1000
Width of plate provided = 150 mm
Length of plate provided = 435 mm

Net C/s area of plate = $(150 - 22 \times 2) \times 15 = 1590 \text{ mm}^2$

Thrfore Tensile strength = $(1590 \times 345 / 1000)$
(of plate) = 548.55 kN
> Actual tensn ∴ ok

Fin Plate

dia of bolt M 20
Net Area of Bolt A_{net} 245 mm^2
Shear capacity of one bolt = $A_{net} \times p_s / 1000$
= $245 \times 375 / 1000$
= 92 kN
No of bolt reqrd $T / 92 = 0.4$ No's
Provide NO of bolts $n = 2$
∴ Shear in each bolt 17.312 kN
Pitch $p = 100$ mm

Trial thickness of plate t 15 mm
Provide length of plate $(0.6 \times d_{section})$ 280 mm
Width of plate = $(2 \times \text{Edg dist} + \text{End projction})$
= $2 \times 2(20) + 10 = 90$

Provide width of plate = 90 mm
Size of plate = 90 x 280
Block shear = $0.6 p_y t (L_v + K_e (L_t - k D_h))$
= 387.504 kN
> Actual shear, ∴ OK !

Bearing strength of plate = $t \times \text{hole dia} \times p_{bs}$
= $15 \times 22 \times 460 / 1000$
= 151.8 kN
> Shear in 1 bolt ∴ ok

Bearing strength of supported beam web = $t \times \text{hole dia} \times p_{bs}$
= $6 \times 22 \times 460 / 1000$
= 60.72 KN

Force in bolt due to moment (F_{sm}) $F_v \times a / Z_{bg}$
Where $Z_{bg} = n(n+1)p/6 = 100$
 $F_{sm} = 8.66 \text{ kN}$
Force acting on bolt group $(F_{sv}^2 + F_{sa}^2 + F_{sm}^2)^{1/2}$
 $= ((34.624/2)^2 + (3.389/2)^2 + 8.66^2)^{0.5}$
 $= 19.44 \text{ kN} \quad \text{OK in Bearing}$
Thickness of fin plate > or = 0.15a $\therefore$ ok in torsional bucking
where a = 2x20+10 = 50 mm

Welding calc

Trial Size of weld a = 6 mm
Design weld stgth $p_w = 220 \text{ N/mm}^2$
Strength of weld/mm = $0.7a \times p_w$
 $= 0.7 \times 6 \times 220$
 $= 924$
Length of weld reqrd = $(T^2 + N^2)^{1/2} \times 10^3 / \text{Strgth of weld/mm}$
 $= 37.7$
Length of plate available for weld = 280 x 2
> reqrd $\therefore$ ok

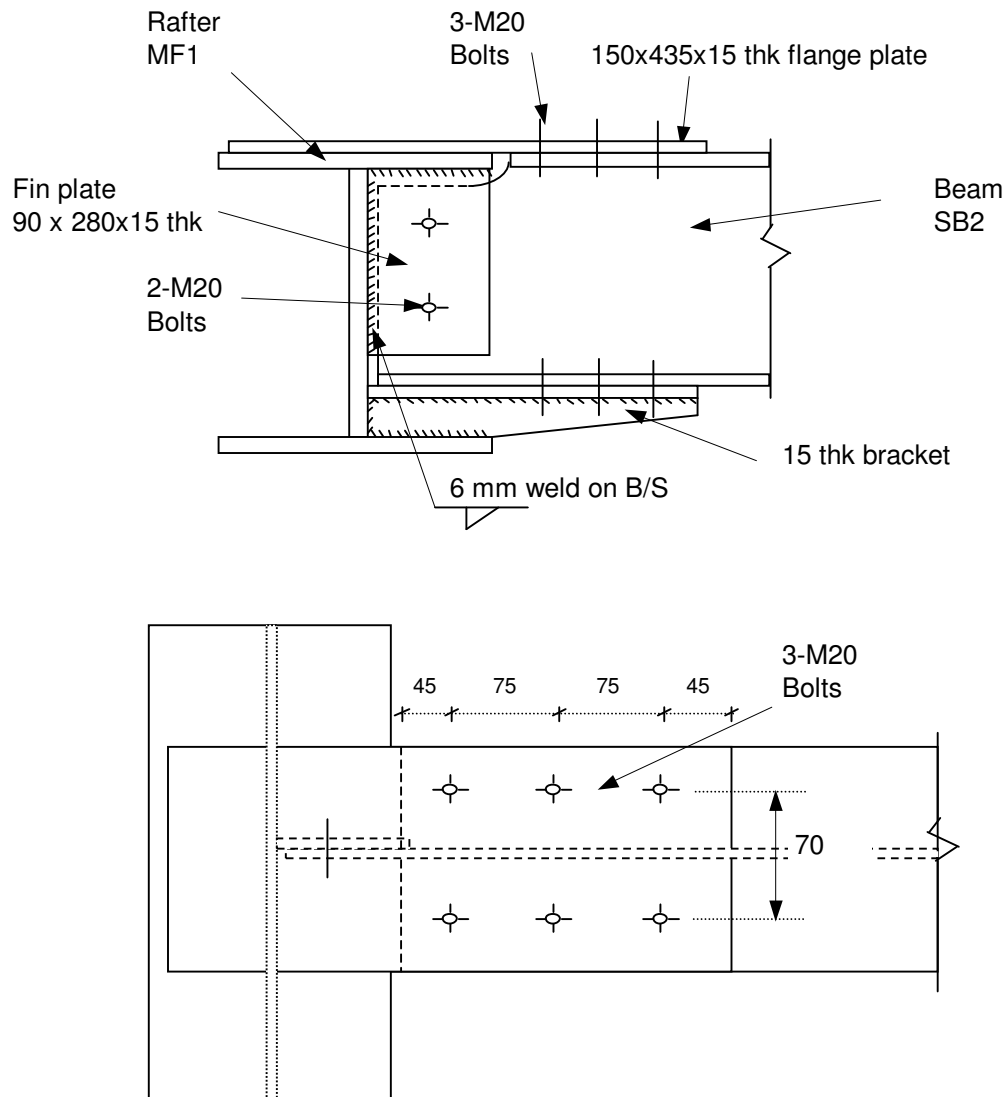
Check for shear capacity of supported Beam flange and web

Block shear capacity of flange = $0.6p_y t(L_v + K_e(L_t - kD_h))$
 $= 0.6 \times 345 \times 10(45 + 1.2(40 - 0.5 \times 22)) / 1000$
 $= 165.19 \text{ kN}$
> Actual shear, $\therefore$ OK !
Block shear capacity of web = $0.6p_y t(L_v + K_e(L_t - kD_h))$
 $= 155.01 \text{ kN}$
> Actual shear, $\therefore$ OK !

Local shear check of Rafter web

I) Local shear failure of the rafter web
Local shear capacity = $0.6SA_v$
 $= 447.12 \text{ kN}$
> Actual beam Reaction $\therefore$ Ok

Figure :



Moment connection of Beam SB2 with Rafter MF1

Rafter to Secondary beam connection

SB2

[Ref. Node on Staad 33]

Data : Load case 207 1.2DL+1.2LL+1.2WL3

Axial force N = 5.396 kN

shear T = 25.9 kN

Section Property	Beam	SB2 Rafter	MF11
h		300 mm	400
b		150 mm	200
t _w		6 mm	6
t _f		10 mm	10
r		0 mm	0

Calculations :

	Bolt grade	8.8	
	Shear stgth of Bolt p_s	375	N/mm ²
	Tension stgth p_t	560	N/mm ²
	Dia of bolt	M 20	
	Dia of hole	22	mm
	Net Area of Bolt A_{net}	245	mm ²
	Steel grade	S 345	N/mm ²
	Bearing stgth p_{bs}	460	N/mm ²
	Ult tensile stgth(U_s)	410	N/mm ²
1	Shear capacity of one bolt	=	$A_{net} \times p_s / 1000$
		=	$345 \times 375 / 1000$
		=	92 kN
	No of bolt reqrd	T/ 92	= 0.3 No's
	Provide NO of bolts n		= 2
	∴ Shear in each bolt		= 12.95 kN
	Pitch p	=	= 100 mm
2	Fin plate		
	Trial thickness of plate t	=	= 15 mm
	Provide length of plate (0.6xd _{section})	=	= 280 mm
	Bearing strength of plate	=	$t \times \text{hole dia} \times p_{bs}$
		=	$15 \times 22 \times 460 / 1000$
		=	151.8 kN
			> Shear in 1 bolt ∴ ok
	Provide width of plate	=	= 90 mm
	Size of plate	=	= 90 x 280

Check

1. Check for bearing capacity of Bolt group

Vertical force on bolt due to shear F_{sv}	=	T/n =	12.95
Force on bolt due to axial F_{sa} (kN)	=	N/n =	2.698
Force on bolt due to moment	=	$F_v \times a / Z_{bg}$	
Where Z_{bg}	=	$n(n+1)p/6$	= 100 mm ³
Force in bolt due to moment F_{sm}	=		12.95
Resultant shear F_s	=	$(F_{sv}^2 + F_{sa}^2 + F_{sm}^2)^{1/2}$	
	=		18.52 kN

$$\begin{aligned}\text{Bearing capa of fin plate } P_{bs} &= k_{bs} d t p_{bs} \\ &= 138 \quad \text{KN} \\ \text{Bearing capa of beam web } P_{bs} &= k_{bs} d t p_{bs} \\ &= 55.2 \quad \text{KN} \\ &\text{Adequate } \therefore \text{Ok}\end{aligned}$$

2. Check for shear & bending capacity of Beam

For shear

$$\begin{aligned}\text{i) Shear capa of the gross section} &= 0.6 p_y A_v \\ &= (0.6 \times 345 \times ((300 - 2 \times (10 + 0)) \times 6)) \\ &= 347.76 \quad \text{kN} \\ \text{ii) Shear capa of the net section} &= 0.7 p_y K_e A_{v,net} \\ &= 0.7 \times 345 \times 1.2 \times (A_v - 2 \times 22 \times 6) \\ &= 410.36 \quad \text{kN} \\ \text{iii) Block shear capacity} &= 0.6 p_y t (L_v + K_e (L_t - k D_h)) \\ &\quad (0.6 \times 345 \times 6 \times ((2 \times 20 + (2 - 1) \times 100 + 1.2 \times ((2 \times 20) - 0.5 \times 22)))) \\ &= 217.1 \quad \text{kN} \\ \text{Therefore Beam shear capacity} &= 217.1 \quad \text{kN} \\ &> \text{Actual Shear } \therefore \text{Ok}\end{aligned}$$

For Bending

$$\begin{aligned}\text{Applied moment } T \times a &= 1.295 \quad \text{kN-m} \\ \text{Moment capacity } M_{cb} = &= p_y Z = 27.05 \quad \text{kN-m} \\ &\text{adequate in bending } \therefore \text{Ok}\end{aligned}$$

3. Check for shear & bending capacity of fin plate

For shear

$$\begin{aligned}\text{i) Shear capa of the gross section} &= 0.6 p_y A_v = 782.46 \quad \text{kN} \\ \text{ii) Shear capa of the net section} &= 0.7 p_y K_e A_{v,net} \\ &= 904.18 \quad \text{kN} \\ \text{iii) Block shear capacity} &= 0.6 p_y t (L_v + K_e (L_t - k D_h)) \\ &= 542.754 \\ \text{Therefore fin plate shear capacity } P_v &= 542.76 \quad \text{kN} \\ &> \text{Actual Shear } \therefore \text{Ok}\end{aligned}$$

For Bending

$$\begin{aligned}\text{Actual shear } T = 25.9 \text{ and } 0.6 \times P_v &= 325.66 \quad \text{kN} \\ T &< 0.6 \times P_v \\ \text{Section is subjected to low shear } \therefore \text{Ok} \\ \text{Moment capacity } M_c &= p_y t [2e + (n - 1)p]^2 / 6 \\ &= 67.62 \quad \text{kN-m} \\ &\text{fin plate adequate } \therefore \text{Ok}\end{aligned}$$

$$\begin{aligned}\text{4. Thickness of fin plate } > \text{ or } &= 0.15a \quad \therefore \text{ok in torsional buckling} \\ \text{where } a &= 2 \times 20 + 15 = 50 \quad \text{mm}\end{aligned}$$

6. Local shear check of Rafter web

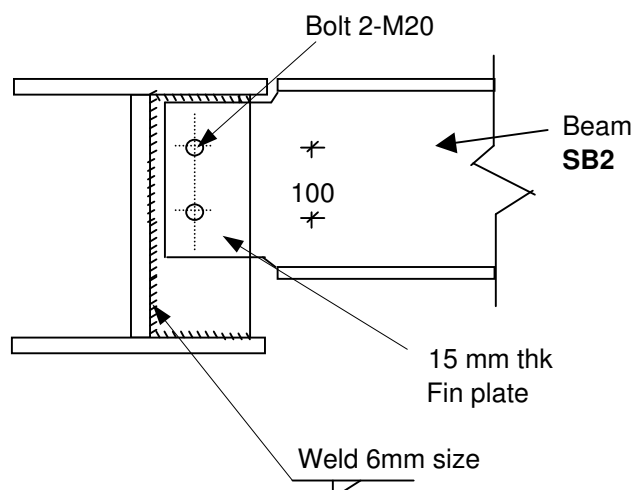
I) Local shear failure of the Rafter web

$$\begin{aligned} \text{Local shear capacity} &= 0.6SA_v \\ &= 312.99 \text{ KN} \\ &> \text{Actual beam Reaction} \therefore \text{Ok} \end{aligned}$$

3 Welding calc

$$\begin{aligned} \text{Trial Size of weld } a &= 6 \text{ mm} \\ \text{Design weld stgth } p_w &= 220 \text{ N/mm}^2 \\ \text{Strength of weld/mm} &= 0.7a \times p_w \\ &= 0.7 \times 6 \times 220 \\ &= 924 \text{ N/mm} \\ \text{Length of weld reqrd} &= (T^2 + N^2)^{1/2} \times 10^3 / \text{Strgth of weld/mm} \\ &= 28.6 \text{ mm} \\ \text{Length of plate available for weld} &= 280 \times 2 \text{ mm} \\ &> \text{reqrd} \therefore \text{ok} \end{aligned}$$

Figure :



Cap Plate Design Type-I

Load case No.206 1DL+1.4WL2

[Ref. Node on Staad 8]

Applied Load

Data :

Section	Stub col.	ST1 Beam	MF1
Properties of section	h	300 mm	400 mm
	b	200 mm	200 mm
	t_w	6 mm	6 mm
	t_f	8 mm	10 mm
	r mm	0 mm	0 mm

Forces :	Axial force N	21.701 kN
	Vert shear force T	3.95 kN
	Hori shear force H	2.6 kN
	Majr axis moment(My)	1.28 kN-m
	Minor axis moment(Mz)	1.98 kN-m

Geometrical data :

Bolt grade	8.8	
Shear stgth of Bolt p_s	375	N/mm ²
Tension stgth p_t	560	N/mm ²
Dia of bolt	M 16	
Net Area of Bolt A_{net}	157	mm ²
Steel grade	S 345	
Bolts Nos in Tension	8	= n
Tension capacity	88	kN

Calculation :

1. Bolt

Tension in each bolt

$$\begin{aligned}
 \text{Due to axial force} &= 21.701/8 \\
 &= 2.72 \text{ kN} \\
 \text{Due to moment} &= 1.28/(100 \times 4) + 1.98/200 \\
 &= 13.1 \text{ kN} \\
 \text{Total tension in each bolt} &= 15.82 \text{ kN} \\
 &< 88 \text{ OK}
 \end{aligned}$$

2. Cap plate

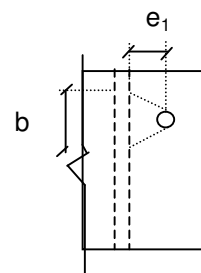
Trial thickness of plate t	20 mm
Provide length of plate ($0.6 \times d_{section}$)	500 mm
Width of Plate	200 mm

$$\text{edge dist } e_1 = 47 \text{ mm}$$

Moment due to tension in each bolt M

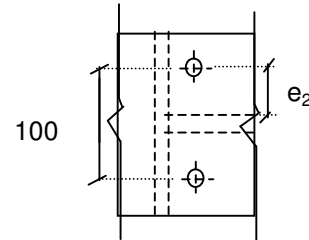
$$\begin{aligned}
 &= 15.82 \times 47 \\
 &= 743.54 \text{ kNm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Resisting width } b &= 16 + 2(47 \tan 30) \\
 \text{By 30deg dissipation} &= 70.28 \text{ mm}
 \end{aligned}$$



$$\begin{aligned}
 \text{Thickness required} &= \sqrt{\frac{6xM/(p_yxb)}{}} \\
 &= \sqrt{\frac{(6x743.54x10^3)/(.66 \times 345 \times 70.28))}{}} \\
 &= 16.697 \quad \text{mm} \\
 &< \text{Plate thk} \quad \text{OK !}
 \end{aligned}$$

3. Check for Beam Flange



$$\text{edge dist from } (e_2) = 47 \quad \text{mm}$$

$$\begin{aligned}
 \text{Moment due to tension in each bolt } M &= 15.82 \times 47 \\
 &= 743.54 \quad \text{kNmm}
 \end{aligned}$$

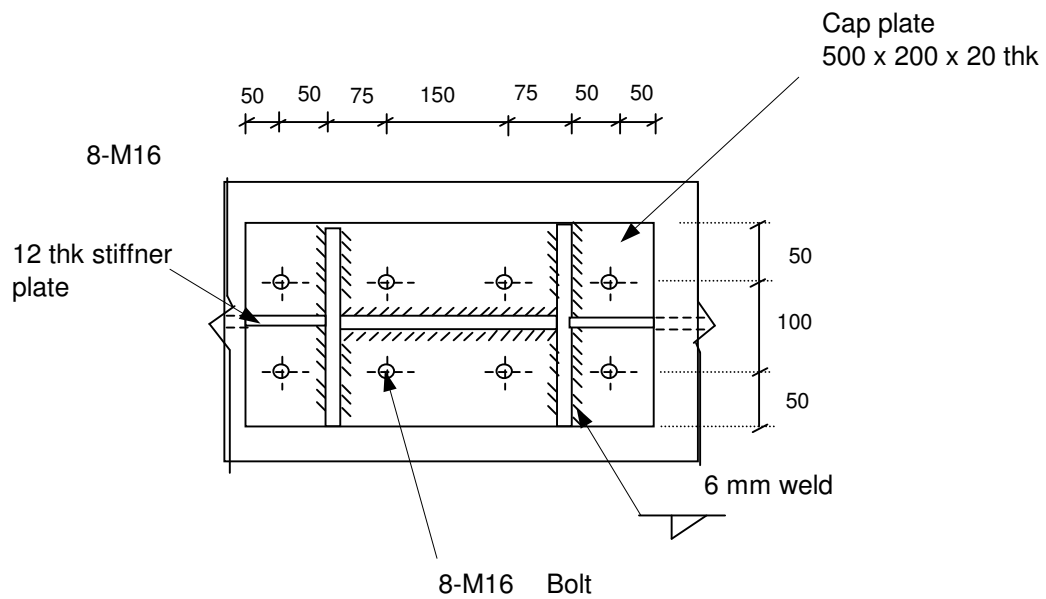
$$\begin{aligned}
 \text{Resisting width } b &= 16 + 2(47 \tan 30) \\
 \text{By 30deg dissperition} &= 70.28 \quad \text{mm} \\
 \text{By three way distribution} &= 3 \times 70.28
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness required} &= \sqrt{\frac{6xM/(p_yxb)}{}} \\
 &= \sqrt{\frac{(6x743.54x10^3)/(.66 \times 345 \times 3 \times 70.28))}{}} \\
 &= 9.64 \quad \text{mm} \\
 &< \text{Flange thk} \quad \text{OK !}
 \end{aligned}$$

4. Welding Calculation

$$\begin{aligned}
 \text{Trial Size of weld } a &= 6 \quad \text{mm} \\
 \text{Design stgth of weld } p_w &= 220 \quad \text{N/mm}^2 \\
 \text{Strength of weld/mm} &= 0.7a \times p_w \\
 &= 0.7 \times 6 \times 220 \\
 &= 924 \quad \text{N/mm} \\
 \text{Resultant force acting on weld (Fr)} &= 56.86 \quad \text{kN} \\
 \text{Length reqd for weld} &= 56.86/924 \\
 &= 61.54 \quad \text{mm} \\
 \text{Length available} &= 1356 \quad \text{mm} \\
 &> \text{reqd,} \quad \text{OK !}
 \end{aligned}$$

Figure:



Purlin Design: (Roof)

	Select a Size	=	Z 200 25	
	Self wt.	=	0.08	kN/m
	Moment of Inertia	=	507.8	cm ⁴
Loadings:				
			(from frame 2 analysis)	
	DL (upper bound)	=	0.4	kPa
	DL (lower bound)	=	0.2	kPa
	LL	=	0.6	kPa
	Maximum Downward wind load, WL3	=	0.93	kN/m
	Maximum Upward wind load, WL4	=	1.83	kN/m
	Span length of purlin	=	3.325	m
	For Purlin spaced at	=	1.5	m c/c
Load Comb. 1	1.4DL + 1.6LL	=	1.52	kN/m
Load Comb. 2	1.7DL + 1.4WL3	=	1.982	kN/m
Load Comb. 3	1.2DL+1.2LL+1.2WL3	=	2.316	kN/m
	Maximum Downward ultimate load	=	0.902	kN/m
	Maximum Upward ultimate load	=	2.162	
	From LYSAGHT catalogue, for		8.1	m
				1.4WL4 - 1.0DL double lapped span
	Purlins with		2	rows of bridging,
	Allowable Downward load	=	3.57	kN/m
	Allowable Upward load	=	3.57	kN/m
				> 0.902
				> 2.162
Deflection Check:				
	0.8(LL+WL3)	=	1.224	kN/m
	WL4 - DL	=	1.43	kN/m
	Design service load	=	1.84	kN/m
	Span of purlin	=	3325	mm
	Permissible Deflection	=	span/200	
		=	16.625	mm
	Deflection	=	5*Load*Span ⁴ /384*E*I	
		=	2.88	mm
		<	16.625	mm

Hence O.K.

Purlin Design: (Side)

	Select a Size	=	Z 200 25	
	Self wt.	=	0.08	kN/m
	Moment of Inertia	=	507.8	cm ⁴
Loadings:				
			(from frame 2 analysis)	
	DL (upper bound)	=	0.4	kPa
	DL (lower bound)	=	0.2	kPa
	LL	=	0	kPa
	Maximum Sidewall wind load, WL3	=	1.7	kN/m
	Maximum Upward wind load, WL4	=	0.94	kN/m
	Span length of purlin	=	3.5	m
	For Purlin spaced at	=	1.5	m c/c
Load Comb. 1	1.4WL3	=	2.38	kN/m
	Maximum Sidewall ultimate load	=	2.38	kN/m
	Maximum Upward ultimate load	=	1.316	
	From LYSAGHT catalogue, for		8.1	m
				1.4WL4 - 1.0DL
	Purlins with		2	rows of bi double lapped span
	Allowable Downward load	=	3.57	kN/m
	Allowable Upward load	=	3.57	kN/m
				> 2.38
				> 1.316
Deflection Check:				
	WL3	=	1.7	kN/m
	Design service load	=	1.84	kN/m
	Span of purlin	=	3500	mm
	Permissible Deflection	=	span/200	
		=	17.5	mm
	Deflection	=	5*Load*Span ⁴ /384*E*I	
		=	3.27	mm
		<	17.5	mm

Hence O.K.

Bracing Connection

UA70x70x6

Load case no. 219	1.2DL+0.5LL+1.0EQX+0.3 EQZ	[Ref. Member on Staad = 522]
Axial force N	Tension 13.992 kN	
	Comp 19.704 kN	
Section Property	Bracing UA70x70x6	<u>safety factor</u>
	h or b 70	Y_m 1.00 material
	A_g 820 mm ²	Y_{m2} 1.20 Net area
	t 6 mm	Y_{Mb} 1.00 Bolts
	r 13.7 mm	Y_{Ov} 1.00 Overstrength
governing radius		

Calculations :

Bolt grade	8.8	
Shear stgth of Bolt p_s	375	N/mm ²
Tension stgth p_t	560	N/mm ²
Steel grade	S 245	N/mm ²
Bearing stgth p_{bs}	460	N/mm ²
Tensile strength f_u	430	N/mm ²
Ult tensile stgth(U_s)	410	N/mm ²
Dia of bolt	M 16	
Net Area of Bolt A_{net}	157	mm ²
Dia of hole	18	mm
Provide NO of bolts n	= 2	nos.
Thk of Gusset plate (t)	= 10	mm

1 Bolt Capacity

Shear capacity of one bolt	=	$A_b \times p_s / 1000$
	=	$157 \times 375 / 1000$
	=	58.88 kN
∴ Shear in each bolt	=	9.85 kN
		< 58.88, ∴ OK !

Bearing stng of 10 mm thk gusset plate

$$= 18 \times 10 \times 460 / 1000$$

$$= 82.8 \text{ kN}$$

2 Checking for compression capacity

Member Length L	=	4450 mm
λ	=	$4450 / 13.7$
	=	324
P_c	=	16.6 kN/mm ² (From curve C)
Compression resistance	=	$820 \times 16.6 / 1000$
	=	27.23 kN
		> 19.704, ∴ OK !

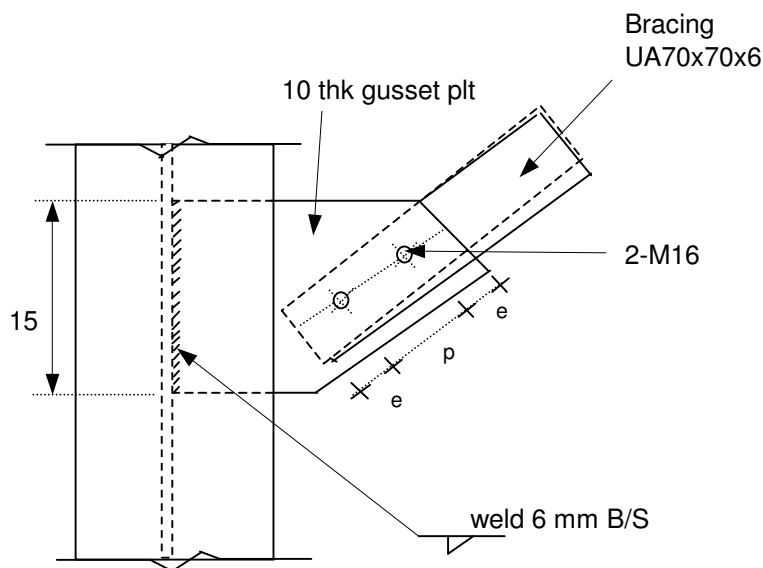
3 Checking for connection design capacity

$$\begin{aligned} \text{Shear stng of member} &= 58.88 \times 2 \\ &= 117.76 \text{ kN} \\ &< \text{Axial load} \\ \text{Bearing stng of plate} &= 82.8 \times 2 \\ &= 165.6 \text{ kN} \\ &< \text{Axial load} \end{aligned}$$

4 Welding calc

$$\begin{aligned} \text{Trial Size of weld } a &= 6 \text{ mm} \\ \text{Design stgth of weld } p_w &= 220 \text{ N/mm}^2 \\ \text{Strength of weld/mm} &= 0.7a \times p_w \\ &= 0.7 \times 6 \times 220 \\ &= 924 \\ \text{Length of weld reqrd} &= \frac{F_r \times 10^3}{\text{Strgth of weld/mm}} \\ &= \frac{19.704 \times 1000}{924} \\ &= 21.3 \text{ mm} \\ &= 10.66 \times 2 \\ \text{Provide} &= 15 \times 2 \end{aligned}$$

Figure :



$$\begin{aligned} e &= 50 \text{ mm} \\ p &= 100 \text{ mm} \end{aligned}$$